UM-3 solution

The approximate solution of the coupled field equations of UM-3: the background and the power series of the metric, the Higgs field in the ad hoc vacuum, the spectra along the hidden direction, the self-consistent Kohn-Sham states of dirac16complex, the residuals of every field equation, the neutrino flavours, the thermal history, and the CMB, the CNB and the BAO

Underlying theory: Patrick L. Nash Solved and verified by Claude (Anthropic)

Solution of UM-3, 2026-10-24 Repository once-ere/Dirac_claude, folder universe_model_3

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1 How to read this document

1.1 What this document contains

This document solves, approximately and step by step, the coupled field equations of the universe model UM-3: Einstein-Lovelock gravity in eight dimensions with the author's metric, the author's field dirac16complex (a 16-component complex spinor field in the bulk), and the Standard Model with three right-handed neutrinos on the fold of the author's metric. It is self-contained: every equation that is solved is written out here (with the tag it carries in every UM-3 document), every new equation of the solution carries a tag S-1, S-2, ..., and every symbol of every display equation is defined right after it. The sections are:

- **The background** (order 0 of the power series of the metric): the expansion histories that the author's metric admits, what they require of the bulk, and the static late-time state.
- **The power series of the metric to first order**: the response of the bulk to the matter of the fold, the radion mode and the condition under which the fold expands as in general relativity.
- **The Higgs field in the ad hoc vacuum**: the author's ad hoc vacuum written as a gauge rotation, the field of W^2 that the Gauss law requires, and the one residual that remains.
- **The spectra along the hidden direction**: the zero modes and the speeds of the bulk fields, the continuum and the bound states of dirac16complex (the x_0 mass).
- **The Kohn-Sham states of dirac16complex**: the self-consistent states of a gas of the bound particles χ with the self-interaction, gravity and the mean fields of the Standard Model, in three regimes, with their residuals.
- **The residuals of every field equation**: how well the approximate solution satisfies each equation.
- **The three neutrino flavours**: the Sun, the envelope of a supernova, the fast instability and the flavour equipartition.
- **The thermal history**: the degrees of freedom, the relic density of χ, its portals to the Standard Model, the right-handed neutrinos and the ad hoc vacuum before nucleosynthesis.
- **The CMB, the CNB and the BAO**: the cosmological perturbations of the metric of the fold, against the measurements.
- **What the solution does not settle**, and **what was verified**.

1.2 The status labels

Every equation and every result carries one of the labels

- **AUTHOR**: from the author (his notebooks `4+4.nb` and `Dirac_Einstein-Lovelock_4+4_Nash.nb`, and his order of 2026-10-24: the coordinates, the metric, the Lovelock weights, the Dirac matrices, the chiral projectors, the list of fields, the Higgs potential and the ad hoc vacuum);
- **RECORD**: from the record of this repository (the Lagrangian of dirac16complex with its canonical spin connection and its contact self-interaction, the Kohn-Sham treatment);
- **STANDARD**: textbook physics, or a published measurement (the source is named);
- **ASSUMED**: a choice made in UM-3 without derivation;
- **INVENTED**: a new interaction of UM-3 (the portals of dirac16complex);
- **DERIVED**: computed and verified in UM-3 (the verification is stated);
- **ESTIMATE**: an order-of-magnitude evaluation;
- **OPEN**: not settled.

1.3 Every equation is followed by its symbols

Every display equation is followed by the list "Symbols in (S-n)" or "Symbols in (FE-n)" of EVERY symbol that appears in it: every quantity, field, coupling, constant and operator, every index letter with the values it takes, every mark (prime, dagger, transpose, star), the imaginary unit, Euler's number and the differential. Each item gives the meaning, the unit (natural units $\hbar = c = 1$), the value where one exists, and the status. A field equation of UM-3 carries the tag FE-1 to FE-62 and a term of the action the tag IN-1 to IN-16; it is written here exactly as in every UM-3 document. A new equation of this document carries the tag S-1, S-2, ...

1.4 Coordinates, signature and units

The author's coordinates are $X = \{x_0, x_1, \ldots, x_7\}$, always with the hidden direction first: x_0 is the hidden space-like coordinate, on which the metric depends periodically with the period $\pi/(6H)$ through the angle $z = 6Hx_0$; x_1, x_2, x_3 are ordinary space, x_4 is the time, and x_5, x_6, x_7 are three extra time-like coordinates. The signature is $(+,+,+,+,-,-,-,-)$ in this order; the four observable coordinates $x_1, \ldots, x_4$ have the signature $(+,+,+,-)$. H is the author's constant (an inverse length), a free parameter of UM-3. Units are natural, $\hbar = c = 1$ ($1\ \mathrm{GeV}^{-1} = 1.973 \times 10^{-14}$ cm $= 6.582 \times 10^{-25}$ s; $1\ \mathrm{GeV}^{-2} = 3.894 \times 10^{-28}\ \mathrm{cm}^2$). A prime on a function of the time x_4 alone is its derivative $\mathrm{d}/\mathrm{d}x_4$, and on a profile (a function of σ alone) its derivative $\mathrm{d}/\mathrm{d}\sigma$. The proper coordinate of the hidden direction is σ (FE-27): $\sigma = 0$ is the fold $z = \pi/2$. The fields live as follows: gravity in the bulk and on the fold; dirac16complex and a medium that keeps the late-time bulk static (the stabiliser) in the bulk; the Standard Model with three right-handed neutrinos, a thin fluid (the fold fluid) and an Einstein-Hilbert term of the fold (the induced gravity, with the coupling κ_4) on the fold $\sigma = 0$.

1.5 The solution at a glance

part	result	status
background	the only histories with constant rates are the author's vacuum $a_4 = Hx_4$ with constant b_4 and its time reverse; a varying b_4 needs bulk sources of second order in $b_4'/(Hb_4)$; the late universe is the static block with frozen extra times, whose bulk media violate the null energy condition	DERIVED
power series	to first order the bulk responds either with a mode that grows away from the fold, or (if its medium responds) with the radion mode, which keeps the fold equations those of general relativity; the latter is required by observation	DERIVED, ASSUMED (the response)
Higgs field	the ad hoc vacuum is an $SU(2)$ rotation of the radial field $\sqrt{v^2 + \mathcal{A}^2}$ with a pure-gauge W^2; one radial residual remains, which vanishes as $\mathcal{A} \to 0$	DERIVED
spectra	bulk fields have zero modes that move faster than light on the fold; dirac16complex has a continuum above m_Ψ and eight bound states at $\omega_\mathrm{b} = \sqrt{m_\Psi^2 - q_\mathrm{b}^2}$, the x_0 mass	DERIVED
Kohn-Sham	KS_GLANCE	DERIVED
residuals	every field equation is satisfied to the stated accuracy, except the radial residual of the ad hoc vacuum and the extra-time component of the junction, which are carried by stated assumptions	DERIVED
neutrinos	the flavour conversion of UM-3 is that of the Standard Model with three Dirac neutrinos: solar survival 0.55 at low and 0.30 at high energy (Borexino 0.57, 0.53, 0.43, 0.37); adiabatic supernova resonances; fast instabilities within centimetres wherever the electron lepton number crosses; equipartition raises or lowers the heating behind the shock depending on the spectra	DERIVED, ESTIMATE
thermal history	χ is a thermal relic for $\langle \sigma_\mathrm{ann} \mathrm{v} \rangle = 2.2 \times 10^{-26}\ \mathrm{cm}^3/\mathrm{s}$; quark portals cannot provide it (direct detection); lepton portals can above about 1.3 TeV; the Higgs portal requires $q_\mathrm{b}/\omega_\mathrm{b} \lesssim 0.03$; the right-handed neutrinos must not thermalise	DERIVED, ESTIMATE
CMB, CNB, BAO	with the Planck 2018 parameters UM-3 reproduces the CMB ($\chi^2 = 584$ for 613 high-l bins) and its derived quantities; the BAO of DESI DR2 show the same 2.3 standard-deviation tension as the cosmological-constant model, since the dark energy of UM-3 has $w = -1$; the CNB of Dirac neutrinos has one helicity per state	DERIVED

2 The background

The order 0 of the power series of the metric is the author's metric itself, with his two functions of the time, a_4 and b_4 (b_4 is a function of x_4 and never identically 1, by the author's order). Its exact curvature and its exact field

equations in the bulk and at the fold are restated here; the solution asks which functions $a_4(x_4)$ and $b_4(x_4)$ they admit and what they require.

$$
\begin{aligned}
g_{MN} &= \mathrm{diag}\Big(\cot^2 z,\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ -1,\\
&\qquad -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6}\Big), \qquad z = 6Hx_0,\\
\sigma &= \mathrm{sgn}\Big(\frac{\pi}{2} - z\Big)\frac{\ln\sin z}{6H}, \qquad \mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0,\\
g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N &= \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\Big[\mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{e}^{-2a_4}b_4^2\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big] - \mathrm{d}x_4^2
\end{aligned}
\tag{FE-27}
$$

Symbols in (FE-27).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

The author's metric (DEFCHART of his notebook) and the proper chart: in the proper coordinate σ of the hidden

direction one period of x_0 is the whole line, the fold $z=\pi/2$ is $\sigma=0$, and ordinary space and the extra times are warped by $\mathrm{e}^{-H|\sigma|}$. Status: AUTHOR (the metric), DERIVED (the chart).

$$
\begin{aligned}
&\mathbf{e}^{(0)}=\mathrm{d}\sigma,\quad \mathbf{e}^{(i)}=\mathrm{e}^{a_4-H|\sigma|}\mathrm{d}x_i,\quad \mathbf{e}^{(4)}=\mathrm{d}x_4,\quad \mathbf{e}^{(r)}=b_4\,\mathrm{e}^{-a_4-H|\sigma|}\mathrm{d}x_r,\\
&\sqrt{|g|}=b_4^3\,\mathrm{e}^{-6H|\sigma|},\qquad \mathsf{A}=\frac{a_4'}{H},\qquad \mathsf{U}=\frac{b_4'}{Hb_4},\qquad \mathsf{A}_1=\frac{\mathsf{A}'}{H},\qquad \mathsf{U}_1=\frac{\mathsf{U}'}{H},\\
&R^{(0)(i)}{}_{(0)(i)}=R^{(0)(r)}{}_{(0)(r)}=-H^2,\qquad R^{(0)(4)}{}_{(0)(4)}=0,\qquad R^{(i)(j)}{}_{(i)(j)}=(\mathsf{A}^2-1)H^2,\\
&R^{(i)(4)}{}_{(i)(4)}=(\mathsf{A}^2+\mathsf{A}_1)H^2,\qquad R^{(i)(r)}{}_{(i)(r)}=-(1+\mathsf{A}^2-\mathsf{AU})H^2,\\
&R^{(4)(r)}{}_{(4)(r)}=\big((\mathsf{U}-\mathsf{A})^2+\mathsf{U}_1-\mathsf{A}_1\big)H^2,\qquad R^{(r)(s)}{}_{(r)(s)}=\big((\mathsf{U}-\mathsf{A})^2-1\big)H^2,\\
&R^{(0)(i)}{}_{(4)(i)}=\mathrm{sgn}(\sigma)\,\mathsf{A}H^2,\qquad R^{(0)(r)}{}_{(4)(r)}=\mathrm{sgn}(\sigma)\,(\mathsf{U}-\mathsf{A})H^2
\end{aligned}
\tag{FE-28}
$$

Symbols in (FE-28).

- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN}=\eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN}=\eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma=|\cot z|\,\mathrm{d}x_0$, $\sigma=\mathrm{sgn}(\pi/2-z)\ln(\sin z)/(6H)$ on one period $0<z<\pi$; $\sigma=0$ is the fold $z=\pi/2$, $\sigma\to\mp\infty$ the tips $z\to0,\pi$; one period of x_0 is the whole line $-\infty<\sigma<\infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda=-10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x: The coordinates of the fold: x_i $(i=1,2,3)$ of ordinary space, x_4 the time, x_r $(r=5,6,7)$ the extra times; a field of the fold is a function of the four observable coordinates $x=(x_1,x_2,x_3,x_4)$, and $\mathrm{d}^4x=\mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- i: A generation index $(i=1,2,3)$, which also labels the neutrino mass states ν_1,ν_2,ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- x_4: The time ($x_4\geq0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c=1$). Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4>0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- r: An extra-time direction of the fold: $r\in\{5,6,7\}$ (x_5,x_6,x_7); not summed (the three components are equal). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A}=a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U}=b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U}=\mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1=\mathsf{A}'/H=a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1=\mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4'=\mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor (convention $[\nabla_P,\nabla_Q]V^M=R^M{}_{NPQ}V^N$), with two indices the Ricci tensor $R_{MN}=R^P{}_{MPN}$, without indices the scalar curvature $R=g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a,b). Unit: 1/length^2. Status: STANDARD.
- j: A second generation or mass-state index (as i). Status: STANDARD.
- s: A second extra-time direction of the fold (as r, different from it). Status: STANDARD.

- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.

The orthonormal frame, the volume element, the dimensionless rates A, U of the two functions of the time, and the exact frame curvature of the author's metric; the components with the extra times follow from the listed ones. Status: DERIVED.

$$
\begin{aligned}
\frac{\kappa\,\mathcal{T}_{\mathrm{bulk}}{}^{(0)}{}_{(0)}}{H^2} &= -\frac{1}{16}\Big(\mathsf{A}^6 - 3\mathsf{A}^5\mathsf{U} + 12\mathsf{A}^4\mathsf{U}^2 + 3\mathsf{A}^4\mathsf{U}_1 + 3\mathsf{A}^4 \\
&\quad + 12\mathsf{A}^3\mathsf{A}_1\mathsf{U} - 19\mathsf{A}^3\mathsf{U}^3 - 12\mathsf{A}^3\mathsf{U}\mathsf{U}_1 - 6\mathsf{A}^3\mathsf{U} - 18\mathsf{A}^2\mathsf{A}_1\mathsf{U}^2 + 9\mathsf{A}^2\mathsf{U}^4 \\
&\quad + 9\mathsf{A}^2\mathsf{U}^2\mathsf{U}_1 - 18\mathsf{A}^2\mathsf{U}^2 - 6\mathsf{A}^2\mathsf{U}_1 - 9\mathsf{A}^2 + 6\mathsf{A}\mathsf{A}_1\mathsf{U}^3 - 12\mathsf{A}\mathsf{A}_1\mathsf{U} \\
&\quad + 21\mathsf{A}\mathsf{U}^3 + 12\mathsf{A}\mathsf{U}\mathsf{U}_1 + 9\mathsf{A}\mathsf{U} + 6\mathsf{A}_1\mathsf{U}^2 + 3\mathsf{U}^4 + 3\mathsf{U}^2\mathsf{U}_1 \\
&\quad + 18\mathsf{U}^2 + 3\mathsf{U}_1 + 5\Big), \\
\frac{\kappa\,\mathcal{T}_{\mathrm{bulk}}{}^{(i)}{}_{(i)}}{H^2} &= -\frac{1}{16}\Big(\mathsf{A}^6 - 6\mathsf{A}^5\mathsf{U} - 5\mathsf{A}^4\mathsf{A}_1 + 12\mathsf{A}^4\mathsf{U}^2 + 3\mathsf{A}^4\mathsf{U}_1 \\
&\quad + 3\mathsf{A}^4 + 12\mathsf{A}^3\mathsf{A}_1\mathsf{U} - 10\mathsf{A}^3\mathsf{U}^3 - 6\mathsf{A}^3\mathsf{U}\mathsf{U}_1 + 12\mathsf{A}^3\mathsf{U} - 9\mathsf{A}^2\mathsf{A}_1\mathsf{U}^2 \\
&\quad + 6\mathsf{A}^2\mathsf{A}_1 + 3\mathsf{A}^2\mathsf{U}^4 + 3\mathsf{A}^2\mathsf{U}^2\mathsf{U}_1 - 36\mathsf{A}^2\mathsf{U}^2 - 6\mathsf{A}^2\mathsf{U}_1 - 9\mathsf{A}^2 \\
&\quad + 2\mathsf{A}\mathsf{A}_1\mathsf{U}^3 - 12\mathsf{A}\mathsf{A}_1\mathsf{U} + 18\mathsf{A}\mathsf{U}^3 + 6\mathsf{A}\mathsf{U}\mathsf{U}_1 - 6\mathsf{A}\mathsf{U} + 3\mathsf{A}_1\mathsf{U}^2 \\
&\quad - \mathsf{A}_1 + 3\mathsf{U}^4 + 3\mathsf{U}^2\mathsf{U}_1 + 24\mathsf{U}^2 + 3\mathsf{U}_1 + 5\Big), \\
\frac{\kappa\,\mathcal{T}_{\mathrm{bulk}}{}^{(4)}{}_{(4)}}{H^2} &= \frac{1}{16}\Big(5\mathsf{A}^6 - 15\mathsf{A}^5\mathsf{U} + 15\mathsf{A}^4\mathsf{U}^2 - 9\mathsf{A}^4 - 5\mathsf{A}^3\mathsf{U}^3 \\
&\quad + 18\mathsf{A}^3\mathsf{U} + 3\mathsf{A}^2 - 9\mathsf{A}\mathsf{U}^3 - 3\mathsf{A}\mathsf{U} - 3\mathsf{U}^2 + 1\Big), \\
\frac{\kappa\,\mathcal{T}_{\mathrm{bulk}}{}^{(r)}{}_{(r)}}{H^2} &= -\frac{1}{16}\Big(\mathsf{A}^6 + 5\mathsf{A}^4\mathsf{A}_1 - 3\mathsf{A}^4\mathsf{U}^2 - 2\mathsf{A}^4\mathsf{U}_1 + 3\mathsf{A}^4 \\
&\quad - 8\mathsf{A}^3\mathsf{A}_1\mathsf{U} + 2\mathsf{A}^3\mathsf{U}^3 + 2\mathsf{A}^3\mathsf{U}\mathsf{U}_1 - 24\mathsf{A}^3\mathsf{U} + 3\mathsf{A}^2\mathsf{A}_1\mathsf{U}^2 - 6\mathsf{A}^2\mathsf{A}_1 \\
&\quad + 18\mathsf{A}^2\mathsf{U}^2 - 9\mathsf{A}^2 + 6\mathsf{A}\mathsf{U}^3 + 6\mathsf{A}\mathsf{U}\mathsf{U}_1 + 24\mathsf{A}\mathsf{U} + 3\mathsf{A}_1\mathsf{U}^2 \\
&\quad + \mathsf{A}_1 + 9\mathsf{U}^2 + 2\mathsf{U}_1 + 5\Big), \\
\frac{\kappa\,\mathcal{T}_{\mathrm{bulk}}{}^{(0)}{}_{(4)}}{H^2} &= \mathrm{sgn}(\sigma)\,\frac{3}{16}\,\mathsf{U}\,\mathcal{G}, \qquad \mathcal{G} = 5\mathsf{A}^2(\mathsf{A}-\mathsf{U})^2 - 6\mathsf{A}^2 + 6\mathsf{A}\mathsf{U} + \mathsf{U}^2 + 1
\end{aligned}
\tag{FE-29}
$$

Symbols in (FE-29).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\mathrm{stab}}$; $\mathcal{T}_{\mathrm{bulk}}{}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

- $\mathcal{G}$: The factor of the (σ, x_4) bulk equation of the author's metric, $\mathcal{G} = 5\mathsf{A}^2(\mathsf{A}-\mathsf{U})^2 - 6\mathsf{A}^2 + 6\mathsf{A}\mathsf{U} + \mathsf{U}^2 + 1$. Unit: dimensionless. Status: DERIVED.

The bulk equations of the author's metric: the energy-momentum tensor $\mathcal{T}_{\text{bulk}}$ that the matter of the bulk must have for the metric to solve the Einstein-Lovelock equations (with the author's Lovelock weights and cosmological constant), as exact polynomials in the rates. Status: AUTHOR (the metric), DERIVED (the polynomials).

$$
\begin{aligned}
&a_4 = Hx_4, \quad b_4 = \text{constant } (\mathsf{A}=1,\ \mathsf{U}=\mathsf{A}_1=\mathsf{U}_1=0): \qquad \sum_{p=1}^{3} \alpha_{(p)}\, {\mathcal{E}_{(p)}}^{M}{}_{N} + \Lambda\, \delta^{M}_{N} = 0 \quad (\sigma \neq 0),\\
&\kappa\, {\mathcal{S}^{(i)}}_{(i)} = \kappa\, {\mathcal{S}^{(r)}}_{(r)} = -10H, \qquad \kappa\, {\mathcal{S}^{(4)}}_{(4)} = -6H,\\
&\kappa\rho_{\text{bulk}} = -H^2\Big[\frac{3}{4}\big((\mathsf{A}-1-\mathsf{U})^2 + (\mathsf{A}-1)^2\big) + \ldots\Big], \qquad \mathcal{G}(1,\mathsf{U}) = 2\mathsf{U}(3\mathsf{U}-2)
\end{aligned}
\tag{FE-31}
$$

Symbols in (FE-31).

- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4} b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number. Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- ${\mathcal{E}_{(p)}}^{M}{}_{N}$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$ (the author writes the pure number $\Lambda_{\mathrm{N}} = 10$ in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}} = 1$ for $X = \mathrm{L}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or −1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- ${\mathcal{S}^{a}}_{b}$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal).

Status: STANDARD.

- $\rho_{\rm bulk}$: The energy density of the bulk sources, $\rho_{\rm bulk} = -{\mathcal{T}_{\rm bulk}}^{(4)}{}_{(4)}$. Unit: GeV^8. Status: DERIVED.
- $\mathcal{G}$: The factor of the (σ, x_4) bulk equation of the author's metric, $\mathcal{G} = 5\mathsf{A}^2(\mathsf{A}-\mathsf{U})^2 - 6\mathsf{A}^2 + 6\mathsf{A}\mathsf{U} + \mathsf{U}^2 + 1$. Unit: dimensionless. Status: DERIVED.

The author's vacuum: with $a_4 = Hx_4$ and a constant b_4 the bulk is empty, and the fold carries a thin source. Status: DERIVED.

$$\begin{aligned} &\mathsf{A}_1 = \mathsf{U}_1 = 0: \qquad {\mathcal{T}_{\rm bulk}}^{a}{}_{b} = 0 \quad\Leftrightarrow\quad \big(\mathsf{A}^2-1\big)^2 = 0, \quad \mathsf{U}\big(\mathsf{A}^2 - \mathsf{A}\mathsf{U} - 1\big) = 0, \quad \mathsf{U}^3 = 0, \\ &\frac{16\,\kappa\,{\mathcal{T}_{\rm bulk}}^{(4)}{}_{(4)}}{H^2}\Big|_{\mathsf{U}=0} = \big(\mathsf{A}^2-1\big)^2\big(5\mathsf{A}^2+1\big) \end{aligned} \tag{S-1}$$

Symbols in (S-1).

- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- $\mathcal{T}_{\rm bulk}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\rm stab}$; ${\mathcal{T}_{\rm bulk}}^{(0)}{}_{b}$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

The histories with constant rates. The first line is the reduced Groebner basis of the polynomial system FE-29 $= 0$ with $\mathsf{A}_1 = \mathsf{U}_1 = 0$: its only solutions are $(\mathsf{A}, \mathsf{U}) = (1, 0)$, the author's vacuum, and $(-1, 0)$, its time reverse. Both are degenerate: $\mathsf{A}^2 = 1$ is a double root and $\mathsf{U} = 0$ a triple root, so the linear response of the bulk equations to a small change of the rates vanishes there (the same degeneracy makes the transverse traceless graviton lose its time-kinetic term at the author's vacuum). Every other history with constant rates needs bulk sources: with a constant b_4 the energy density $-{\mathcal{T}_{\rm bulk}}^{(4)}{}_{(4)}$ that the second line requires is negative for every $\mathsf{A}^2 \neq 1$. Status: DERIVED (`um3_background.py`, BG1).

$$\begin{aligned} &\mathsf{A} = 1,\ \mathsf{A}_1 = \mathsf{U}_1 = 0: \qquad \frac{\kappa\,{\mathcal{T}_{\rm bulk}}^{(0)}{}_{(0)}}{H^2} = -\frac{\mathsf{U}^2\big(6\mathsf{U}^2+\mathsf{U}+6\big)}{8}, \qquad \frac{\kappa\,{\mathcal{T}_{\rm bulk}}^{(i)}{}_{(i)}}{H^2} = -\frac{\mathsf{U}^3\big(3\mathsf{U}+4\big)}{8}, \\ &\frac{\kappa\,{\mathcal{T}_{\rm bulk}}^{(4)}{}_{(4)}}{H^2} = -\frac{\mathsf{U}^2\big(7\mathsf{U}-6\big)}{8}, \qquad \frac{\kappa\,{\mathcal{T}_{\rm bulk}}^{(r)}{}_{(r)}}{H^2} = -\frac{\mathsf{U}^2\big(\mathsf{U}+3\big)}{2}, \qquad \frac{\kappa\,{\mathcal{T}_{\rm bulk}}^{(0)}{}_{(4)}}{H^2} = \mathrm{sgn}(\sigma)\,\frac{3\mathsf{U}^2\big(3\mathsf{U}-2\big)}{8} \end{aligned} \tag{S-2}$$

Symbols in (S-2).

- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\rm bulk}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\rm stab}$; ${\mathcal{T}_{\rm bulk}}^{(0)}{}_{b}$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The primordial era with a varying b_4: the author's expansion $a_4 = Hx_4$ with $b_4' = \mathsf{U}Hb_4$ for a constant U (so $b_4 \propto \mathrm{e}^{\mathsf{U}Hx_4}$). The bulk must then carry a medium whose energy density, pressures and energy flux along σ (the last expression) all start at second order in U: a slowly varying b_4 costs little. The flux vanishes for $\mathsf{U} = 2/3$, where the medium is static with $\kappa\mathcal{T}_{\mathrm{bulk}}/H^2 = (-14/27, -2/9, 2/27, -22/27)$ for (σ, ordinary space, time, extra times): the flux-free primordial branch $b_4 \propto \mathrm{e}^{2Hx_4/3}$. Status: DERIVED (BG3).

$$\begin{aligned}
&\mathsf{A} = \mathsf{U} = \mathsf{A}_1 = \mathsf{U}_1 = 0: \qquad \kappa\,{\mathcal{T}_{\mathrm{bulk}}}^{a}{}_{b} = H^2\,\mathrm{diag}\Big(-\frac{5}{16}, -\frac{5}{16}, -\frac{5}{16}, -\frac{5}{16}, \frac{1}{16}, -\frac{5}{16}, -\frac{5}{16}, -\frac{5}{16}\Big),\\
&\kappa\,{\mathcal{S}^{(i)}}_{(i)} = \kappa\,{\mathcal{S}^{(r)}}_{(r)} = -\frac{61}{8}H, \qquad \kappa\,{\mathcal{S}^{(4)}}_{(4)} = -\frac{31}{4}H
\end{aligned} \tag{FE-32}$$

Symbols in (FE-32).

- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\mathrm{stab}}$; ${\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_{b}$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- $\mathcal{S}^a{}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.

$$\begin{aligned}
&\mathsf{U} = \mathsf{A}, \quad \mathsf{U}_1 = \mathsf{A}_1 \ \ \big(\mathrm{e}^{-a_4}b_4 = \text{constant},\ \mathcal{Z} = 1\big):\\
&\frac{\kappa\rho_{\mathrm{bulk}}}{H^2} = \frac{3\mathsf{A}^2 - 1}{16}, \qquad \frac{\kappa\,{\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_{(0)}}{H^2} = -\frac{3\mathsf{A}^4 + 3\mathsf{A}^2\mathsf{A}_1 + 18\mathsf{A}^2 + 3\mathsf{A}_1 + 5}{16},\\
&\frac{\kappa\,{\mathcal{T}_{\mathrm{bulk}}}^{(i)}{}_{(i)}}{H^2} = -\frac{9\mathsf{A}^2 + 2\mathsf{A}_1 + 5}{16}, \qquad \frac{\kappa\,{\mathcal{T}_{\mathrm{bulk}}}^{(r)}{}_{(r)}}{H^2} = -\frac{3\mathsf{A}^4 + 3\mathsf{A}^2\mathsf{A}_1 + 24\mathsf{A}^2 + 3\mathsf{A}_1 + 5}{16},\\
&\frac{\kappa\,{\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_{(4)}}{H^2} = \mathrm{sgn}(\sigma)\,\frac{3\mathsf{A}(1 + \mathsf{A}^2)}{16}, \qquad \frac{\kappa\,{\mathcal{S}^{(i)}}_{(i)}}{H} = -\frac{57\mathsf{A}^2 + 40\mathsf{A}_1 + 183}{24},\\
&\frac{\kappa\,{\mathcal{S}^{(4)}}_{(4)}}{H} = -\frac{10\mathsf{A}^2 + 31}{4}, \qquad \frac{\kappa\,{\mathcal{S}^{(r)}}_{(r)}}{H} = -\frac{2\mathsf{A}^4 + 2\mathsf{A}^2\mathsf{A}_1 + 37\mathsf{A}^2 + 20\mathsf{A}_1 + 61}{8}
\end{aligned} \tag{FE-33}$$

Symbols in (FE-33).

- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- ρ_{bulk}: The energy density of the bulk sources, $\rho_{\mathrm{bulk}} = -{\mathcal{T}_{\mathrm{bulk}}}^{(4)}{}_{(4)}$. Unit: GeV^8. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\mathrm{stab}}$; ${\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- ${\mathcal{S}^a}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.

The static block of the late universe (a_4 and b_4 constant, FE-32) and the branch with frozen extra times ($\mathrm{e}^{-a_4}b_4$ constant, FE-33), on which ordinary space expands while the proper size of the extra times at the fold stays fixed. These are the backgrounds on which the late universe is solved. Status: DERIVED.

$$\begin{aligned}
&\kappa\big(\rho_{\mathrm{s}} + p_{\mathrm{s}}\big) = -\frac{3}{8}H^2, \qquad \kappa\big(\rho_{\mathrm{bulk}} + {\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_{(0)}\big) = -\frac{3\big(\mathsf{A}^4 + \mathsf{A}^2\mathsf{A}_1 + 5\mathsf{A}^2 + \mathsf{A}_1 + 2\big)}{16}H^2, \\
&\kappa\big(\rho_{\mathrm{bulk}} + {\mathcal{T}_{\mathrm{bulk}}}^{(i)}{}_{(i)}\big) = -\frac{3\mathsf{A}^2 + \mathsf{A}_1 + 3}{8}H^2 \qquad (\mathsf{U} = \mathsf{A},\ \mathsf{U}_1 = \mathsf{A}_1)
\end{aligned} \tag{S-3}$$

Symbols in (S-3).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- ρ_{s}: The energy density of the stabiliser. Unit: GeV^8. Value: $\kappa\rho_{\mathrm{s}} = -H^2/16$. Status: ASSUMED.
- p_{s}: The pressure of the stabiliser (equal along σ, ordinary space and the extra times). Unit: GeV^8. Value: $\kappa p_{\mathrm{s}} = -5H^2/16$. Status: ASSUMED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- ρ_{bulk}: The energy density of the bulk sources, $\rho_{\mathrm{bulk}} = -{\mathcal{T}_{\mathrm{bulk}}}^{(4)}{}_{(4)}$. Unit: GeV^8. Status: DERIVED.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\mathrm{stab}}$; ${\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.

- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.

The null energy condition of the bulk media of the late universe: for the stabiliser of the static block (energy density ρ_s, pressure p_s) and for the medium of the frozen branch (energy density $\rho_\mathrm{bulk} = -{\mathcal{T}_\mathrm{bulk}}^{(4)}{}_{(4)}$, pressures ${\mathcal{T}_\mathrm{bulk}}^{(0)}{}_{(0)}$ along σ and ${\mathcal{T}_\mathrm{bulk}}^{(i)}{}_{(i)}$ along ordinary space), the sum of the energy density and the pressure along a space-like direction is negative for every $|\mathsf{A}| < 1$ (sampled: at most $-0.375H^2/\kappa$). The bulk media that the late universe of UM-3 needs violate the null energy condition: they are not ordinary matter (a scalar field cannot provide them). Status: DERIVED (BG2).

$$
\begin{aligned}
&g_{1,\mathrm{eff}} = \frac{g_1}{\sqrt{\mathcal{Z}}}, \quad g_{2,\mathrm{eff}} = \frac{g_2}{\sqrt{\mathcal{Z}}}, \quad g_{3,\mathrm{eff}} = \frac{g_3}{\sqrt{\mathcal{Z}}}, \qquad \lambda_\mathrm{eff} = \frac{\lambda}{\mathcal{Z}}, \\
&\mathcal{Y}_\mathrm{eff} = \frac{\mathcal{Y}}{\sqrt{\mathcal{Z}}}, \qquad G_{\mathrm{N,eff}} = \frac{G_\mathrm{N}}{\mathcal{Z}}, \qquad \mathcal{Z} = \frac{V_\mathrm{t}\left(\mathrm{e}^{-a_4}b_4\right)^3}{V_*}
\end{aligned}
\tag{FE-44}
$$

Symbols in (FE-44).

- $g_{k,\mathrm{eff}}$: The effective (canonically normalised) gauge couplings of the fold when the fold factor is $\mathcal{Z}$, $g_{k,\mathrm{eff}} = g_k/\sqrt{\mathcal{Z}}$ ($k = 1, 2, 3$). Unit: dimensionless. Status: DERIVED.
- g_1: The $U(1)_Y$ gauge coupling (usually g'; not normalised for grand unification). Unit: dimensionless. Value: $g_1(m_Z) = 0.3574$. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- g_3: The $SU(3)_C$ gauge coupling (the strong coupling g_s). Unit: dimensionless. Value: $g_3(m_Z) = 1.2177$ ($\alpha_s = 0.1180$). Status: STANDARD.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- λ_eff: The effective quartic Higgs coupling, $\lambda/\mathcal{Z}$. Unit: dimensionless. Status: DERIVED.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- $\mathcal{Y}_\mathrm{eff}$: The effective Yukawa matrices, $\mathcal{Y}/\sqrt{\mathcal{Z}}$ (for each of $\mathcal{Y}^u, \mathcal{Y}^d, \mathcal{Y}^e, \mathcal{Y}^\nu$). Unit: dimensionless. Status: DERIVED.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $G_{\mathrm{N,eff}}$: The effective Newton constant of the fold, $G_\mathrm{N}/\mathcal{Z}$. Unit: GeV^{-2}. Status: DERIVED.
- G_N: Newton's constant of four-dimensional gravity on the fold, $G_\mathrm{N} = 1/M_\mathrm{Pl}^2$. Unit: GeV^{-2}. Value: $6.708{\times}10^{-39}$ GeV^{-2}. Status: STANDARD.
- V_t: The coordinate volume of the (compact) extra times, $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$. Unit: length^3. Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.

$$
\frac{\mathrm{d}\ln\mathcal{Z}}{\mathrm{d}x_4} = 3H\left(\mathsf{U} - \mathsf{A}\right) \tag{S-4}
$$

Symbols in (S-4).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_\mathrm{t}(\mathrm{e}^{-a_4} b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.

The factor $\mathcal{Z}$ that the volume of the extra times gives the four-dimensional couplings of the fold (FE-44) changes at the rate S-4: with the author's vacuum and a constant b_4 it falls as e^{-3Hx_4}, so every coupling of the Standard Model grows as $\mathrm{e}^{3Hx_4/2}$ (and Newton's constant as e^{3Hx_4}); on the flux-free branch $\mathsf{U} = 2/3$ the couplings grow as $\mathrm{e}^{Hx_4/2}$; with frozen extra times they are constant. The late universe, whose couplings are measured to be constant, must therefore have frozen extra times. Status: DERIVED (BG4).

3 The power series of the metric to first order

The matter of the fold and of dirac16complex perturbs the background. The metric is written as a power series of terms of decreasing size, and each order is solved with the sources of that order.

$$
\begin{aligned}
&g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = \mathrm{d}\sigma^2 + \mathrm{e}^{2\mathfrak{a}}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{e}^{2\mathfrak{n}}\,\mathrm{d}x_4^2 - \mathrm{e}^{2\mathfrak{b}}\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big),\\
&R^{(0)(i)}{}_{(0)(i)} = -\partial_\sigma^2\mathfrak{a} - (\partial_\sigma\mathfrak{a})^2, \qquad R^{(0)(4)}{}_{(0)(4)} = -\partial_\sigma^2\mathfrak{n} - (\partial_\sigma\mathfrak{n})^2, \qquad R^{(i)(j)}{}_{(i)(j)} = -(\partial_\sigma\mathfrak{a})^2 + \mathrm{e}^{-2\mathfrak{n}}(\partial_{x_4}\mathfrak{a})^2,\\
&R^{(i)(r)}{}_{(i)(r)} = -\partial_\sigma\mathfrak{a}\,\partial_\sigma\mathfrak{b} + \mathrm{e}^{-2\mathfrak{n}}\,\partial_{x_4}\mathfrak{a}\,\partial_{x_4}\mathfrak{b}, \qquad R^{(i)(4)}{}_{(i)(4)} = -\partial_\sigma\mathfrak{a}\,\partial_\sigma\mathfrak{n} + \mathrm{e}^{-2\mathfrak{n}}\big(\partial_{x_4}^2\mathfrak{a} + (\partial_{x_4}\mathfrak{a})^2 - \partial_{x_4}\mathfrak{a}\,\partial_{x_4}\mathfrak{n}\big),\\
&R^{(0)(i)}{}_{(4)(i)} = -\mathrm{e}^{-\mathfrak{n}}\big(\partial_\sigma\partial_{x_4}\mathfrak{a} + \partial_\sigma\mathfrak{a}\,\partial_{x_4}\mathfrak{a} - \partial_{x_4}\mathfrak{a}\,\partial_\sigma\mathfrak{n}\big)
\end{aligned}
\tag{FE-34}
$$

Symbols in (FE-34).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $\mathfrak{a}$, $\mathfrak{a}_1$: The logarithm of the scale factor of ordinary space in the general warped metric FE-34, a function of σ and x_4; for the author's metric $\mathfrak{a} = a_4 - H|\sigma|$; $\mathfrak{a}_n$ its term of order n in the power series FE-35 ($\mathfrak{a}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.

- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $\mathfrak{n}$, $\mathfrak{n}_1$: The logarithm of the lapse in the general warped metric FE-34 ($g_{x_4x_4} = -\mathrm{e}^{2\mathfrak{n}}$); for the author's metric $\mathfrak{n} = 0$; $\mathfrak{n}_n$ its term of order n ($\mathfrak{n}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\mathfrak{b}$, $\mathfrak{b}_1$: The logarithm of the scale factor of the extra times in the general warped metric FE-34; for the author's metric $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$; $\mathfrak{b}_n$ its term of order n ($\mathfrak{b}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor (convention $[\nabla_P, \nabla_Q]V^M = R^M{}_{NPQ}V^N$), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\text{length}^2$. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- j: A second generation or mass-state index (as i). Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.

$$\begin{aligned} &\mathfrak{a} = a_4 - H|\sigma| + \sum_{n\geq 1} \epsilon_{\mathrm{r}}^n\, \mathfrak{a}_n(\sigma, x_4), \qquad \mathfrak{b} = \ln b_4 - a_4 - H|\sigma| + \sum_{n\geq 1} \epsilon_{\mathrm{r}}^n\, \mathfrak{b}_n(\sigma, x_4), \\ &\mathfrak{n} = \sum_{n\geq 1} \epsilon_{\mathrm{r}}^n\, \mathfrak{n}_n(\sigma, x_4), \qquad \hat{h}_{\mu\nu} \to \hat{h}_{\mu\nu} + \epsilon_{\mathrm{c}}\, \hat{h}^{(1)}_{\mu\nu} + \ldots \end{aligned} \tag{FE-35}$$

Symbols in (FE-35).

- $\mathfrak{a}$, $\mathfrak{a}_1$: The logarithm of the scale factor of ordinary space in the general warped metric FE-34, a function of σ and x_4; for the author's metric $\mathfrak{a} = a_4 - H|\sigma|$; $\mathfrak{a}_n$ its term of order n in the power series FE-35 ($\mathfrak{a}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- n: The order $n = 1, 2, \ldots$ of a term of the power series of the metric (also its exponent of ϵ_{r}); $\mathfrak{a}_n$, $\mathfrak{b}_n$, $\mathfrak{n}_n$ are the terms of order n. Status: STANDARD.
- ϵ_{r}: The bookkeeping parameter of the response of the bulk to the matter of the fold: the ratio of the bulk to the fold gravity, $\epsilon_{\mathrm{r}} = \kappa_4 V_*/(\kappa H)$. Unit: dimensionless. Value: $\lesssim 10^{-15}$. Status: DERIVED.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\mathfrak{b}$, $\mathfrak{b}_1$: The logarithm of the scale factor of the extra times in the general warped metric FE-34; for the author's metric $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$; $\mathfrak{b}_n$ its term of order n ($\mathfrak{b}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- $\mathfrak{n}$, $\mathfrak{n}_1$: The logarithm of the lapse in the general warped metric FE-34 ($g_{x_4x_4} = -\mathrm{e}^{2\mathfrak{n}}$); for the author's metric $\mathfrak{n} = 0$; $\mathfrak{n}_n$ its term of order n ($\mathfrak{n}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its

first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.

- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- ϵ_{c}: The bookkeeping parameter of the cosmological perturbations of the metric of the fold (the size of the CMB anisotropies). Unit: dimensionless. Value: $\sim 10^{-5}$. Status: STANDARD.

The general warped metric with the symmetries of the author's metric (FE-34) and its power series (FE-35): the bulk response is counted by ϵ_{r}, the ratio of bulk to fold gravity, and the cosmological perturbations of the fold by ϵ_{c}. Status: DERIVED.

$$\begin{aligned}
\kappa\,\Delta{\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_{(0)} &= -\frac{75H}{16}\left(\mathfrak{a}_1' + \mathfrak{b}_1'\right) - \frac{3H}{8}\,\mathfrak{n}_1', \\
\kappa\,\Delta{\mathcal{T}_{\mathrm{bulk}}}^{(i)}{}_{(i)} &= -\frac{15H}{4}\mathfrak{a}_1' - \frac{45H}{8}\mathfrak{b}_1' - \frac{5H}{16}\mathfrak{n}_1' + \frac{5}{8}\mathfrak{a}_1'' + \frac{15}{16}\mathfrak{b}_1'' + \frac{1}{16}\mathfrak{n}_1'', \\
\kappa\,\Delta{\mathcal{T}_{\mathrm{bulk}}}^{(4)}{}_{(4)} &= -\frac{21H}{16}\left(\mathfrak{a}_1' + \mathfrak{b}_1'\right) + \frac{3}{16}\left(\mathfrak{a}_1'' + \mathfrak{b}_1''\right), \\
\kappa\,\Delta{\mathcal{T}_{\mathrm{bulk}}}^{(r)}{}_{(r)} &= -\frac{45H}{8}\mathfrak{a}_1' - \frac{15H}{4}\mathfrak{b}_1' - \frac{5H}{16}\mathfrak{n}_1' + \frac{15}{16}\mathfrak{a}_1'' + \frac{5}{8}\mathfrak{b}_1'' + \frac{1}{16}\mathfrak{n}_1'' \qquad (\sigma > 0)
\end{aligned} \tag{S-5}$$

Symbols in (S-5).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- Δ: A change, difference or small variation of the quantity that follows ($\Delta\rho$ a perturbation, Δm^2 a difference of squared masses, ΔN_{eff} an addition to N_{eff}). Status: STANDARD.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\mathrm{stab}}$; ${\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- $\mathfrak{a}$, $\mathfrak{a}_1$: The logarithm of the scale factor of ordinary space in the general warped metric FE-34, a function of σ and x_4; for the author's metric $\mathfrak{a} = a_4 - H|\sigma|$; $\mathfrak{a}_n$ its term of order n in the power series FE-35 ($\mathfrak{a}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- $\mathfrak{b}$, $\mathfrak{b}_1$: The logarithm of the scale factor of the extra times in the general warped metric FE-34; for the author's metric $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$; $\mathfrak{b}_n$ its term of order n ($\mathfrak{b}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- $\mathfrak{n}$, $\mathfrak{n}_1$: The logarithm of the lapse in the general warped metric FE-34 ($g_{x_4x_4} = -\mathrm{e}^{2\mathfrak{n}}$); for the author's metric $\mathfrak{n} = 0$; $\mathfrak{n}_n$ its term of order n ($\mathfrak{n}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The first-order terms of the bulk equations around the static block for static perturbations $\mathfrak{a} = -H|\sigma| + \epsilon_{\mathrm{r}}\mathfrak{a}_1(\sigma)$, $\mathfrak{b} = -H|\sigma| + \epsilon_{\mathrm{r}}\mathfrak{b}_1(\sigma)$, $\mathfrak{n} = \epsilon_{\mathrm{r}}\mathfrak{n}_1(\sigma)$ ($a_4 = 0$ and $b_4 = 1$ by normalisation of the coordinates): the energy-momentum tensor of the bulk changes by $\epsilon_{\mathrm{r}}\,\Delta\mathcal{T}_{\mathrm{bulk}}$, whose components are linear in the first and second derivatives of the three functions with constant coefficients (the side $\sigma < 0$ is the mirror image). Status: DERIVED (`um3_series.py`, SR1, from the exact equations of the general metric FE-34).

$$\mathfrak{a}_1' = -\mathfrak{b}_1' = C_{\mathrm{d}}\,\mathrm{e}^{6H|\sigma|}, \qquad \mathfrak{n}_1' = 0 \tag{S-6}$$

Symbols in (S-6).

- $\mathfrak{a}$, $\mathfrak{a}_1$: The logarithm of the scale factor of ordinary space in the general warped metric FE-34, a function of σ and x_4; for the author's metric $\mathfrak{a} = a_4 - H|\sigma|$; $\mathfrak{a}_n$ its term of order n in the power series FE-35 ($\mathfrak{a}_1$ the

first-order term). Unit: dimensionless. Status: DERIVED.

- $\mathfrak{b}$, $\mathfrak{b}_1$: The logarithm of the scale factor of the extra times in the general warped metric FE-34; for the author's metric $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$; $\mathfrak{b}_n$ its term of order n ($\mathfrak{b}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- $\mathfrak{n}$, $\mathfrak{n}_1$: The logarithm of the lapse in the general warped metric FE-34 ($g_{x_4x_4} = -\mathrm{e}^{2\mathfrak{n}}$); for the author's metric $\mathfrak{n} = 0$; $\mathfrak{n}_n$ its term of order n ($\mathfrak{n}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- C_d: The amplitude of the difference mode S-6, the only static first-order correction of the metric when the bulk medium does not respond: $\mathfrak{a}_1' = -\mathfrak{b}_1' = C_\mathrm{d}\mathrm{e}^{6H|\sigma|}$. Unit: 1/length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

If the medium of the bulk does not respond ($\Delta\mathcal{T}_\mathrm{bulk} = 0$), the four equations S-5 for the three functions have exactly one solution of exponential form, the difference mode S-6 (the only root of all the 3×3 minors is the exponent $6H$): ordinary space and the extra times change in opposite directions, with an amplitude C_d that grows away from the fold. No bounded static correction to the metric exists then. Status: DERIVED (SR2).

$$\begin{aligned}
&\kappa\,\Delta\mathcal{S}^{(i)}{}_{(i)} = \frac{5}{8}\big[\mathfrak{a}_1'\big] + \frac{15}{16}\big[\mathfrak{b}_1'\big] + \frac{1}{16}\big[\mathfrak{n}_1'\big], \qquad \kappa\,\Delta\mathcal{S}^{(4)}{}_{(4)} = \frac{3}{16}\Big(\big[\mathfrak{a}_1'\big] + \big[\mathfrak{b}_1'\big]\Big), \\
&\kappa\,\Delta\mathcal{S}^{(r)}{}_{(r)} = \frac{15}{16}\big[\mathfrak{a}_1'\big] + \frac{5}{8}\big[\mathfrak{b}_1'\big] + \frac{1}{16}\big[\mathfrak{n}_1'\big], \qquad \Big(\big[\mathfrak{a}_1'\big], \big[\mathfrak{b}_1'\big], \big[\mathfrak{n}_1'\big]\Big) = c_\mathrm{rad}\,(1, -1, 5): \quad \kappa\,\Delta\mathcal{S}^{(r)}{}_{(r)} = \frac{5}{8}\,c_\mathrm{rad}
\end{aligned} \tag{S-7}$$

Symbols in (S-7).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- Δ: A change, difference or small variation of the quantity that follows ($\Delta\rho$ a perturbation, Δm^2 a difference of squared masses, ΔN_eff an addition to N_eff). Status: STANDARD.
- $\mathcal{S}^a{}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- $\mathfrak{a}$, $\mathfrak{a}_1$: The logarithm of the scale factor of ordinary space in the general warped metric FE-34, a function of σ and x_4; for the author's metric $\mathfrak{a} = a_4 - H|\sigma|$; $\mathfrak{a}_n$ its term of order n in the power series FE-35 ($\mathfrak{a}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- $\mathfrak{b}$, $\mathfrak{b}_1$: The logarithm of the scale factor of the extra times in the general warped metric FE-34; for the author's metric $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$; $\mathfrak{b}_n$ its term of order n ($\mathfrak{b}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- $\mathfrak{n}$, $\mathfrak{n}_1$: The logarithm of the lapse in the general warped metric FE-34 ($g_{x_4x_4} = -\mathrm{e}^{2\mathfrak{n}}$); for the author's metric $\mathfrak{n} = 0$; $\mathfrak{n}_n$ its term of order n ($\mathfrak{n}_1$ the first-order term). Unit: dimensionless. Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- c_rad: The amplitude of the radion mode: the first-order jumps at the fold $([\mathfrak{a}_1'], [\mathfrak{b}_1'], [\mathfrak{n}_1']) = c_\mathrm{rad}(1, -1, 5)$, which change only the extra-time component of the junction. Unit: 1/length. Value: $3.806\,\mathsf{A}^2 H/\epsilon_\mathrm{r}$ today. Status: DERIVED.

The first-order junction conditions at the fold: the jumps $[X] = X(0^+) - X(0^-)$ of the first derivatives of the three functions give the first-order thin source $\epsilon_\mathrm{r}\,\Delta\mathcal{S}$ (computed from the junction potential of FE-4 and, independently, from the coefficients of the second derivatives in S-5, with the same result). The difference mode carries no energy and opposite stresses, $(-5/16, 0, 5/16)$ per unit jump; the combination of jumps in the ratio $1 : -1 : 5$, the radion mode of amplitude c_rad, changes only the extra-time component. Status: DERIVED (SR3).

$$\frac{1}{\kappa_4}\,\hat{G}^{\mu}{}_{\nu} = \mathcal{T}_{\rm SM}{}^{\mu}{}_{\nu} + \mathcal{T}_{\rm P}{}^{\mu}{}_{\nu} + \mathcal{T}_{\rm ff}{}^{\mu}{}_{\nu} - \frac{V_*}{\kappa}\,\mathcal{J}^{\mu}{}_{\nu}, \qquad \mathcal{J}^{a}{}_{b} = \kappa\,\mathcal{S}^{a}{}_{b} \text{ of FE-30},$$
$$H_4^2 = \frac{8\pi G_{\rm N}}{3}\big(\rho_{\rm SM} + \rho_\chi + \rho_\Lambda\big), \qquad 8\pi G_{\rm N} = \frac{\kappa_4}{1+5\epsilon_{\rm r}/6}, \qquad \epsilon_{\rm r} = \frac{\kappa_4 V_*}{\kappa H}, \qquad \rho_\Lambda = \rho_{\rm ff} - \frac{31 H V_*}{4\kappa}, \tag{FE-59}$$
$$\frac{\hat{R}}{2\kappa_4} + q_{\rm SM} + q_\chi + q_{\rm P} + q_{\rm ff} = \frac{V_*}{\kappa}\,\mathcal{J}^{(r)}{}_{(r)}, \qquad q_{\rm SM} = \big\langle \mathcal{L}_{\rm H} + \mathcal{L}_{\rm gauge} + \mathcal{L}_{\rm f} + \mathcal{L}_{\rm Y}\big\rangle, \qquad q_{\rm P} = \langle \mathcal{L}_{\rm P}\rangle$$

Symbols in (FE-59).

- κ_4: The coupling of the induced gravity of the fold, $\kappa_4 = 8\pi G_{\rm ind}$; four-dimensional gravity on the fold has $8\pi G_{\rm N} = \kappa_4/(1+5\epsilon_{\rm r}/6)$ (FE-59). Unit: GeV^{-2}. Value: $\kappa_4 \simeq 8\pi G_{\rm N} = 8\pi/M_{\rm Pl}^2$. Status: ASSUMED.
- $\hat{G}^{\mu}{}_{\nu}$: The Einstein tensor of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold. Unit: 1/length^2. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- $\mathcal{T}_{\rm SM}$: The energy-momentum tensor of the Standard Model on the fold, $\mathcal{T}_{\rm H} + \mathcal{T}_{\rm gauge} + \mathcal{T}_{\rm f}$. Unit: GeV^4. Status: STANDARD.
- $\mathcal{T}_{\rm P}$: The energy-momentum tensor of the portals of dirac16complex. Unit: GeV^4. Status: DERIVED.
- $\mathcal{T}_{\rm ff}{}^{a}{}_{b}$: The energy-momentum tensor of the fold fluid (frame components of the fold, four-dimensional normalisation). Unit: GeV^4. Status: ASSUMED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\rm t}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{J}^{a}{}_{b}$: The jump of the junction potential across the fold computed from the metric (the right-hand side of FE-30, or of FE-33 with frozen extra times): the geometric side of the junction condition, $\mathcal{J} = \kappa\mathcal{S}$. Unit: GeV. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- $\mathcal{S}^{a}{}_{b}$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_{\rm N}$: Newton's constant of four-dimensional gravity on the fold, $G_{\rm N} = 1/M_{\rm Pl}^2$. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.
- $\rho_{\rm SM}$: The energy density of the Standard-Model particles. Unit: GeV^4. Status: STANDARD.
- ρ_χ: The energy density of the dark matter, the bound states χ of dirac16complex; $\rho_\chi^{(0)}$ its value today. Unit: GeV^4. Status: DERIVED.
- ρ_Λ: The vacuum energy density of the fold, the dark energy of UM-3: $\rho_\Lambda = \rho_{\rm ff} - 31HV_*/(4\kappa)$ (FE-59). Unit: GeV^4. Status: DERIVED.
- $\epsilon_{\rm r}$: The bookkeeping parameter of the response of the bulk to the matter of the fold: the ratio of the bulk to the fold gravity, $\epsilon_{\rm r} = \kappa_4 V_*/(\kappa H)$. Unit: dimensionless. Value: $\lesssim 10^{-15}$. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- $\rho_{\rm ff}$: The energy density of the fold fluid. Unit: GeV^4. Value: $31HV_*/(4\kappa)$ at the static block. Status: ASSUMED.
- $\hat{R}$: The scalar curvature of the metric $\hat{h}_{\mu\nu}$ of the fold. Unit: 1/length^2. Status: STANDARD.
- $q_{\rm SM}$: The pressure of the Standard Model along the extra times at the fold: the mean on-shell Lagrangian density of the Standard-Model fields (FE-5: the extra-time component of the energy-momentum tensor of every term of the fold is its Lagrangian). Unit: GeV^4. Status: DERIVED.

- q_χ: The pressure of the bound states of dirac16complex along the extra times, integrated across the fold (their on-shell Lagrangian, the self-interaction energy per volume; zero for a dilute gas). Unit: GeV^4. Status: DERIVED.
- q_P: The pressure of the portals along the extra times: their mean on-shell Lagrangian density. Unit: GeV^4. Status: DERIVED.
- q_ff: The pressure of the fold fluid along the extra times. Unit: GeV^4. Value: $-61HV_*/(8\kappa)$ at the static block. Status: ASSUMED.
- r: An extra-time direction of the fold: $r \in \{5,6,7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\mathcal{L}_\mathrm{H}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\mathcal{L}_\mathrm{gauge}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{f}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{Y}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{P}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The fold equations to leading order in ϵ_r (FE-59): Einstein's equations of the fold with the sources of the fold, and the additional extra-time component. Status: DERIVED.

$$\begin{aligned} &3H_4^2 = \kappa_4\,\rho, \qquad 2H_4' + 3H_4^2 = -\kappa_4\,\mathcal{P}, \qquad q_\mathrm{SM} + q_\chi + q_\mathrm{P} + q_\mathrm{ff} + \frac{\rho - 3\mathcal{P}}{2} = 0, \\ &\text{rigid bulk medium:} \qquad 5H_4' + 9H_4^2 = -\kappa_4\big(\mathcal{P} + q_\mathrm{SM} + q_\chi + q_\mathrm{P} + q_\mathrm{ff}\big) \end{aligned} \tag{S-8}$$

Symbols in (S-8).

- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- κ_4: The coupling of the induced gravity of the fold, $\kappa_4 = 8\pi G_\mathrm{ind}$; four-dimensional gravity on the fold has $8\pi G_\mathrm{N} = \kappa_4/(1 + 5\epsilon_\mathrm{r}/6)$ (FE-59). Unit: GeV^{-2}. Value: $\kappa_4 \simeq 8\pi G_\mathrm{N} = 8\pi/M_\mathrm{Pl}^2$. Status: ASSUMED.
- ρ: The total energy density of the matter of the fold (all species); $\Delta\rho$ its perturbation. Unit: GeV^4. Status: STANDARD.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\mathcal{P}$: The total pressure of the matter of the fold; $\Delta\mathcal{P}$ its perturbation. Unit: GeV^4. Status: STANDARD.
- q_SM: The pressure of the Standard Model along the extra times at the fold: the mean on-shell Lagrangian density of the Standard-Model fields (FE-5: the extra-time component of the energy-momentum tensor of every term of the fold is its Lagrangian). Unit: GeV^4. Status: DERIVED.
- q_χ: The pressure of the bound states of dirac16complex along the extra times, integrated across the fold (their on-shell Lagrangian, the self-interaction energy per volume; zero for a dilute gas). Unit: GeV^4. Status: DERIVED.
- q_P: The pressure of the portals along the extra times: their mean on-shell Lagrangian density. Unit: GeV^4. Status: DERIVED.
- q_ff: The pressure of the fold fluid along the extra times. Unit: GeV^4. Value: $-61HV_*/(8\kappa)$ at the static block. Status: ASSUMED.

To leading order in ϵ_r the time and space components of the junction at the fold are the Friedmann and the acceleration equation of general relativity (ρ and $\mathcal{P}$ the total energy density and pressure of the matter of the fold, $H_4 = a_4'$), and the extra-time component adds the radion condition (the first line, last equation): the sum over the matter of the fold of its pressure q along the extra times (for every term of the fold its on-shell Lagrangian density) and $(\rho - 3\mathcal{P})/2$ must vanish. Radiation satisfies it ($q = 0$, $\rho = 3\mathcal{P}$); non-relativistic matter ($q = 0$ for the bound states χ of dirac16complex, $q = -\rho$ for classical particles of the fold) and vacuum energy ($\mathcal{P} = q = -\rho$) do not. If the bulk medium is rigid, the only first-order mode, the difference mode, must absorb the mismatch, and the space and extra-time components combine into the second line: radiation still expands as $x_4^{1/2}$, but dust would expand as $x_4^{5/6}$ ($q = -\rho$) or $x_4^{5/9}$ ($q = 0$), and vacuum energy would give $H_4' = -\frac{3}{5}H_4^2$ instead of a constant expansion rate, which

observation excludes. UM-3 therefore requires that the bulk medium respond with the radion mode, which keeps the four-dimensional equations those of general relativity. Status: DERIVED (SR4, SR5), ASSUMED (the response).

$$\frac{c_{\mathrm{rad}}}{H}=\frac{8}{5}\,\frac{3\mathsf{A}^2}{\epsilon_{\mathrm{r}}}\,\frac{\Delta_{\mathrm{r}}}{\rho_{\mathrm{c}}^{(0)}}=3.806\,\frac{\mathsf{A}^2}{\epsilon_{\mathrm{r}}},\qquad \Delta_{\mathrm{r}}=q_{\mathrm{SM}}+q_\chi+q_{\mathrm{P}}+q_{\mathrm{ff}}+\frac{\rho-3\mathcal{P}}{2}=0.793\,\rho_{\mathrm{c}}^{(0)} \tag{S-9}$$

Symbols in (S-9).

- c_{rad}: The amplitude of the radion mode: the first-order jumps at the fold $([\mathfrak{a}_1'],[\mathfrak{b}_1'],[\mathfrak{n}_1'])=c_{\mathrm{rad}}(1,-1,5)$, which change only the extra-time component of the junction. Unit: 1/length. Value: $3.806\,\mathsf{A}^2H/\epsilon_{\mathrm{r}}$ today. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda=-10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A}=a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- ϵ_{r}: The bookkeeping parameter of the response of the bulk to the matter of the fold: the ratio of the bulk to the fold gravity, $\epsilon_{\mathrm{r}}=\kappa_4V_*/(\kappa H)$. Unit: dimensionless. Value: $\lesssim 10^{-15}$. Status: DERIVED.
- Δ_{r}: The mismatch of the radion condition: the sum over the matter of the fold of its pressure along the extra times and $(\rho-3\mathcal{P})/2$ (zero for radiation). Unit: GeV^4. Value: $0.793\,\rho_{\mathrm{c}}^{(0)}$ today. Status: DERIVED.
- ρ_{c}: The critical density, the energy density of a spatially flat fold at the expansion rate H_4; $\rho_{\mathrm{c}}^{(0)}$ its value today. Unit: GeV^4. Value: $\rho_{\mathrm{c}}^{(0)}=1.0537\times10^{-5}\,h_{100}^2$ GeV/cm^3. Status: STANDARD.
- q_{SM}: The pressure of the Standard Model along the extra times at the fold: the mean on-shell Lagrangian density of the Standard-Model fields (FE-5: the extra-time component of the energy-momentum tensor of every term of the fold is its Lagrangian). Unit: GeV^4. Status: DERIVED.
- q_χ: The pressure of the bound states of dirac16complex along the extra times, integrated across the fold (their on-shell Lagrangian, the self-interaction energy per volume; zero for a dilute gas). Unit: GeV^4. Status: DERIVED.
- q_{P}: The pressure of the portals along the extra times: their mean on-shell Lagrangian density. Unit: GeV^4. Status: DERIVED.
- q_{ff}: The pressure of the fold fluid along the extra times. Unit: GeV^4. Value: $-61HV_*/(8\kappa)$ at the static block. Status: ASSUMED.
- ρ: The total energy density of the matter of the fold (all species); $\Delta\rho$ its perturbation. Unit: GeV^4. Status: STANDARD.
- $\mathcal{P}$: The total pressure of the matter of the fold; $\Delta\mathcal{P}$ its perturbation. Unit: GeV^4. Status: STANDARD.

The amplitude of the radion response today: the mismatch Δ_{r} of the radion condition with the present content of the fold (dark matter χ with $q=0$: $0.265\rho_{\mathrm{c}}^{(0)}$; baryons with $q=-\rho$: $0.049\rho_{\mathrm{c}}^{(0)}$; vacuum energy: $0.685\rho_{\mathrm{c}}^{(0)}$; $\rho_{\mathrm{c}}^{(0)}$ the critical density) must be carried by the jump $\frac{5}{8}c_{\mathrm{rad}}$ of S-7, which gives S-9 ($\mathsf{A}=H_4/H$). It is a small term of the power series when $\mathsf{A}^2\ll\epsilon_{\mathrm{r}}$, that is when the bulk scale H exceeds the Hubble rate $H_4=1.4\times10^{-42}$ GeV by far more than $\epsilon_{\mathrm{r}}^{-1/2}\gtrsim3\times10^7$: for any H above 10^{-33} GeV. Status: DERIVED (SR6).

4 The Higgs field in the ad hoc vacuum

$$\begin{aligned}&\frac{1}{\sqrt{-\hat h}\,\mathcal{Z}}\,D_\mu\big(\sqrt{-\hat h}\,\mathcal{Z}\,\hat h^{\mu\nu}D_\nu\Phi\big)-\mu^2\Phi-2\lambda\big(\Phi^\dagger\Phi\big)\Phi-c_\Phi\,\bar\Psi P_{\mathrm{L}}\Psi\,\Phi=\Theta,\\&\Theta=\begin{pmatrix}\bar d_{\mathrm{R}}(\mathcal{Y}^d)^\dagger u_{\mathrm{L}}+\bar e_{\mathrm{R}}(\mathcal{Y}^e)^\dagger\nu_{\mathrm{L}}-\bar d_{\mathrm{L}}\mathcal{Y}^u u_{\mathrm{R}}-\bar e_{\mathrm{L}}\mathcal{Y}^\nu\nu_{\mathrm{R}}\\ \bar d_{\mathrm{R}}(\mathcal{Y}^d)^\dagger d_{\mathrm{L}}+\bar e_{\mathrm{R}}(\mathcal{Y}^e)^\dagger e_{\mathrm{L}}+\bar u_{\mathrm{L}}\mathcal{Y}^u u_{\mathrm{R}}+\bar\nu_{\mathrm{L}}\mathcal{Y}^\nu\nu_{\mathrm{R}}\end{pmatrix}\end{aligned} \tag{FE-15}$$

Symbols in (FE-15).

- $\hat h_{\mu\nu}$: The four-dimensional metric of the fold, $\hat h_{\mu\nu}=\mathrm{diag}(\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},\mathrm{e}^{2a_4},-1)$ in the coordinates $x_1,\dots,x_4$ (the induced metric at $\sigma=0$ without the extra times); $\hat h^{\mu\nu}$ its inverse, $\hat h$ its determinant, $\sqrt{-\hat h}=\mathrm{e}^{3a_4}$; $\hat h_{\mu\nu}^{(1)}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4)=V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z}=1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.

- $D_M,\ D_\mu$: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m_h^2/2 = -(88.6\ \mathrm{GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- $\Psi,\ \bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- Θ: The Yukawa source of the Higgs equation, $\Theta = -\partial\mathcal{L}_\mathrm{Y}/\partial\Phi^\dagger$ (a doublet of fermion bilinears). Unit: GeV^3. Status: STANDARD.
- d_R: The right-handed down-type quarks d, s, b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\mathcal{Y}^u,\ \mathcal{Y}^d,\ \mathcal{Y}^e,\ \mathcal{Y}^\nu$: The Yukawa matrices ($3\times 3$ complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- u_L: The upper (up-type, left-handed) components of the quark doublets. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- e_R: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- ν_L: The upper (neutral, left-handed) components of the lepton doublets: the left-handed neutrinos. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_L: The lower (down-type, left-handed) components of the quark doublets. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- u_R: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- e_L: The lower (charged, left-handed) components of the lepton doublets. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.

$$\frac{1}{\sqrt{-\hat h}\,\mathcal{Z}}\Big(\partial_\mu\big[\sqrt{-\hat h}\,\mathcal{Z}\,\mathcal{K}_W\,W^{I\mu\nu}\big] + g_2\,\epsilon^{IJK}W^J_\mu\sqrt{-\hat h}\,\mathcal{Z}\,\mathcal{K}_W\,W^{K\mu\nu}\Big) = -g_2\big(J^{I\nu}_\mathrm{w} + J^{I\nu}_\Phi\big),$$
$$\mathcal{K}_W = 1 + \sum_X c^W_X\,\bar\Psi P_X\Psi, \qquad J^{I\nu}_\mathrm{w} = \bar Q\,\hat\gamma^\nu\frac{\tau^I}{2}Q + \bar\ell\,\hat\gamma^\nu\frac{\tau^I}{2}\ell, \qquad J^{I\nu}_\Phi = \mathrm{i}\Big[(D^\nu\Phi)^\dagger\frac{\tau^I}{2}\Phi - \Phi^\dagger\frac{\tau^I}{2}D^\nu\Phi\Big] \tag{FE-17}$$

Symbols in (FE-17).

- $\hat h_{\mu\nu}$: The four-dimensional metric of the fold, $\hat h_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat h^{\mu\nu}$ its inverse, $\hat h$ its determinant, $\sqrt{-\hat h} = \mathrm{e}^{3a_4}$; $\hat h^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\mathcal{K}_G,\ \mathcal{K}_W,\ \mathcal{K}_B$: The portal factors of the gauge kinetic terms, $\mathcal{K}_G = 1 + \sum_X c^G_X\bar\Psi P_X\Psi$ (likewise for W and B): the gauge couplings at the fold are divided by $\sqrt{\mathcal{K}}$. Unit: dimensionless. Value: 1 where dirac16complex vanishes. Status: DERIVED.
- $W^I_\mu,\ W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.

- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- ϵ^{IJK}: The Levi-Civita symbol of three indices (the structure constants of $SU(2)$), $\epsilon^{123} = 1$. Unit: dimensionless. Status: STANDARD.
- J: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- K: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- $J_{\mathrm{w}}^{I\nu}$: The weak-isospin current of the left-handed fermion doublets. Unit: GeV^3. Status: STANDARD.
- $J_{\Phi}^{I\nu}$: The weak-isospin current of the Higgs doublet (it contains the mass terms of the W fields); $J_{\Phi}^{2\,x_4}$ its isospin charge density with $I = 2$. Unit: GeV^3. Status: STANDARD.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu \hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.

The Higgs equation and the equation of the $SU(2)$ gauge fields on the fold (with the factor $\mathcal{Z}$ of the extra times and the portal factor $\mathcal{K}_W$). Status: DERIVED.

$$\begin{aligned}
&\mathcal{R}_\mathcal{A} = \mathcal{A}'' + 3\frac{b_4'}{b_4}\mathcal{A}' + \lambda\mathcal{A}^3 + c_\Phi\,\bar{\Psi}P_\mathrm{L}\Psi\,\mathcal{A} = \Big(4a_{40}^2 - 6a_{40}\frac{b_4'}{b_4}\Big)\mathcal{A} + \lambda\mathcal{A}^3 + c_\Phi\,\bar{\Psi}P_\mathrm{L}\Psi\,\mathcal{A}, \\
&\mathcal{R}_v = \lambda\mathcal{A}^2 v + c_\Phi\,\bar{\Psi}P_\mathrm{L}\Psi\,v, \qquad J_\Phi^{2\,x_4} = a_{40}\,v\,\mathcal{A}, \qquad J_\Phi^{1\,x_4} = J_\Phi^{3\,x_4} = 0
\end{aligned} \tag{FE-26}$$

Symbols in (FE-26).

- $\mathcal{R}_\mathcal{A}$, $\mathcal{R}_v$: The residuals of the ad hoc Higgs vacuum in the upper ($\mathcal{R}_\mathcal{A}$) and lower ($\mathcal{R}_v$) components of the Higgs equation: what the ad hoc vacuum leaves unbalanced. Unit: GeV^3. Status: DERIVED.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_\mathcal{A}\mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.

- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: GeV$^{7/2}$. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- a_{40}: The decay constant of almightyAdHoc, $a_{40} > 0$: $\mathcal{A}$ falls by e in the time $1/(2a_{40})$. Unit: GeV (1/length). Status: AUTHOR.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV. Status: STANDARD.
- $J_\Phi^{I\nu}$: The weak-isospin current of the Higgs doublet (it contains the mass terms of the W fields); $J_\Phi^{2\,x_4}$ its isospin charge density with $I = 2$. Unit: GeV^3. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.

The author's ad hoc vacuum $\Phi = (\mathcal{A}, v + h)/\sqrt{2}$ with $\mathcal{A} = v_\mathcal{A}\mathrm{e}^{-2a_{40}x_4}$, inserted literally into FE-15 and FE-17 (with $h = 0$ and no gauge field): it leaves the residuals $\mathcal{R}_\mathcal{A}$ and $\mathcal{R}_v$ in the two components of the Higgs equation, and it carries a homogeneous charge density of the second isospin direction, which the Gauss law of FE-17 forbids for homogeneous gauge fields. Status: DERIVED.

$$
\begin{aligned}
&\Phi = \frac{1}{\sqrt{2}}\begin{pmatrix}\mathcal{A}\\ v\end{pmatrix} = \mathcal{O}(\phi_\mathrm{rot})\,\frac{1}{\sqrt{2}}\begin{pmatrix}0\\ V_\mathrm{rad}\end{pmatrix}, \qquad V_\mathrm{rad} = \sqrt{v^2 + \mathcal{A}^2}, \qquad \tan\frac{\phi_\mathrm{rot}}{2} = \frac{\mathcal{A}}{v},\\
&\mathcal{O}(\phi_\mathrm{rot}) = \begin{pmatrix}\cos\frac{\phi_\mathrm{rot}}{2} & \sin\frac{\phi_\mathrm{rot}}{2}\\ -\sin\frac{\phi_\mathrm{rot}}{2} & \cos\frac{\phi_\mathrm{rot}}{2}\end{pmatrix}, \qquad W_{x_4}^2 = \frac{\phi_\mathrm{rot}'}{g_2} = -\frac{4a_{40}\,v\,\mathcal{A}}{g_2\big(v^2 + \mathcal{A}^2\big)}, \qquad J_\Phi^{2\,x_4} = 0,\\
&\text{FE-15} \;\Leftrightarrow\; \mathcal{O}(\phi_\mathrm{rot})\,\frac{1}{\sqrt{2}}\begin{pmatrix}0\\ -\mathcal{R}_\mathrm{rad}\end{pmatrix} = 0, \qquad \mathcal{R}_\mathrm{rad} = V_\mathrm{rad}'' + 3\,\frac{b_4'}{b_4}\,V_\mathrm{rad}' + \lambda\,\mathcal{A}^2\,V_\mathrm{rad} + c_\Phi\,\bar\Psi P_\mathrm{L}\Psi\,V_\mathrm{rad}
\end{aligned}
\tag{S-10}
$$

Symbols in (S-10).

- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_\mathcal{A}\mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV. Status: STANDARD.
- $\mathcal{O}(\phi_\mathrm{rot})$: The 2×2 isospin rotation by the angle ϕ_rot about the second isospin axis (a real element of $SU(2)$). Unit: dimensionless. Status: DERIVED.
- V_rad: The modulus of the Higgs doublet in the ad hoc vacuum, $V_\mathrm{rad} = \sqrt{v^2 + \mathcal{A}^2}$ ($V_\mathrm{rad}^2 = 2\Phi^\dagger\Phi$). Unit: GeV. Status: DERIVED.
- ϕ_rot: The angle of the isospin rotation that turns $(0, V_\mathrm{rad})/\sqrt{2}$ into the ad hoc vacuum: $\tan(\phi_\mathrm{rot}/2) = \mathcal{A}/v$. Unit: dimensionless. Status: DERIVED.
- W_μ^I, $W_{\mu\nu}^I$: The gauge field of $SU(2)_L$ on the fold, $W_\mu^1, W_\mu^2, W_\mu^3$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- a_{40}: The decay constant of almightyAdHoc, $a_{40} > 0$: $\mathcal{A}$ falls by e in the time $1/(2a_{40})$. Unit: GeV (1/length). Status: AUTHOR.
- $J_\Phi^{I\nu}$: The weak-isospin current of the Higgs doublet (it contains the mass terms of the W fields); $J_\Phi^{2\,x_4}$ its isospin charge density with $I = 2$. Unit: GeV^3. Status: STANDARD.
- $\mathcal{R}_\mathrm{rad}$: The radial residual of the Higgs equation FE-15 in the ad hoc vacuum written as a gauge rotation (S-10): the amount by which V_rad fails to solve the radial equation; zero for a solution. Unit: GeV^3. Status: DERIVED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.

- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: GeV$^{7/2}$. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.

The ad hoc vacuum is a rotation of the doublet $(0, V_\mathrm{rad})/\sqrt{2}$ by the angle $\phi_\mathrm{rot}(x_4)$ about the second isospin axis. Together with the pure-gauge field $W^2_{x_4} = \phi'_\mathrm{rot}/g_2$, which undoes the rotation in the covariant derivative, the isospin charge density of the Higgs field vanishes, so the Gauss law of FE-17 holds with homogeneous fields; the masses of FE-45 are unchanged ($v^2 + \mathcal{A}^2 = V^2_\mathrm{rad}$). The Higgs equation FE-15 then reduces exactly to one radial residual $\mathcal{R}_\mathrm{rad}$ (at 20 random points of the parameters the reduction holds to 10^{-15}). With $\mu^2 = -\lambda v^2$ the ad hoc vacuum is a solution only in the limit $\mathcal{A} \to 0$, which it reaches exponentially. Status: DERIVED (BG5).

a_{40}/v	$\mathcal{R}_\mathrm{rad}(x_4 = 0)/(\lambda v^3)$	$\rho_\mathrm{H}(x_4 = 0)/v^4$	time until $\mathcal{A} < 10^{-3}v$
0.1	1.74	0.042	$34.5/v$
1	34.2	1.03	$3.45/v$
10	3280	100	$0.345/v$

The radial residual at $x_4 = 0$ and the energy density of the Higgs field, $\rho_\mathrm{H} = \frac{1}{2}V'^2_\mathrm{rad} + \frac{\lambda}{4}\mathcal{A}^4$, for $v_\mathcal{A} = v$, a constant b_4 and no dirac16complex at the fold, for three values of a_{40} (in units of $v = 246.22$ GeV); the ad hoc component is gone after a time of order $3.5/a_{40}$. Before nucleosynthesis the Fermi constant $G_\mathrm{F,loc} = 1/(\sqrt{2}(v^2 + \mathcal{A}^2))$ must be within one per cent of its value: $\mathcal{A}^2/v^2 < 0.01$ at $x_4 = 1$ s requires $a_{40} > \ln(v_\mathcal{A}/(0.1v))/(2\text{ s})$, a weak bound ($a_{40} > 2.3\text{ s}^{-1} = 1.5 \times 10^{-24}$ GeV for $v_\mathcal{A} = 10v$). Status: DERIVED (BG6, `um3_thermal.py`, T6).

5 The spectra along the hidden direction

Every field of the bulk is a superposition of modes along the hidden direction σ; their frequencies are the masses and energies that an observer on the fold sees (the "x_0 masses"). This section gives the spectra of the bulk fields and of dirac16complex on the static block (a_4, b_4 constant, normalised to $\mathrm{e}^{a_4} = b_4 = 1$ at the fold).

5.1 The bulk fields and why the Standard Model lives on the fold

$$u_\mathrm{s}'' + \left(\omega^2 - 9H^2\right)u_\mathrm{s} + 6H\,\delta_\mathrm{D}(\sigma)\,u_\mathrm{s} = 0, \qquad u_\mathrm{s} = \mathrm{e}^{-3H|\sigma|}\ \ (\omega = 0), \qquad \omega \geq 3H\ \ \text{(continuum)} \tag{S-11}$$

Symbols in (S-11).

- u_s: The rescaled profile of a massless scalar field of the bulk at zero momentum along ordinary space: the field is $\mathrm{e}^{3H|\sigma|}u_\mathrm{s}(\sigma)\,\mathrm{e}^{-\mathrm{i}\omega x_4}$. Unit: as the field. Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a'_4 = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- δ_D: The Dirac delta function: $\int \mathrm{d}\sigma\,\delta_\mathrm{D}(\sigma)F(\sigma) = F(0)$. Unit: 1/length (for $\delta_\mathrm{D}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

A massless scalar field of the bulk at zero momentum along ordinary space, with its profile written as $\mathrm{e}^{3H|\sigma|}u_\mathrm{s}$ (which removes the first derivative): the kink of the warp acts as an attractive delta function at the fold, which binds exactly one state, the zero mode (a constant profile, $\omega = 0$), separated by the gap $3H$ from the continuum of the bulk. Status: DERIVED (`um3_bulk.py`, B1).

$$c_{\rm sc}^2 = \left\langle \mathrm{e}^{2H|\sigma|} \right\rangle = \frac{3}{2}, \qquad c_{\rm gauge}^2 = 2, \qquad c_{\rm fer} = \frac{2M_{\rm k}}{2M_{\rm k} - H}, \qquad c_{\rm g}^2 = 5 \cdot \frac{3}{2} = \frac{15}{2} \tag{S-12}$$

Symbols in (S-12).

- $c_{\rm sc}$: The limiting speed of the zero mode of a scalar field of the bulk along ordinary space, in units of the speed of light on the fold. Unit: dimensionless. Value: $\sqrt{3/2}$. Status: DERIVED.
- $\langle \cdots \rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $c_{\rm gauge}$: The limiting speed of the zero mode of a gauge field (a photon) of the bulk, in units of the speed of light on the fold. Unit: dimensionless. Value: $\sqrt{2}$. Status: DERIVED.
- $c_{\rm fer}$: The limiting speed of the zero mode of a fermion of the bulk with the kink mass $M_{\rm k}\,\mathrm{sgn}(\sigma)$, in units of the speed of light on the fold. Unit: dimensionless. Status: DERIVED.
- $M_{\rm k}$: The mass of a fermion of the bulk that changes sign at the fold (a kink mass $M_{\rm k}\,\mathrm{sgn}(\sigma)$), used to illustrate the speeds of bulk fermions. Unit: GeV. Status: ASSUMED.
- $c_{\rm g}$: The limiting speed of the zero mode of the transverse traceless graviton of the bulk at the static block, in units of the speed of light on the fold. Unit: dimensionless. Value: $\sqrt{15/2} = 2.74$. Status: DERIVED.

The limiting speeds, in units of the speed of light on the fold, of the zero modes of the bulk fields when they move along ordinary space: a zero mode samples the warp $\mathrm{e}^{-H|\sigma|}$ of ordinary space over its profile, so its speed is the average of $\mathrm{e}^{2H|\sigma|}$ with the weight of its profile. A bulk scalar moves $\sqrt{3/2} = 1.22$ times faster than light on the fold, a bulk gauge field (photon) $\sqrt{2} = 1.41$ times, a bulk fermion with the kink mass $M_{\rm k}\,\mathrm{sgn}(\sigma)$ (zero mode $\propto \mathrm{e}^{(3H - M_{\rm k})|\sigma|}$, normalisable for $M_{\rm k} > 0$) $2M_{\rm k}/(2M_{\rm k} - H)$ times (2 for $M_{\rm k} = H$, 1.0005 for $M_{\rm k} = 1000H$), and the transverse traceless graviton of the static block, whose local cone is $\sqrt{5}$ times the light cone because of the Gauss-Bonnet and cubic Lovelock terms, $\sqrt{15/2} = 2.74$ times (the numerical eigenvalues agree to 10^{-4}). Species in the bulk would therefore have different limiting speeds, which measurements exclude at the level 10^{-15} to 10^{-20} for photons, electrons and nucleons, and three neutrino flavours with three profiles would oscillate with phases proportional to the energy, which is excluded too: the Standard Model lives on the fold, where every field has the one light cone of the fold. Bulk gravitons are excluded as the carriers of four-dimensional gravity by GW170817 (the speed of gravitational waves equals that of light to 10^{-15}), and at the author's vacuum the graviton has no time-kinetic term at all: UM-3 gives the fold its own Einstein-Hilbert term. Status: DERIVED (B1 to B6).

5.2 dirac16complex: the continuum and the bound states

$$\Psi = \mathrm{e}^{3H|\sigma|}\,\chi(\sigma)\,\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x} - \omega x_4)}, \qquad \gamma^{(0)}\chi' - \mathrm{i}\omega\,\gamma^{(4)}\chi + \mathrm{i}\,\mathrm{e}^{H|\sigma|}\,\vec{k}\cdot\vec{\gamma}\,\chi - m_\Psi\chi = 0 \qquad (\sigma \neq 0) \tag{FE-36}$$

Symbols in (FE-36).

- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x} - \omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.

- $\vec{k}$: The (comoving) momentum along ordinary space, $\vec{k}^2$ its square. Unit: GeV. Status: STANDARD.
- $\vec{x}$: The position in ordinary space, $\vec{x} = (x_1, x_2, x_3)$. Unit: length. Status: AUTHOR.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\vec{\gamma}$: The three frame Dirac matrices of dirac16complex along ordinary space, $\vec{\gamma} = (\gamma^{(1)}, \gamma^{(2)}, \gamma^{(3)})$. Unit: dimensionless. Status: AUTHOR.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.

$$\chi' = \mathbb{K}\,\chi, \qquad \mathbb{K} = \gamma^{(0)}\big(m_\Psi + \mathrm{i}\omega\gamma^{(4)}\big), \qquad \mathbb{K}^2 = \big(m_\Psi^2 - \omega^2\big)\mathbf{1}, \qquad \chi \propto \mathrm{e}^{\pm q\sigma}, \qquad q = \sqrt{m_\Psi^2 - \omega^2} \tag{FE-37}$$

Symbols in (FE-37).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x} - \omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\mathbb{K}$: The transfer matrix of the profile equation at zero momentum, $\chi' = \mathbb{K}\chi$. Unit: GeV. Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- q: The decay rate of a profile along σ at frequency ω, $q = \sqrt{m_\Psi^2 - \omega^2}$ (imaginary in the continuum). Unit: GeV. Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

$$\omega^2 = m_\Psi^2 + p_\sigma^2 + \mathrm{e}^{2H|\sigma| - 2a_4}\,\vec{k}^2 - \mathrm{e}^{2H|\sigma| + 2a_4}\,b_4^{-2}\,\vec{\kappa}^2 \tag{FE-38}$$

Symbols in (FE-38).

- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- p_σ: The momentum along the hidden direction (local wave number of a profile along σ). Unit: GeV. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.

- $\vec{k}$: The (comoving) momentum along ordinary space, $\vec{k}^2$ its square. Unit: GeV. Status: STANDARD.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4} b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- $\vec{\kappa}$: A momentum along the extra times (three components); the fields of UM-3 have $\vec{\kappa} = 0$. Unit: GeV. Status: DERIVED.

The field equation of dirac16complex on the static block (FE-36), its solutions at zero momentum along ordinary space (FE-37): the continuum $|\omega| \geq m_\Psi$ and the exponentials below it, and the local dispersion relation (FE-38), which shows that a dependence on the extra times would make every field tachyonic. Status: DERIVED.

$$\chi(0^+) = \exp\Big(\gamma^{(0)}\big(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R}\big)\Big)\,\chi(0^-),$$
$$\mu_X = \frac{1}{V_*}\Big[c_\Phi\,\langle\Phi^\dagger\Phi\rangle\,\delta_{X\mathrm{L}} + \frac{1}{4}\big(c_X^G\langle G^2\rangle + c_X^W\langle W^2\rangle + c_X^B\langle B^2\rangle\big) + 2\omega\sum_f c_X^f\,n_f\Big], \qquad \langle\Phi^\dagger\Phi\rangle = \frac{v^2 + \mathcal{A}^2}{2} \tag{FE-39}$$

Symbols in (FE-39).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x} - \omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- μ_X, μ_L, μ_R: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex (FE-39): the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- P_R: The author's right-handed chiral projector, $P_\mathrm{R} = (\mathbf{1} + \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4} b_4)^3$ at a time when $\mathrm{e}^{-a_4} b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta_N^M = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}} = 1$ for $X = \mathrm{L}$); with upper and lower index strings the generalised Kronecker delta $\delta_{N_1\cdots N_n}^{M_1\cdots M_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta_{N_b}^{M_a}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- $\langle G^2\rangle$: The mean square of the gluon field strength, $\langle G^A_{\mu\nu} G^{A\mu\nu}\rangle$ (the gluon condensate in the QCD vacuum, a thermal value in the plasma). Unit: GeV^4. Status: STANDARD.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- $\langle W^2\rangle$: The mean square of the $SU(2)$ field strength, $\langle W^I_{\mu\nu} W^{I\mu\nu}\rangle$. Unit: GeV^4. Status: STANDARD.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- $\langle B^2\rangle$: The mean square of the hypercharge field strength, $\langle B_{\mu\nu} B^{\mu\nu}\rangle$. Unit: GeV^4. Status: STANDARD.

- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- n_f: The number density of the Standard-Model fermions of kind f (particles minus antiparticles, summed over generations and gauge components), the time component of the mean current $\langle J_f^\mu \rangle$ in the rest frame of the medium. Unit: GeV^3. Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV. Status: STANDARD.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_{\mathcal{A}} \mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.

$$q_\mathrm{b} = -m_\Psi \left(\mu_\mathrm{L} + \mu_\mathrm{R}\right) \frac{\tanh \varsigma}{2\varsigma}, \qquad \varsigma = \sqrt{\mu_\mathrm{L}\mu_\mathrm{R}}, \qquad \omega_\mathrm{b} = \sqrt{m_\Psi^2 - q_\mathrm{b}^2}, \qquad |\chi_\mathrm{b}(\sigma)| \propto \mathrm{e}^{-q_\mathrm{b}|\sigma|} \tag{FE-40}$$

Symbols in (FE-40).

- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- μ_X, μ_L, μ_R: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex (FE-39): the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_\mathrm{L}P_\mathrm{L} + \mu_\mathrm{R}P_\mathrm{R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- ς: The auxiliary variable of the bound-state condition, $\varsigma = \sqrt{\mu_\mathrm{L}\mu_\mathrm{R}}$ ($\varsigma_- = \sqrt{-\mu_\mathrm{L}\mu_\mathrm{R}}$ when $\mu_\mathrm{L}\mu_\mathrm{R} < 0$). Unit: dimensionless. Status: DERIVED.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- χ_b: The profile of a state of dirac16complex bound to the fold. Unit: GeV^2. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The fold acts on dirac16complex as a chirality-dependent mass concentrated at $\sigma = 0$, with strengths μ_L and μ_R set by the mean fields of the Standard Model (FE-39); the bound states (FE-40) exist for an attractive fold. Status: DERIVED.

$(\mu_\mathrm{L}, \mu_\mathrm{R})$	$(-0.4, 0)$	$(-1, 0)$	$(-0.6, -0.3)$	$(-0.5, 0.2)$	$(-1.5, -0.5)$	$(0.4, 0)$	$(0.3, 0.2)$
ω_b/m_Ψ	0.979796	0.866025	0.905282	0.987882	0.589814	none	none

The bound states computed with the author's exact 16×16 matrices: the roots of the 8×8 matching determinant agree with FE-40 to 10^{-7} in all seven cases, each bound level is eight-fold degenerate, and there is none for a repulsive fold. Of the eight states at $+\omega_\mathrm{b}$, four have the Krein charge $+1$ and four -1 (no null states), so four carry positive and four negative energy (FE-43). The bound-state energy $\omega_\mathrm{b} = \sqrt{m_\Psi^2 - q_\mathrm{b}^2}$ is the x_0 mass of dirac16complex: the rest mass of a particle χ that lives near the fold with the profile $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$ of width $1/q_\mathrm{b}$; the more tightly it is bound, the lighter it is. Status: DERIVED (`um3_spectra.py`, S1 to S4).

$$c_\mathrm{b}^2 = \left\langle \mathrm{e}^{2H|\sigma|} \right\rangle = \frac{q_\mathrm{b}}{q_\mathrm{b} - H} \qquad (q_\mathrm{b} > H) \tag{S-13}$$

Symbols in (S-13).

- c_b: The limiting speed of the bound particles χ of dirac16complex along ordinary space, in units of the speed of light on the fold (first order in the momentum). Unit: dimensionless. Status: DERIVED.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.

The limiting speed of the bound particles χ along ordinary space, to first order in their momentum: the average of the warp over the density $\mathrm{e}^{-2q_\mathrm{b}|\sigma|}$ of their profile ($c_\mathrm{b}^2 = 1.25$, 2 and 5 for $H = 0.1m_\Psi$, $0.25m_\Psi$ and $0.4m_\Psi$ with $q_\mathrm{b} = m_\Psi/2$). The particles χ are slower than the bulk continuum but faster than light on the fold, and for $q_\mathrm{b} \le H$ the average diverges: a bound state of dirac16complex has a sensible dispersion relation only if it is bound more tightly than the warp, $q_\mathrm{b} > H$. Status: DERIVED (S5), ESTIMATE (first order in the momentum).

6 The residuals of every field equation

The approximate solution of UM-3 consists of the parts above and below. This table states, for every field equation, how the solution satisfies it.

equation	how it is solved	residual	status
bulk gravity FE-1, order 0	the energy-momentum tensor of the bulk that the author's metric requires is computed exactly (FE-29, S-1 to S-3); the author's vacuum needs none	0 (exact polynomials)	DERIVED
bulk gravity FE-1, order ϵ_r	S-5: the difference mode with a rigid medium, the radion mode with a responding one	0 at first order; the profile of the responding medium is not computed	DERIVED, ASSUMED
junction FE-4, FE-5: time and space components	S-8: Friedmann and acceleration equations of general relativity	relative order $\epsilon_\mathrm{r} \lesssim 10^{-15}$	DERIVED
junction FE-4: extra-time component	S-8, S-9: the radion condition	the mismatch $\Delta_\mathrm{r} = 0.793\rho_\mathrm{c}^{(0)}$, carried by the radion mode	DERIVED, ASSUMED
conservation at the fold FE-11	the fold loses exactly the energy that the bulk flux carries away	0 (an identity for every a_4, b_4)	DERIVED
dirac16complex FE-12 to FE-14, Kohn-Sham FE-41	the Kohn-Sham states above, lifted to sixteen components	KS_RESIDUALS	DERIVED
Higgs doublet FE-15	S-10: the ad hoc vacuum as a gauge rotation	the radial residual $\mathcal{R}_\mathrm{rad}$ (table of S-10), zero once $\mathcal{A} = 0$	DERIVED
$SU(2)$ gauge fields FE-17	S-10: the pure-gauge $W_{x_4}^2$	0: the isospin charge density vanishes exactly	DERIVED
gluons FE-16 and hypercharge field FE-18	no mean field strengths (the gluon condensate enters FE-39 as a mean value)	0	DERIVED

equation	how it is solved	residual	status
quarks and leptons FE-19 to FE-24	no mean fermion fields; the plasma and the neutrinos are treated statistically (FE-48 to FE-62)	0 in the vacuum	STANDARD
neutrino flavours FE-48 to FE-50	the Sun and the supernova envelope (S-14, S-15)	halving the step changes P_{ee} by less than 10^{-6} (Sun) and 10^{-9} (envelope)	DERIVED
fast instability FE-51	S-16	the dispersion relation holds to 10^{-14} at the computed frequencies	DERIVED
relic density FE-61	S-18	relative tolerance 10^{-8} of the implicit integration	DERIVED
cosmological perturbations FE-54, FE-55	CAMB with the content of UM-3	the accuracy of CAMB (about 10^{-3} in C_l); background and ruler checked independently to 10^{-6}	STANDARD, DERIVED

7 The three neutrino flavours

The author asks for a correct flavour conversion in UM-3, because the flavour conversion of neutrinos shapes the explosion of massive stars (Gogilashvili and Tamborra, "Neutrino Flavor Conversion Shapes the Rate of Failed Core-collapse Supernovae", arXiv:2605.16504). This section solves the flavour equations of UM-3 in the Sun and in core-collapse supernovae.

$$
\begin{aligned}
&\mathrm{i}\big(\partial_{x_4} + \hat{v}\cdot\vec{\nabla}\big)\,\varrho_{\vec{p}} = \big[\mathsf{H}_{\vec{p}},\ \varrho_{\vec{p}}\big] + \mathrm{i}\,\mathcal{C}_{\vec{p}}, \qquad \mathrm{i}\big(\partial_{x_4} + \hat{v}\cdot\vec{\nabla}\big)\,\bar{\varrho}_{\vec{p}} = \big[\bar{\mathsf{H}}_{\vec{p}},\ \bar{\varrho}_{\vec{p}}\big] + \mathrm{i}\,\bar{\mathcal{C}}_{\vec{p}},\\
&\mathsf{H}_{\vec{p}} = \frac{v_{\mathrm{loc}}^2}{v^2}\,\frac{U_{\mathrm{PMNS}}\,\mathsf{M}^2\,U_{\mathrm{PMNS}}^{\dagger}}{2E} + \mathsf{V}_{\mathrm{m}} + \mathsf{V}_{\nu\nu} + V_{\Psi}\mathbf{1}, \qquad \mathsf{M}^2 = \mathrm{diag}\big(m_1^2,\ m_2^2,\ m_3^2\big),\\
&\bar{\mathsf{H}}_{\vec{p}} = -\frac{v_{\mathrm{loc}}^2}{v^2}\,\frac{U_{\mathrm{PMNS}}\,\mathsf{M}^2\,U_{\mathrm{PMNS}}^{\dagger}}{2E} + \mathsf{V}_{\mathrm{m}} + \mathsf{V}_{\nu\nu} + V_{\Psi}\mathbf{1}
\end{aligned}
\tag{FE-48}
$$

Symbols in (FE-48).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\hat{v}$, $\hat{v}^\mu$: The direction of motion of a neutrino: the unit vector $\hat{v} = \vec{p}/|\vec{p}|$ ($\hat{v}_{\vec{p}}$, $\hat{v}_{\vec{q}}$ for the momenta $\vec{p}$, $\vec{q}$), and the four-vector $\hat{v}^\mu = (1, \hat{v})$ in the order $(x_4; x_1, x_2, x_3)$; $\mathrm{d}^2\hat{v}$ is the element of solid angle of the directions. Unit: dimensionless. Status: STANDARD.
- $\vec{\nabla}$: The gradient along ordinary space, $\vec{\nabla} = (\partial_{x_1}, \partial_{x_2}, \partial_{x_3})$ (in the local frame of the medium). Unit: 1/length. Status: STANDARD.
- $\varrho_{\vec{p}}$, $\bar{\varrho}_{\vec{p}}$: The 3×3 flavour density matrices of the neutrinos and of the antineutrinos of momentum $\vec{p}$ in the basis e, μ, τ: the diagonal entries are the occupation numbers of the three flavours, the off-diagonal entries the coherence between the flavours; $\bar{\varrho}$ is defined with the flavour indices in the reversed order, so that it obeys an equation of the same form. Unit: dimensionless. Status: STANDARD.
- $\vec{p}$: The momentum of a neutrino (three components along ordinary space, in the rest frame of the medium); $|\vec{p}| = E$ for a neutrino of negligible mass. Unit: GeV. Status: STANDARD.
- $\mathsf{H}_{\vec{p}}$, $\bar{\mathsf{H}}_{\vec{p}}$: The flavour Hamiltonians (3×3 Hermitian matrices) of the neutrinos and of the antineutrinos of momentum $\vec{p}$. Unit: GeV. Status: STANDARD.
- $\mathcal{C}_{\vec{p}}$, $\bar{\mathcal{C}}_{\vec{p}}$: The collision terms ($3 \times 3$ matrices) of the neutrinos and the antineutrinos of momentum $\vec{p}$: the rates of change of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$ by emission, absorption, scattering and pair processes with the matter. Unit: GeV. Status: STANDARD.
- v_{loc}: The local value of the electroweak vacuum, $v_{\mathrm{loc}}^2 = 2\langle\Phi^\dagger\Phi\rangle$: it sets the local masses of the quarks and leptons (each mass times v_{loc}/v) and the local Fermi constant; $v_{\mathrm{loc}} = v$ in the Standard-Model vacuum. Unit: GeV. Value: v in the late universe outside dense dirac16complex. Status: DERIVED.

- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV. Status: STANDARD.
- U_{PMNS}: The lepton mixing matrix (Pontecorvo-Maki-Nakagawa-Sakata) of the three Dirac neutrinos; $U_{\alpha i}$ are its entries. Unit: dimensionless. Value: $\theta_{12} = 33.4°$, $\theta_{23} = 49.0°$, $\theta_{13} = 8.57°$. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- M^2: The diagonal matrix of the squared masses of the three neutrino mass states, $\mathsf{M}^2 = \mathrm{diag}(m_1^2, m_2^2, m_3^2)$ (M appears only squared). Unit: GeV^2 (of M^2). Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- V_{m}: The matter potential of the neutrinos (a 3×3 matrix in flavour space) from their forward scattering on the electrons, positrons and neutrons of the medium. Unit: GeV. Status: STANDARD.
- $\mathsf{V}_{\nu\nu}$: The neutrino-neutrino potential (a 3×3 matrix in flavour space) from the forward scattering of a neutrino on the other neutrinos and antineutrinos of the medium; it depends on their flavour content and their directions. Unit: GeV. Status: STANDARD.
- V_Ψ: The potential energy that dirac16complex gives every neutrino through the current portal of the lepton doublets, $V_\Psi = \mathcal{V}^\ell_{x_4}$ (the same for the three flavours). Unit: GeV. Status: DERIVED.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- $m_u, \ldots, m_3$: The masses of the quarks (m_u, m_c, m_t; m_d, m_s, m_b), of the charged leptons (m_e, m_μ, m_τ) and of the three neutrino mass states (m_1, m_2, m_3, Dirac masses). Unit: GeV. Value: measured values; $\Delta m^2_{21} = 7.41 \times 10^{-5}$ eV^2, $|\Delta m^2_{31}| = 2.51 \times 10^{-3}$ eV^2. Status: STANDARD.

$$
\begin{aligned}
&\mathsf{V}_{\mathrm{m}} = \sqrt{2}\, G_{\mathrm{F,loc}}\, \mathrm{diag}\big(n_e - n_{\bar e},\ 0,\ 0\big) - \frac{G_{\mathrm{F,loc}}}{\sqrt{2}}\, n_n \mathbf{1}, \qquad \mathsf{V}_{\nu\nu} = \sqrt{2}\, G_{\mathrm{F,loc}} \int \frac{\mathrm{d}^3\vec{q}}{(2\pi)^3} \big(1 - \hat{v}_{\vec{p}} \cdot \hat{v}_{\vec{q}}\big)\big(\varrho_{\vec{q}} - \bar{\varrho}_{\vec{q}}\big), \\
&V_\Psi = \mathcal{V}^\ell_{x_4} = \sum_X c^\ell_X\, j_{X\, x_4}, \qquad G_{\mathrm{F,loc}} = \frac{1}{\sqrt{2}\, v^2_{\mathrm{loc}}}, \\
&v^2_{\mathrm{loc}} = 2\langle \Phi^\dagger \Phi \rangle, \qquad v^2_{\mathrm{loc}} = v^2 + \mathcal{A}^2 \ \text{ (the ad hoc vacuum)}, \\
&v^2_{\mathrm{loc}} = v^2 - \frac{c_\Phi}{\lambda} \langle \bar\Psi P_{\mathrm{L}} \Psi \rangle \ \text{ (the relaxed vacuum with dirac16complex)}
\end{aligned}
\tag{FE-49}
$$

Symbols in (FE-49).

- V_{m}: The matter potential of the neutrinos (a 3×3 matrix in flavour space) from their forward scattering on the electrons, positrons and neutrons of the medium. Unit: GeV. Status: STANDARD.
- $G_{\mathrm{F,loc}}$: The local Fermi constant, $G_{\mathrm{F,loc}} = 1/(\sqrt{2}\, v^2_{\mathrm{loc}})$ ($= G_{\mathrm{F}}$ where $v_{\mathrm{loc}} = v$). Unit: GeV^{-2}. Status: DERIVED.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- n_e: The number density of electrons (bound and free; in a supernova all are free, in the recombining universe $x_e n_e$ is the density of the free electrons). Unit: GeV^3 (cm^{-3}). Status: STANDARD.
- $n_{\bar e}$: The number density of positrons. Unit: GeV^3. Status: STANDARD.
- n_n: The number density of neutrons. Unit: GeV^3. Status: STANDARD.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- $\mathsf{V}_{\nu\nu}$: The neutrino-neutrino potential (a 3×3 matrix in flavour space) from the forward scattering of a neutrino on the other neutrinos and antineutrinos of the medium; it depends on their flavour content and their directions. Unit: GeV. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $\vec{q}$: The momentum of another neutrino of the medium (the integration variable of the neutrino-neutrino potential); $\mathrm{d}^3\vec{q}$ its volume element. Unit: GeV. Status: STANDARD.

- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\hat{v}$, $\hat{v}^\mu$: The direction of motion of a neutrino: the unit vector $\hat{v} = \vec{p}/|\vec{p}|$ ($\hat{v}_{\vec{p}}$, $\hat{v}_{\vec{q}}$ for the momenta $\vec{p}$, $\vec{q}$), and the four-vector $\hat{v}^\mu = (1, \hat{v})$ in the order $(x_4; x_1, x_2, x_3)$; $\mathrm{d}^2\hat{v}$ is the element of solid angle of the directions. Unit: dimensionless. Status: STANDARD.
- $\vec{p}$: The momentum of a neutrino (three components along ordinary space, in the rest frame of the medium); $|\vec{p}| = E$ for a neutrino of negligible mass. Unit: GeV. Status: STANDARD.
- $\varrho_{\vec{p}}$, $\bar{\varrho}_{\vec{p}}$: The 3×3 flavour density matrices of the neutrinos and of the antineutrinos of momentum $\vec{p}$ in the basis e, μ, τ: the diagonal entries are the occupation numbers of the three flavours, the off-diagonal entries the coherence between the flavours; $\bar{\varrho}$ is defined with the flavour indices in the reversed order, so that it obeys an equation of the same form. Unit: dimensionless. Status: STANDARD.
- V_Ψ: The potential energy that dirac16complex gives every neutrino through the current portal of the lepton doublets, $V_\Psi = \mathcal{V}^\ell_{x_4}$ (the same for the three flavours). Unit: GeV. Status: DERIVED.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c^f_X: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c^ℓ_X and c^ν_X are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- j^μ_X: The convective current of the half X of dirac16complex, $j^\mu_X = \mathrm{i}(\bar{\Psi} P_X D^\mu \Psi - (D^\mu \bar{\Psi}) P_X \Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- v_{loc}: The local value of the electroweak vacuum, $v^2_{\mathrm{loc}} = 2\langle \Phi^\dagger \Phi \rangle$: it sets the local masses of the quarks and leptons (each mass times v_{loc}/v) and the local Fermi constant; $v_{\mathrm{loc}} = v$ in the Standard-Model vacuum. Unit: GeV. Value: v in the late universe outside dense dirac16complex. Status: DERIVED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\langle \cdots \rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV. Status: STANDARD.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_{\mathcal{A}} \mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_{L}: The author's left-handed chiral projector, $P_{\mathrm{L}} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.

The flavour equations of the neutrinos of momentum $\vec{p}$ and the potentials (FE-48, FE-49). Two facts make them those of the Standard Model with three Dirac neutrinos wherever $v_{\mathrm{loc}} = v$: the portal potential V_Ψ is the same for the three flavours and drops out of every commutator, and the Higgs portal moves v_{loc} by $\Delta v^2/v^2 \approx 10^{-50}\, \mathfrak{g}_{h\chi}$ at the local dark-matter density. Status: DERIVED (`um3_neutrino.py`, N1: $V_\Psi \mathbf{1}$ changes the evolution by at most 7×10^{-13}, a change of v_{loc} leaves the mixing matrix unchanged to 10^{-15} and scales the masses by v_{loc}/v). For a single neutrino along its path (no collisions, no neutrino-neutrino potential) FE-48 is a Schrodinger equation $\mathrm{i}\,\mathrm{d}\psi_\nu/\mathrm{d}r = \mathsf{H}\psi_\nu$ for the flavour amplitudes ψ_ν; for antineutrinos the vacuum term has U^*_{PMNS} and the matter potential the opposite sign (the convention of FE-48, with the flavour indices of $\bar{\varrho}$ reversed, gives the same probabilities). It is integrated here with the fourth-order commutator-free Magnus method (the exponentials of the Hamiltonian at the two Gauss points of each step, exact for a constant density), and every result is recomputed with half the step.

$$\tan 2\theta_{\rm m} = \frac{\sin 2\theta}{\cos 2\theta - 2EV_{\rm CC}/\Delta m^2}, \qquad V_{\rm CC} = \sqrt{2}\, G_{\rm F,loc}\, n_e, \qquad n_{e,\rm res} = \frac{\Delta m^2 \cos 2\theta}{2\sqrt{2}\, G_{\rm F,loc}\, E},$$
$$P_{\rm LZ} = \exp\Big(-\frac{\pi}{2}\, \frac{\Delta m^2 \sin^2 2\theta}{2E \cos 2\theta}\, \Big| \frac{{\rm d} \ln n_e}{{\rm d}r} \Big|^{-1}_{\rm res} \Big) \tag{FE-50}$$

Symbols in (FE-50).

- $\theta_{\rm m}$: The mixing angle of two neutrino flavours in matter. Unit: dimensionless. Status: STANDARD.
- θ: The vacuum mixing angle of two neutrino flavours (in the two-flavour description of a resonance, one of the three angles $\theta_{12}, \theta_{13}, \theta_{23}$ of $U_{\rm PMNS}$). Unit: dimensionless. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- $V_{\rm CC}$: The charged-current matter potential of the electron neutrinos (forward scattering on electrons through W exchange). Unit: GeV. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows ($\Delta\rho$ a perturbation, Δm^2 a difference of squared masses, $\Delta N_{\rm eff}$ an addition to $N_{\rm eff}$). Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; in Δm^2 the squared-mass difference of two neutrino mass states ($m_2^2 - m_1^2$ for the two states of a resonance, multiplied by $v_{\rm loc}^2/v^2$ where this differs from 1). Unit: GeV. Status: STANDARD.
- $G_{\rm F,loc}$: The local Fermi constant, $G_{\rm F,loc} = 1/(\sqrt{2}\, v_{\rm loc}^2)$ ($= G_{\rm F}$ where $v_{\rm loc} = v$). Unit: GeV^{-2}. Status: DERIVED.
- n_e: The number density of electrons (bound and free; in a supernova all are free, in the recombining universe $x_{\rm e} n_e$ is the density of the free electrons). Unit: GeV^3 (cm^{-3}). Status: STANDARD.
- $n_{e,\rm res}$: The electron density at which the flavour conversion in matter is resonant (the Mikheyev-Smirnov-Wolfenstein resonance). Unit: GeV^3. Status: STANDARD.
- $P_{\rm LZ}$: The probability that a neutrino jumps between the eigenstates of matter at the resonance (Landau-Zener); $P_{\rm LZ} = 0$ is an adiabatic resonance. Unit: dimensionless. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: ${\rm d}X$ is an infinitesimal change of X, ${\rm d}X/{\rm d}Y$ the derivative of X with respect to Y, ${\rm d}^n X$ the volume element of n variables. Status: STANDARD.
- r: The distance from the centre of the star along the path of a neutrino. Unit: length. Status: STANDARD.
- res: The label of a quantity evaluated at the resonance, where $n_e = n_{e,\rm res}$. Status: STANDARD.

The mixing angle in matter, the resonance density and the Landau-Zener jump probability (FE-50). Status: STANDARD (`um3_neutrino.py`, N2: the formula for $\theta_{\rm m}$ agrees with an exact diagonalisation to 10^{-16}, the Landau-Zener formula with an exact integration through a linear profile to 10^{-6}).

7.1 The Sun

$$\mathsf{H}_2 = \frac{1}{2}\begin{pmatrix} -\frac{\Delta m^2_{21}}{2E}\cos 2\theta_{12} + \cos^2\theta_{13}\, V_{\rm CC} & \frac{\Delta m^2_{21}}{2E}\sin 2\theta_{12} \\ \frac{\Delta m^2_{21}}{2E}\sin 2\theta_{12} & \frac{\Delta m^2_{21}}{2E}\cos 2\theta_{12} - \cos^2\theta_{13}\, V_{\rm CC} \end{pmatrix}, \qquad n_e(r) = 245\, N_{\rm A}\, {\rm e}^{-10.54\, r/R_\odot}\ {\rm cm}^{-3},$$
$$\mathsf{P}_{\rm ee} = \cos^4\theta_{13}\, \mathsf{P}_2 + \sin^4\theta_{13}, \qquad \mathsf{P}_2 = \frac{1}{2} + \Big(\frac{1}{2} - \mathsf{P}_{\rm c}\Big)\cos 2\theta_{12}\cos 2\theta_{\rm m}(r_{\rm p}),$$
$$\mathsf{P}_{\rm c} = \frac{{\rm e}^{-2\pi r_\odot \frac{\Delta m^2_{21}}{2E}\sin^2\theta_{12}} - {\rm e}^{-2\pi r_\odot \frac{\Delta m^2_{21}}{2E}}}{1 - {\rm e}^{-2\pi r_\odot \frac{\Delta m^2_{21}}{2E}}}, \qquad r_\odot = \frac{R_\odot}{10.54} \tag{S-14}$$

Symbols in (S-14).

- H_2: The two-flavour Hamiltonian of the electron neutrinos in the Sun (the mass states 1 and 2 with the potential $\cos^2\theta_{13} V_{\rm CC}$; the third state decouples). Unit: GeV (or 1/km). Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows ($\Delta\rho$ a perturbation, Δm^2 a difference of squared masses, $\Delta N_{\rm eff}$ an addition to $N_{\rm eff}$). Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; in Δm^2 the squared-mass difference of two neutrino mass states ($m_2^2 - m_1^2$ for the two states of a resonance, multiplied by $v_{\rm loc}^2/v^2$ where this differs from 1). Unit: GeV. Status: STANDARD.

- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- θ_{12}, θ_{13}, θ_{23}: The three mixing angles of U_{PMNS} (the Dirac phase is set to zero here). Unit: dimensionless. Value: 33.4°, 8.57°, 49.0°. Status: STANDARD.
- V_{CC}: The charged-current matter potential of the electron neutrinos (forward scattering on electrons through W exchange). Unit: GeV. Status: STANDARD.
- n_e: The number density of electrons (bound and free; in a supernova all are free, in the recombining universe $x_e n_e$ is the density of the free electrons). Unit: GeV^3 (cm^{-3}). Status: STANDARD.
- r: The distance from the centre of the star along the path of a neutrino. Unit: length. Status: STANDARD.
- N_{A}: Avogadro's number: $245 N_{\mathrm{A}}$ cm^{-3} is 245 moles of electrons per cubic centimetre. Unit: per mole. Value: $6.02214076 \times 10^{23}$. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $R_\odot$: The radius of the Sun. Unit: length. Value: 6.957×10^5 km. Status: STANDARD.
- P_{ee}: The survival probability of electron neutrinos: the probability that an electron neutrino is detected as an electron neutrino. Unit: dimensionless. Status: DERIVED.
- P_2: The survival probability of the two-flavour problem of the mass states 1 and 2 (S-14). Unit: dimensionless. Status: DERIVED.
- P_{c}: The crossing probability: the probability that a neutrino jumps from one eigenstate of matter to the other on its way through the Sun (exact for an exponential profile, Petcov). Unit: dimensionless. Status: STANDARD.
- θ_{m}: The mixing angle of two neutrino flavours in matter. Unit: dimensionless. Status: STANDARD.
- r_{p}: The radius at which the solar neutrinos of a given reaction are produced. Unit: length. Value: $0.05 R_\odot$ to $0.10 R_\odot$. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $r_\odot$: The scale height of the electron density of the Sun, $r_\odot = R_\odot/10.54$. Unit: length. Value: 6.6×10^4 km. Status: STANDARD.

The solar neutrinos: the third mass state decouples (it oscillates too fast and mixes only through θ_{13}), so the electron neutrinos evolve with the two-flavour Hamiltonian H_2 (with the potential $\cos^2\theta_{13} V_{\mathrm{CC}}$) through the electron density of the Sun in its exponential approximation (Bahcall), from the radius r_{p} of their production to the surface; the phases average out over the production region and the distance to the Earth, so the survival probability P_{ee} follows from the probabilities of the mass states at the surface. The second line is the adiabatic solution with the exact crossing probability P_{c} of an exponential profile (Petcov). Status: STANDARD (the equations), DERIVED (the solution below).

source	E (MeV)	$r_{\mathrm{p}}/R_\odot$	P_{ee} (exact integration)	P_{ee} (S-14)	Borexino
pp	0.267	0.10	0.5452	0.5452	0.57 ± 0.09
^{7}Be	0.862	0.06	0.5124	0.5124	0.53 ± 0.05
pep	1.44	0.10	0.5081	0.5081	0.43 ± 0.11
^{8}B	5.0	0.05	0.3361	0.3361	
^{8}B	8.1	0.05	0.3079	0.3079	0.37 ± 0.08
^{8}B	10.0	0.05	0.3016	0.3016	
^{8}B	15.0	0.05	0.2951	0.2951	

The survival probability of solar electron neutrinos in UM-3 (with $\theta_{12} = 33.4°$, $\theta_{13} = 8.57°$, $\Delta m^2_{21} = 7.41 \times 10^{-5}$ eV^2): the exact integration and the adiabatic formula agree to 10^{-6}, $P_{\mathrm{c}} < 10^{-10}$ at every energy (the Sun is adiabatic), and halving the step changes nothing at the level 10^{-6}. At low energy P_{ee} approaches the vacuum average $\cos^4\theta_{13}(1 - \frac{1}{2}\sin^2 2\theta_{12}) + \sin^4\theta_{13} = 0.553$, at high energy the full matter conversion $\cos^4\theta_{13}\sin^2\theta_{12} + \sin^4\theta_{13} = 0.290$; the Borexino measurements (Nature 562, 505 (2018), with the high-metallicity solar model) lie within one standard deviation. Status: DERIVED (`um3_flavour.py`, F1).

7.2 The envelope of a supernova

$$\rho_{\rm b}(r) = 10^4\ {\rm g/cm^3}\Big(\frac{r}{10^9\ {\rm cm}}\Big)^{-3}, \qquad n_e = \frac{Y_{\rm e}\,\rho_{\rm b}}{m_{\rm u}}, \qquad \gamma_{\rm LZ} = \frac{\Delta m^2 \sin^2 2\theta}{2E\cos 2\theta}\,\frac{r_{\rm res}}{3} \tag{S-15}$$

Symbols in (S-15).

- $\rho_{\rm b}$: The (rest-mass) energy density of the baryons: in a supernova the local density of the matter (quoted in g/cm^3), in cosmology the mean density of the baryons ($\rho_{\rm b}^{(0)}$ today). Unit: GeV^4. Status: STANDARD.
- r: The distance from the centre of the star along the path of a neutrino. Unit: length. Status: STANDARD.
- n_e: The number density of electrons (bound and free; in a supernova all are free, in the recombining universe $x_e n_e$ is the density of the free electrons). Unit: GeV^3 (cm^{-3}). Status: STANDARD.
- $Y_{\rm e}$: The electron fraction of stellar matter: the number of electrons per nucleon. Unit: dimensionless. Value: 0.5 in the envelope. Status: STANDARD.
- $m_{\rm u}$: The mass per nucleon of stellar matter (the atomic mass unit). Unit: g. Value: 1.6726×10^{-24} g. Status: STANDARD.
- $\gamma_{\rm LZ}$: The adiabaticity parameter of a resonance: the jump probability of FE-50 is $\exp(-\pi\gamma_{\rm LZ}/2)$; $\gamma_{\rm LZ} \gg 1$ is adiabatic. Unit: dimensionless. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows ($\Delta\rho$ a perturbation, Δm^2 a difference of squared masses, $\Delta N_{\rm eff}$ an addition to $N_{\rm eff}$). Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; in Δm^2 the squared-mass difference of two neutrino mass states ($m_2^2 - m_1^2$ for the two states of a resonance, multiplied by $v_{\rm loc}^2/v^2$ where this differs from 1). Unit: GeV. Status: STANDARD.
- θ: The vacuum mixing angle of two neutrino flavours (in the two-flavour description of a resonance, one of the three angles $\theta_{12}, \theta_{13}, \theta_{23}$ of $U_{\rm PMNS}$). Unit: dimensionless. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- $r_{\rm res}$: The radius of a resonance: where the electron density equals $n_{e,{\rm res}}$ of FE-50 ($r_{\rm H}$ for Δm^2_{31}, $r_{\rm L}$ for Δm^2_{21}). Unit: length. Status: STANDARD.

The outer layers of the progenitor star (density $\rho_{\rm b} \propto r^{-3}$, electron fraction $Y_e = 0.5$, $m_{\rm u}$ the atomic mass unit), through which the neutrinos of a supernova pass after they have left the dense core; $\gamma_{\rm LZ}$ is the adiabaticity of a resonance at the radius $r_{\rm res}$ (the exponent of FE-50 is $-\frac{\pi}{2}\gamma_{\rm LZ}$, since $|{\rm d}\ln n_e/{\rm d}r| = 3/r$). The three-flavour equation with the full mixing matrix ($\theta_{23} = 49.0°$, $\Delta m^2_{31} = 2.511 \times 10^{-3}$ eV^2) is integrated from $0.2r_{\rm H}$ to $5r_{\rm L}$ at $E = 15$ MeV, starting with an electron neutrino (antineutrino). Status: STANDARD (the profile and the equations), DERIVED (the solution).

ordering	particle	P_{ee}	expected for adiabatic resonances	probabilities of the mass states at the surface
normal	ν_e	0.022207	$\lvert U_{\rm e3}\rvert^2 = 0.022206$	ν_3: $1 - 1.5 \times 10^{-6}$
normal	$\bar\nu_e$	0.681493	$\lvert U_{\rm e1}\rvert^2 = 0.681494$	$\bar\nu_1$: $1 - 1.5 \times 10^{-6}$
inverted	ν_e	0.296300	$\lvert U_{\rm e2}\rvert^2 = 0.296300$	ν_2: $1 - 1.5 \times 10^{-6}$
inverted	$\bar\nu_e$	0.022207	$\lvert U_{\rm e3}\rvert^2 = 0.022206$	$\bar\nu_3$: $1 - 1.5 \times 10^{-6}$

The resonance of Δm^2_{31} lies at $r_{\rm H} = 16\,794$ km and that of Δm^2_{21} at $r_{\rm L} = 73\,022$ km; both are adiabatic ($\gamma_{\rm LZ} = 216$ and 653, so $P_{\rm LZ} = 0$). An electron neutrino, produced in the heaviest state of matter, leaves the star as ν_3 in the normal ordering and as ν_2 in the inverted one; an electron antineutrino leaves as $\bar\nu_1$ or $\bar\nu_3$. The exact integration agrees with these limits to 10^{-6} (and with half the step to 10^{-9}). Shock waves and turbulence in the envelope, which can make the high resonance non-adiabatic, and the collective conversion deeper inside are not included in this static profile. Status: DERIVED (F2).

7.3 The fast instability near the neutrinosphere

$$\mathcal{G}_{\rm ELN}(\hat{v}) = \sqrt{2}\, G_{\rm F,loc} \int_0^\infty \frac{E^2\, {\rm d}E}{(2\pi)^3} \Big[\mathcal{F}_e - \bar{\mathcal{F}}_e - \big(\mathcal{F}_x - \bar{\mathcal{F}}_x \big) \Big],$$
$$\det \Big[\eta^{\mu\nu}_{\rm M} + \int {\rm d}^2\hat{v}\; \mathcal{G}_{\rm ELN}(\hat{v})\, \frac{\hat{v}^\mu \hat{v}^\nu}{\Omega_{\rm fw} - \vec{K}\cdot\hat{v}} \Big] = 0, \qquad \hat{v}^\mu = (1,\ \hat{v}) \tag{FE-51}$$

Symbols in (FE-51).

- $\mathcal{G}_{\rm ELN}(\hat{v})$: The angular distribution of the electron lepton number carried by the neutrinos minus that carried by the heavy-lepton neutrinos, per unit solid angle of the direction $\hat{v}$, multiplied by $\sqrt{2}G_{\rm F,loc}$. Unit: GeV. Status: STANDARD.
- $\hat{v}$, $\hat{v}^\mu$: The direction of motion of a neutrino: the unit vector $\hat{v} = \vec{p}/|\vec{p}|$ ($\hat{v}_{\vec{p}}$, $\hat{v}_{\vec{q}}$ for the momenta $\vec{p}$, $\vec{q}$), and the four-vector $\hat{v}^\mu = (1, \hat{v})$ in the order $(x_4; x_1, x_2, x_3)$; ${\rm d}^2\hat{v}$ is the element of solid angle of the directions. Unit: dimensionless. Status: STANDARD.
- $G_{\rm F,loc}$: The local Fermi constant, $G_{\rm F,loc} = 1/(\sqrt{2}\, v_{\rm loc}^2)$ ($= G_{\rm F}$ where $v_{\rm loc} = v$). Unit: GeV^{-2}. Status: DERIVED.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- d: The differential: ${\rm d}X$ is an infinitesimal change of X, ${\rm d}X/{\rm d}Y$ the derivative of X with respect to Y, ${\rm d}^n X$ the volume element of n variables. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\mathcal{F}_e$, $\bar{\mathcal{F}}_e$, $\mathcal{F}_x$, $\bar{\mathcal{F}}_x$: The occupation numbers of the electron neutrinos, of the electron antineutrinos, and of one heavy-lepton neutrino and antineutrino ($x = \mu$ or τ), as functions of the energy E and the direction $\hat{v}$ (the diagonal entries of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$); in a transport code, the same for the number of particles in an energy bin. Unit: dimensionless. Status: STANDARD.
- $\eta^{\mu\nu}_{\rm M}$: The Minkowski metric of the four-vectors $\hat{v}^\mu$, $\eta_{\rm M} = {\rm diag}(1, -1, -1, -1)$ in the order $(x_4; x_1, x_2, x_3)$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- $\Omega_{\rm fw}$: The (complex) frequency of a flavour wave of the neutrino gas, measured from the shift caused by the potentials; an imaginary part greater than zero is an instability. Unit: GeV. Status: STANDARD.
- $\vec{K}$: The (real) wave vector of a flavour wave of the neutrino gas, measured from the shift caused by the potentials. Unit: GeV. Status: STANDARD.

$$\Big(\Omega_{\rm fw} - K_{\rm r}\, \mathsf{u}\Big) \mathsf{Q}(\mathsf{u}) = -2\pi \int_{-1}^{1} {\rm d}\mathsf{u}' \left(1 - \mathsf{u}\mathsf{u}'\right) \mathcal{G}_{\rm ELN}(\mathsf{u}')\, \mathsf{Q}(\mathsf{u}') \qquad \text{(axially symmetric mode)},$$
$$\Big(\Omega_{\rm fw} - K_{\rm r}\, \mathsf{u}\Big) \mathsf{Q}(\mathsf{u}) = \pi \int_{-1}^{1} {\rm d}\mathsf{u}'\, \sqrt{1 - \mathsf{u}^2}\sqrt{1 - \mathsf{u}'^2}\, \mathcal{G}_{\rm ELN}(\mathsf{u}')\, \mathsf{Q}(\mathsf{u}') \qquad \text{(first azimuthal mode)} \tag{S-16}$$

Symbols in (S-16).

- $\Omega_{\rm fw}$: The (complex) frequency of a flavour wave of the neutrino gas, measured from the shift caused by the potentials; an imaginary part greater than zero is an instability. Unit: GeV. Status: STANDARD.
- $K_{\rm r}$: The wave number of a flavour wave along the radial direction (the radial component of $\vec{K}$). Unit: GeV (1/length). Status: STANDARD.
- u: The cosine of the angle between the direction of motion of a neutrino and the radial direction (u_e, $\mathsf{u}_{\bar{e}}$ the edges of the emission cones of the electron neutrinos and antineutrinos). Unit: dimensionless. Status: STANDARD.
- $\mathsf{Q}(\mathsf{u})$: The amplitude of a small flavour coherence (the off-diagonal entry of ϱ) of the neutrinos moving with the direction cosine u, in a flavour wave. Unit: dimensionless. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: ${\rm d}X$ is an infinitesimal change of X, ${\rm d}X/{\rm d}Y$ the derivative of X with respect to Y, ${\rm d}^n X$ the volume element of n variables. Status: STANDARD.

- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\mathcal{G}_{\mathrm{ELN}}(\hat{v})$: The angular distribution of the electron lepton number carried by the neutrinos minus that carried by the heavy-lepton neutrinos, per unit solid angle of the direction $\hat{v}$, multiplied by $\sqrt{2}G_{\mathrm{F,loc}}$. Unit: GeV. Status: STANDARD.

The fast instability as an eigenvalue problem: for an electron-lepton-number distribution that depends only on the cosine u of the angle to the radial direction, a flavour wave with the wave number K_{r} along the radius and the amplitude $\mathsf{Q}(\mathsf{u})$ (axially symmetric, or proportional to the cosine of the azimuth) has the frequencies Ω_{fw} that are the eigenvalues of S-16 (FE-48 linearised; equivalent to the dispersion relation FE-51, which the solutions satisfy to 10^{-14}). The distribution is taken from the neutrinospheres: electron neutrinos emitted uniformly (half-isotropically) from a sphere of radius $R_e = 30$ km and antineutrinos from $R_{\bar{e}} = 27$ km (they decouple deeper and are more forward peaked), seen at $r = 40$ km, where each fills the cone $\mathsf{u} > \sqrt{1-(R/r)^2}$; α is the ratio of their number densities, the heavy flavours cancel in the electron lepton number, and 200 Gauss-Legendre nodes discretise the angle. Status: STANDARD (the equations), DERIVED (the solution).

$\alpha = n_{\bar{\nu}_e}/n_{\nu_e}$	0.50	0.75	0.80	0.95	1.20
angular crossing	no	no	yes	yes	yes
growth rate $\mathrm{Im}\,\Omega_{\mathrm{fw}}$ ($\sqrt{2}G_{\mathrm{F}}n_{\nu_e}$)	0	0	0.0137	0.0556	0.0925
e-folding length (cm)	–	–	26	6.5	3.9

The electron lepton number crosses zero as a function of the angle exactly when $\alpha > (1-\mathsf{u}_{\bar{e}})/(1-\mathsf{u}_e) = 0.7744$ (u_e, $\mathsf{u}_{\bar{e}}$ the edges of the two cones); without a crossing every frequency is real (growth below 10^{-17}), with a crossing the axially symmetric mode grows. For a luminosity of 5×10^{52} erg/s and a mean energy of 12 MeV the density of electron neutrinos at 40 km is 4.3×10^{32} cm^{-3}, $\sqrt{2}G_{\mathrm{F}}n_{\nu_e} = 2.77$ cm^{-1}, and the instability grows within centimetres: much faster than anything else in the star, it drives the flavours toward equipartition wherever the crossing exists. Status: DERIVED (F3; N3 of `um3_neutrino.py` for the crossing criterion with general distributions).

7.4 The flavour equipartition and the heating behind the shock

$$\begin{aligned}
&\mathcal{F}_e \to \mathcal{F}_e - \delta_{\mathrm{FC}}, \qquad \bar{\mathcal{F}}_e \to \bar{\mathcal{F}}_e - \delta_{\mathrm{FC}}, \qquad \mathcal{F}_x = \bar{\mathcal{F}}_x \to \mathcal{F}_x + \frac{\delta_{\mathrm{FC}}}{2},\\
&\delta_{\mathrm{FC}} = \max\Big(\frac{2}{3}\Big(\min\left(\mathcal{F}_e,\ \bar{\mathcal{F}}_e\right) - \mathcal{F}_x\Big),\ \max\left(\mathcal{F}_e,\ \bar{\mathcal{F}}_e\right) - 1\Big),\\
&\text{applied where}\ \ \rho_{\mathrm{b}} < \rho_{\mathrm{FC}}\ \ \text{and}\ \ x_4 - x_{4,\mathrm{b}} > t_{\mathrm{FC}}
\end{aligned} \tag{FE-52}$$

Symbols in (FE-52).

- $\mathcal{F}_e$, $\bar{\mathcal{F}}_e$, $\mathcal{F}_x$, $\bar{\mathcal{F}}_x$: The occupation numbers of the electron neutrinos, of the electron antineutrinos, and of one heavy-lepton neutrino and antineutrino ($x = \mu$ or τ), as functions of the energy E and the direction $\hat{v}$ (the diagonal entries of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$); in a transport code, the same for the number of particles in an energy bin. Unit: dimensionless. Status: STANDARD.
- δ_{FC}: The occupation (or number per energy bin) of electron neutrino-antineutrino pairs converted into heavy-lepton pairs by the flavour equipartition FE-52; a negative value converts heavy-lepton pairs into electron pairs. Unit: dimensionless. Status: ASSUMED.
- ρ_{b}: The (rest-mass) energy density of the baryons: in a supernova the local density of the matter (quoted in g/cm^3), in cosmology the mean density of the baryons ($\rho_{\mathrm{b}}^{(0)}$ today). Unit: GeV^4. Status: STANDARD.
- ρ_{FC}: The baryon density below which the flavour equipartition FE-52 is applied (the critical density ρ_{c} of Gogilashvili and Tamborra). Unit: GeV^4 (g/cm^3). Value: 10^9, 10^{10}, 10^{11} or 10^{13} g/cm^3. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $x_{4,\mathrm{b}}$: The time of the bounce of the collapsing stellar core. Unit: length (= time). Status: STANDARD.

- t_{FC}: The time after the bounce from which the flavour equipartition FE-52 is applied. Unit: time. Value: 0.02 s. Status: STANDARD.

The flavour equipartition of Gogilashvili and Tamborra (FE-52), applied in every energy bin. Status: STANDARD (the constraints), ASSUMED (this form). On 10^5 random occupations in $[0,1]^3$ it conserves the electron lepton number and the numbers of neutrinos and of antineutrinos to 10^{-15}, keeps every occupation between 0 and 1 (the Pauli bound acts in 22% of the cases), and a second application changes nothing (10^{-16}). Status: DERIVED (F4, N4).

$$\dot{Q}_{\mathrm{heat}} \propto \int_0^\infty \mathrm{d}E\, E^3 \big(\mathcal{F}_e + \bar{\mathcal{F}}_e\big), \qquad \mathcal{F}_e,\ \bar{\mathcal{F}}_e,\ \mathcal{F}_x \propto \frac{L_{\mathrm{s}}}{E_{\mathrm{s}}} \Big(\frac{E}{E_{\mathrm{s}}}\Big)^{\alpha_{\mathrm{s}}} \mathrm{e}^{-(\alpha_{\mathrm{s}}+1)E/E_{\mathrm{s}}} \tag{S-17}$$

Symbols in (S-17).

- $\dot{Q}_{\mathrm{heat}}$: The rate at which the absorption of electron neutrinos and antineutrinos heats the matter of the gain region (energy per time and mass). Unit: energy per time and mass. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.
- $\mathcal{F}_e,\ \bar{\mathcal{F}}_e,\ \mathcal{F}_x,\ \bar{\mathcal{F}}_x$: The occupation numbers of the electron neutrinos, of the electron antineutrinos, and of one heavy-lepton neutrino and antineutrino ($x = \mu$ or τ), as functions of the energy E and the direction $\hat{v}$ (the diagonal entries of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$); in a transport code, the same for the number of particles in an energy bin. Unit: dimensionless. Status: STANDARD.
- L_{s}: The luminosity (energy per time) of one neutrino species leaving the neutrinosphere. Unit: erg/s. Value: (2 to 5) $\times 10^{52}$ erg/s. Status: STANDARD.
- E_{s}: The mean energy of one neutrino species. Unit: GeV (MeV). Value: 10 to 16 MeV. Status: STANDARD.
- α_{s}: The shape (pinching) parameter of a neutrino spectrum (Keil, Raffelt and Janka). Unit: dimensionless. Value: 3. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

The charged-current heating of the gain region behind the stalled shock ($\nu_e n \to e^- p$ and $\bar{\nu}_e p \to e^+ n$: the cross-section grows as E^2 and the deposited energy is about E), with the number spectra of the neutrinos (luminosity L_ν, mean energy $\langle E \rangle$, shape parameter $\alpha_{\mathrm{s}} = 3$). Status: STANDARD, ESTIMATE (equal weights of the two reactions).

phase	$L_{\nu_e}, L_{\bar{\nu}_e}, L_{\nu_x}$ (10^{52} erg/s)	$\langle E \rangle$ (MeV)	$\langle E \rangle$ after (MeV)	heating after / before
accretion (0.2 s)	5, 5, 2	11, 14, 16	11.3, 16.1, 14.1	0.96
accretion, softer ν_x	5, 5, 2	11, 14, 14	10.9, 15.3, 13.2	0.86
end of accretion	4, 4, 2.5	11, 13, 15	11.9, 14.7, 13.6	1.12
cooling (comparable luminosities)	3, 3, 3	10, 12, 14	11.4, 13.8, 13.0	1.55

Equipartition removes absorbable electron neutrinos and antineutrinos and hardens their spectra. When the heavy-lepton luminosity is well below the electron-flavour one (the accretion phase) the loss of numbers wins and the heating falls; when they are comparable the harder spectra win and it rises. The sign depends on the spectra, as in the transport simulations of Gogilashvili and Tamborra, where flavour conversion near the stalled shock accelerates the revival of the shock, while conversion triggered deeper (below baryon densities of 10^{10} to 10^{13} g/cm^3) cools the core more efficiently and on the whole reduces the energy deposited behind the shock, increasing the fraction of failed explosions (especially for 16 to 30 solar masses). This estimate includes neither the feedback on the cooling nor the hydrodynamics. Since the flavour physics of UM-3 is that of the Standard Model with three Dirac neutrinos (FE-49), their results apply to UM-3 unchanged. Status: ESTIMATE (F4).

8 The thermal history

$$\rho_{\rm SM} = \frac{\pi^2}{30}\, g_\rho(\mathsf{T})\, \mathsf{T}^4, \qquad s = \frac{2\pi^2}{45}\, g_{\rm s}(\mathsf{T})\, \mathsf{T}^3, \qquad H_4 = \sqrt{\frac{8\pi^3 g_\rho}{90}}\, \frac{\mathsf{T}^2}{M_{\rm Pl}}, \qquad M_{\rm Pl} = G_{\rm N}^{-1/2} \tag{FE-60}$$

Symbols in (FE-60).

- $\rho_{\rm SM}$: The energy density of the Standard-Model particles. Unit: GeV^4. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- g_ρ: The effective number of relativistic degrees of freedom of the energy density. Unit: dimensionless. Value: 106.75 in the Standard Model above the electroweak scale. Status: STANDARD.
- T: The temperature of the Standard-Model plasma (of the photons after the annihilation of electrons and positrons). Unit: GeV. Status: STANDARD.
- s: The entropy density of the Standard-Model plasma. Unit: GeV^3. Status: STANDARD.
- $g_{\rm s}$: The effective number of relativistic degrees of freedom of the entropy density. Unit: dimensionless. Value: 3.909 today. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\, h_{100}$ km/s/Mpc. Status: STANDARD.
- $M_{\rm Pl}$: The Planck mass, $M_{\rm Pl} = G_{\rm N}^{-1/2}$. Unit: GeV. Value: 1.221×10^{19} GeV. Status: STANDARD.
- $G_{\rm N}$: Newton's constant of four-dimensional gravity on the fold, $G_{\rm N} = 1/M_{\rm Pl}^2$. Unit: GeV^{-2}. Value: $6.708{\times}10^{-39}$ GeV^{-2}. Status: STANDARD.

The radiation of the Standard Model in thermal equilibrium and the expansion rate of the radiation era (FE-60). The effective numbers of degrees of freedom are computed here from the exact Bose and Fermi integrals of every particle of the Standard Model with its mass (quarks and gluons above the QCD transition, taken at $\mathsf{T} = 0.155$ GeV, the light hadrons below; the neutrinos decouple at 1 MeV and afterwards have the temperature $(4/11)^{1/3}\mathsf{T}$): $g_\rho = g_{\rm s} = 106.75$ above the electroweak scale, 97.4 and 96.6 at 50 GeV, 85.6 and 85.4 at 5 GeV, 76.4 and 75.5 at 1 GeV, 10.76 at 10 MeV, and today $g_{\rm s} = 3.909$, $g_\rho = 3.363$ (the transition region near 0.155 GeV and the annihilation of electrons and positrons are approximated). Status: STANDARD, DERIVED (`um3_thermal.py`, T1).

8.1 The relic density of the particles χ

$$\frac{{\rm d}n_\chi}{{\rm d}x_4} + 3H_4\, n_\chi = -\langle\sigma_{\rm ann}\mathsf{v}\rangle\big(n_\chi^2 - n_{\chi,{\rm eq}}^2\big), \qquad n_{\chi,{\rm eq}} = g_\chi\Big(\frac{\omega_{\rm b}\mathsf{T}}{2\pi}\Big)^{3/2} {\rm e}^{-\omega_{\rm b}/\mathsf{T}}, \qquad \Omega_\chi h_{100}^2 \simeq \frac{1.07\times 10^9\ {\rm GeV}^{-1}\, x_{\rm f}}{\sqrt{g_*}\, M_{\rm Pl}\, \langle\sigma_{\rm ann}\mathsf{v}\rangle} \tag{FE-61}$$

Symbols in (FE-61).

- d: The differential: dX is an infinitesimal change of X, dX/dY the derivative of X with respect to Y, d$^n X$ the volume element of n variables. Status: STANDARD.
- n_χ: The number density of the dark-matter particles χ (particles plus antiparticles). Unit: GeV^3. Status: DERIVED.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is d/dx_4. Unit: length (= time, $c = 1$). Status: AUTHOR.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\, h_{100}$ km/s/Mpc. Status: STANDARD.
- $\sigma_{\rm ann}$: The cross-section of the annihilation of χ with its antiparticle into Standard-Model particles (through the portals). Unit: GeV^{-2}. Status: DERIVED.
- v: The relative velocity of two annihilating particles; $\langle\sigma_{\rm ann}\mathsf{v}\rangle$ is the thermal average of the product. Unit: dimensionless. Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $n_{\chi,{\rm eq}}$: The number density of χ in thermal equilibrium with the plasma (non-relativistic). Unit: GeV^3. Status: STANDARD.
- g_χ: The number of internal states of the bound level of χ that are in thermal equilibrium. Unit: dimensionless. Status: DERIVED.

- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- T: The temperature of the Standard-Model plasma (of the photons after the annihilation of electrons and positrons). Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- Ω_χ: The density parameter of the dark matter today. Unit: dimensionless. Value: $\Omega_\chi h_{100}^2 = 0.120$ (Planck 2018). Status: DERIVED.
- h_{100}: The present expansion rate of the fold in units of 100 km/s/Mpc. Unit: dimensionless. Value: 0.674 (Planck 2018). Status: STANDARD.
- x_f: The ratio of the mass of χ to the temperature at which its annihilations stop (freeze-out). Unit: dimensionless. Value: 20 to 25. Status: STANDARD.
- g_*: The effective number of relativistic degrees of freedom at the freeze-out (g_ρ there). Unit: dimensionless. Status: STANDARD.
- M_Pl: The Planck mass, $M_\mathrm{Pl} = G_\mathrm{N}^{-1/2}$. Unit: GeV. Value: 1.221×10^{19} GeV. Status: STANDARD.

$$\frac{\mathrm{d}\mathsf{Y}}{\mathrm{d}\mathsf{x}} = -\frac{s\,\langle\sigma_\mathrm{ann}\mathsf{v}\rangle}{\mathsf{x}\,H_4}\Big(1 + \frac{1}{3}\frac{\mathrm{d}\ln g_\mathrm{s}}{\mathrm{d}\ln\mathsf{T}}\Big)\big(\mathsf{Y}^2 - \mathsf{Y}_\mathrm{eq}^2\big), \qquad \mathsf{Y} = \frac{n_\chi}{s}, \qquad \mathsf{Y}_\mathrm{eq} = \frac{n_{\chi,\mathrm{eq}}}{s}, \qquad \mathsf{x} = \frac{\omega_\mathrm{b}}{\mathsf{T}},$$
$$n_{\chi,\mathrm{eq}} = \frac{g_\chi\,\omega_\mathrm{b}^2\,\mathsf{T}}{2\pi^2}\,K_2\Big(\frac{\omega_\mathrm{b}}{\mathsf{T}}\Big), \qquad \Omega_\chi h_{100}^2 = \frac{\omega_\mathrm{b}\,\mathsf{Y}(\infty)\,s^{(0)}}{\rho_\mathrm{c}^{(0)}/h_{100}^2} \tag{S-18}$$

Symbols in (S-18).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- Y: The yield of the particles χ: their number density divided by the entropy density, $\mathsf{Y} = n_\chi/s$ (constant when they neither annihilate nor are produced). Unit: dimensionless. Status: STANDARD.
- x: The ratio of the mass of χ to the temperature, $\mathsf{x} = \omega_\mathrm{b}/\mathsf{T}$ (the time variable of the freeze-out). Unit: dimensionless. Status: STANDARD.
- s: The entropy density of the Standard-Model plasma. Unit: GeV^3. Status: STANDARD.
- σ_ann: The cross-section of the annihilation of χ with its antiparticle into Standard-Model particles (through the portals). Unit: GeV^{-2}. Status: DERIVED.
- v: The relative velocity of two annihilating particles; $\langle\sigma_\mathrm{ann}\mathsf{v}\rangle$ is the thermal average of the product. Unit: dimensionless. Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- g_s: The effective number of relativistic degrees of freedom of the entropy density. Unit: dimensionless. Value: 3.909 today. Status: STANDARD.
- T: The temperature of the Standard-Model plasma (of the photons after the annihilation of electrons and positrons). Unit: GeV. Status: STANDARD.
- Y_eq: The yield of χ in thermal equilibrium, $n_{\chi,\mathrm{eq}}/s$. Unit: dimensionless. Status: STANDARD.
- n_χ: The number density of the dark-matter particles χ (particles plus antiparticles). Unit: GeV^3. Status: DERIVED.
- $n_{\chi,\mathrm{eq}}$: The number density of χ in thermal equilibrium with the plasma (non-relativistic). Unit: GeV^3. Status: STANDARD.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- g_χ: The number of internal states of the bound level of χ that are in thermal equilibrium. Unit: dimensionless. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

- K_2: The modified Bessel function of the second kind of order 2 ($K_2(y) \approx \sqrt{\pi/(2y)}\,\mathrm{e}^{-y}$ for large y). Unit: dimensionless. Status: STANDARD.
- Ω_χ: The density parameter of the dark matter today. Unit: dimensionless. Value: $\Omega_\chi h_{100}^2 = 0.120$ (Planck 2018). Status: DERIVED.
- h_{100}: The present expansion rate of the fold in units of 100 km/s/Mpc. Unit: dimensionless. Value: 0.674 (Planck 2018). Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.
- ρ_c: The critical density, the energy density of a spatially flat fold at the expansion rate H_4; $\rho_\mathrm{c}^{(0)}$ its value today. Unit: GeV^4. Value: $\rho_\mathrm{c}^{(0)} = 1.0537 \times 10^{-5}\, h_{100}^2$ GeV/cm^3. Status: STANDARD.

FE-61 rewritten for the yield Y (the number density per entropy, constant without annihilation) as a function of $\mathsf{x} = \omega_\mathrm{b}/\mathsf{T}$, with the equilibrium density valid also for relativistic χ (K_2 the modified Bessel function of the second kind), and the relic density from the yield after freeze-out ($s^{(0)} = 2891.2$ cm^{-3} the entropy density today, $\rho_\mathrm{c}^{(0)}/h_{100}^2 = 1.0537 \times 10^{-5}$ GeV/cm^3). The bound level of dirac16complex holds four particle states (one multiplet of the rotations of ordinary space and of the extra times: two Dirac fermions χ_a, $a = 1, 2$) and four antiparticle states, so $g_\chi = 8$ and FE-61, written for the total density, has $\langle\sigma_\mathrm{ann}\mathsf{v}\rangle = (\sigma\mathsf{v})_{\chi\bar{\chi}}/4$, where $(\sigma\mathsf{v})_{\chi\bar{\chi}}$ is the annihilation of one χ_a with one $\bar{\chi}_a$ summed over the final states. The equation is integrated from $\mathsf{x} = 1$ to $\mathsf{x} = 1000$ with an implicit method. Status: STANDARD (the equation), DERIVED (the solution).

ω_b (GeV)	10	30	100	300	1000	3000	10000
$\langle\sigma_\mathrm{ann}\mathsf{v}\rangle$ for $\Omega_\chi h_{100}^2 = 0.120$ (10^{-26} cm^3/s)	2.31	2.19	2.17	2.25	2.28	2.26	2.31
freeze-out x_f	23.1	24.0	25.1	26.2	27.4	28.4	29.6
g_* at freeze-out	67.9	78.6	85.2	86.4	93.8	103.9	106.5
estimate of FE-61 ($\Omega_\chi h_{100}^2$)	0.124	0.127	0.128	0.128	0.127	0.126	0.127

The particles χ are a thermal relic with the measured dark-matter density (Planck 2018: $\Omega_\chi h_{100}^2 = 0.1200$) if their annihilation, in the normalisation of FE-61, is $\langle\sigma_\mathrm{ann}\mathsf{v}\rangle \approx 2.2 \times 10^{-26}$ cm^3/s for every mass from 10 GeV to 10 TeV (so $(\sigma\mathsf{v})_{\chi\bar{\chi}} \approx 9 \times 10^{-26}$ cm^3/s); the analytic estimate of FE-61 agrees within 7%. Status: DERIVED (T2).

8.2 The portals of χ to the Standard Model

$$\mathfrak{g}_\Phi = \frac{\partial\omega_\mathrm{b}}{\partial\mu_\mathrm{L}}\,\frac{c_\Phi}{V_*}, \qquad \mathfrak{g}_G = \sum_X \frac{\partial\omega_\mathrm{b}}{\partial\mu_X}\,\frac{c_X^G}{4V_*}, \qquad \mathfrak{g}_f = \sum_X \frac{\partial\omega_\mathrm{b}}{\partial\mu_X}\,\frac{2\omega_\mathrm{b}\,c_X^f}{V_*}, \qquad \frac{\partial\omega_\mathrm{b}}{\partial\mu_X} = -\frac{q_\mathrm{b}}{\omega_\mathrm{b}}\,\frac{\partial q_\mathrm{b}}{\partial\mu_X},$$
$$\Delta\omega_\mathrm{b} = \mathfrak{g}_\Phi\,\Delta\langle\Phi^\dagger\Phi\rangle + \mathfrak{g}_G\,\Delta\langle G^2\rangle + \sum_f \mathfrak{g}_f\,\Delta n_f, \qquad \mathfrak{g}_{h\chi} = \mathfrak{g}_\Phi\,v = -\frac{m_\Psi^2\,\mu_\mathrm{L}^2}{2\,\omega_\mathrm{b}\,v} \quad (\mu_\mathrm{R} = 0) \tag{FE-47}$$

Symbols in (FE-47).

- $\mathfrak{g}_\Phi$, $\mathfrak{g}_G$, $\mathfrak{g}_f$, $\mathfrak{g}_{h\chi}$: The four-dimensional couplings of the bound states χ of dirac16complex to the Standard Model (FE-47): to $\Phi^\dagger\Phi$, to the gluon kinetic density, to the number densities of the fermions of kind f, and the Yukawa coupling of the Higgs boson to χ. Unit: $\mathfrak{g}_\Phi$: GeV^{-1}; $\mathfrak{g}_G$: GeV^{-3}; $\mathfrak{g}_f$: GeV^{-2}; $\mathfrak{g}_{h\chi}$: dimensionless. Status: DERIVED.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.

- μ_X, μ_{L}, μ_{R}: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex (FE-39): the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_{\mathrm{L}}P_{\mathrm{L}} + \mu_{\mathrm{R}}P_{\mathrm{R}})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- q_{b}: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_{\mathrm{b}}|\sigma|}$. Unit: GeV. Status: DERIVED.
- Δ: A change, difference or small variation of the quantity that follows ($\Delta\rho$ a perturbation, Δm^2 a difference of squared masses, ΔN_{eff} an addition to N_{eff}). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\langle G^2\rangle$: The mean square of the gluon field strength, $\langle G^A_{\mu\nu}G^{A\mu\nu}\rangle$ (the gluon condensate in the QCD vacuum, a thermal value in the plasma). Unit: GeV^4. Status: STANDARD.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- n_f: The number density of the Standard-Model fermions of kind f (particles minus antiparticles, summed over generations and gauge components), the time component of the mean current $\langle J^\mu_f\rangle$ in the rest frame of the medium. Unit: GeV^3. Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.

The couplings of the bound states χ to the Standard Model (FE-47): the Higgs-portal Yukawa coupling $\mathfrak{g}_{h\chi}$ is fixed by the binding, the current-portal couplings $\mathfrak{g}_f$ are free (through the INVENTED c_X^f). Status: DERIVED.

$$\sigma_{\mathrm{pair}}\mathsf{v} = \sum_f \frac{\mathfrak{g}_f^2\,\omega_{\mathrm{b}}^2}{2\pi}, \qquad \langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle = \frac{\sigma_{\mathrm{pair}}\mathsf{v}}{4}, \qquad \sigma_{\mathrm{SI}} = \frac{\mu_{\mathrm{red}}^2\,f_{\mathrm{N}}^2}{\pi}, \qquad f_{\mathrm{N}} = 3\,\mathfrak{g}_{\mathrm{q}} \ \text{(quark portals)},$$
$$f_{\mathrm{N}} = \mathfrak{g}_{\mathrm{p}} = \frac{\alpha_{\mathrm{em}}}{3\pi}\,\mathfrak{g}_{\mathrm{l}} \sum_l \ln\frac{\omega_{\mathrm{b}}^2}{\max\left(m_l^2,\ q_{\mathrm{t}}^2\right)} \ \text{(lepton portals)}, \qquad f_{\mathrm{N}} = \frac{\mathfrak{g}_{h\chi}\,f_{\mathrm{H}}\,m_{\mathrm{N}}^2}{v\,m_h^2} \ \text{(Higgs portal)} \tag{S-19}$$

Symbols in (S-19).

- σ_{pair}: The cross-section of the annihilation of one particle χ_a with one antiparticle $\bar\chi_a$ of the same species, summed over the final states; $\sigma_{\mathrm{pair}}\mathsf{v}$ its product with the relative velocity (a constant for s-wave annihilation). Unit: GeV^{-2}. Status: DERIVED.
- v: The relative velocity of two annihilating particles; $\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle$ is the thermal average of the product. Unit: dimensionless. Status: STANDARD.
- $\mathfrak{g}_\Phi$, $\mathfrak{g}_G$, $\mathfrak{g}_f$, $\mathfrak{g}_{h\chi}$: The four-dimensional couplings of the bound states χ of dirac16complex to the Standard Model (FE-47): to $\Phi^\dagger\Phi$, to the gluon kinetic density, to the number densities of the fermions of kind f, and the Yukawa coupling of the Higgs boson to χ. Unit: $\mathfrak{g}_\Phi$: GeV^{-1}; $\mathfrak{g}_G$: GeV^{-3}; $\mathfrak{g}_f$: GeV^{-2}; $\mathfrak{g}_{h\chi}$: dimensionless.

Status: DERIVED.

- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- σ_ann: The cross-section of the annihilation of χ with its antiparticle into Standard-Model particles (through the portals). Unit: GeV^{-2}. Status: DERIVED.
- $\langle \cdots \rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- σ_SI: The spin-independent cross-section of the scattering of χ on a nucleon (normalised per nucleon, as the direct-detection limits are quoted). Unit: GeV^{-2} (cm^2). Status: ESTIMATE.
- μ_red: The reduced mass of χ and a nucleon, $\omega_\mathrm{b} m_\mathrm{N}/(\omega_\mathrm{b} + m_\mathrm{N})$. Unit: GeV. Status: STANDARD.
- f_N: The effective coupling of χ to a nucleon in the coherent scattering (a vector or scalar contact interaction). Unit: GeV^{-2}. Status: ESTIMATE.
- $\mathfrak{g}_\mathrm{q}$, $\mathfrak{g}_\mathrm{l}$, $\mathfrak{g}_\mathrm{p}$: The current-portal couplings $\mathfrak{g}_f$ of FE-47 taken equal for all chiral quark states ($\mathfrak{g}_\mathrm{q}$) or for all twelve chiral lepton states ($\mathfrak{g}_\mathrm{l}$), and the coupling to the proton that the lepton loop induces ($\mathfrak{g}_\mathrm{p}$). Unit: GeV^{-2}. Status: ESTIMATE.
- α_em: The fine-structure constant. Unit: dimensionless. Value: 1/137.036. Status: STANDARD.
- l: The multipole index and number ($l = 0, 1, 2, \ldots$) of an expansion in Legendre polynomials of the direction of a photon or a neutrino relative to the wave vector, or of the CMB sky in spherical harmonics. Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; in Δm^2 the squared-mass difference of two neutrino mass states ($m_2^2 - m_1^2$ for the two states of a resonance, multiplied by v_loc^2/v^2 where this differs from 1). Unit: GeV. Status: STANDARD.
- q_t: The momentum transfer of the scattering of χ on a xenon nucleus in a detector. Unit: GeV. Value: 0.05 GeV. Status: STANDARD.
- m_N: The mass of a nucleon. Unit: GeV. Value: 0.939 GeV. Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV. Status: STANDARD.
- m_h: The mass of the Higgs boson. Unit: GeV. Value: 125.25 GeV. Status: STANDARD.
- f_H: The nucleon matrix element of the Higgs coupling: the fraction of the nucleon mass carried by the quark masses and, through the trace anomaly, by the gluons. Unit: dimensionless. Value: 0.30. Status: STANDARD.

The annihilation and the scattering on nucleons through the portals, for a vector contact interaction $\mathfrak{g}_f(\bar{\chi}\hat{\gamma}^\mu\chi)(\bar{\psi}_f\hat{\gamma}_\mu\psi_f)$ of χ with a chiral fermion state f of the Standard Model: $\sigma_\mathrm{pair}\mathsf{v}$ is the annihilation of one χ_a with one $\bar{\chi}_a$ into massless final states (s-wave, summed over the chiral states f), and FE-61 has a quarter of it because n_χ counts the four particle and the four antiparticle states. The coherent spin-independent cross-section per nucleon σ_SI has the reduced mass μ_red of χ and the nucleon and the nucleon coupling f_N: for quark portals of equal strength $\mathfrak{g}_\mathrm{q}$ the vector charge of the nucleon is three times the quark coupling; for lepton portals ($\mathfrak{g}_\mathrm{l}$ for the twelve chiral lepton states: six of the doublets, three right-handed charged leptons, three right-handed neutrinos) the scattering arises at one loop through the photon, with the coupling $\mathfrak{g}_\mathrm{p}$ to the proton (the sum runs over $l = e, \mu, \tau$ with their masses m_l; $\alpha_\mathrm{em} = 1/137.036$, the momentum transfer $q_\mathrm{t} = 50$ MeV of a xenon detector, the cut-off of the logarithm taken at ω_b; per nucleon of xenon σ_SI carries the factor $(54/131)^2$); for the Higgs portal the nucleon (mass $m_\mathrm{N} = 0.939$ GeV) couples through $f_\mathrm{H} = 0.30$, the nucleon matrix element of the quark masses and the gluons. Status: STANDARD (the formulas), ESTIMATE (the loop and the matrix elements).

ω_b (GeV)	10	30	100	300	1000	3000	10000
quark portals: σ_SI (cm^2)	1.1×10^{-38}	1.3×10^{-39}	1.3×10^{-40}	1.5×10^{-41}	1.3×10^{-42}	1.5×10^{-43}	1.4×10^{-44}
lepton portals: σ_SI (cm^2)	2.1×10^{-43}	4.0×10^{-44}	5.8×10^{-45}	9.4×10^{-46}	1.2×10^{-46}	1.7×10^{-47}	1.9×10^{-48}

$\omega_{\rm b}$ (GeV)	10	30	100	300	1000	3000	10000
LZ limit (cm^2)	2.1×10^{-48}	2.1×10^{-48}	5.8×10^{-48}	1.8×10^{-47}	5.8×10^{-47}	1.8×10^{-46}	5.8×10^{-46}
Higgs portal: largest $q_{\rm b}/\omega_{\rm b}$			0.063	0.048	0.035	0.027	0.020

The direct-detection test of a thermal relic χ against the LUX-ZEPLIN limit (LZ 2024, 4.2 tonne-years: 2.1×10^{-48} cm^2 at its minimum of 36 GeV; above it the limit is scaled in proportion to the mass, which is weaker than the measured curve between the minimum and a few hundred GeV, so the exclusions drawn here are conservative). Quark portals that set the relic density are excluded at every mass up to 10 TeV, by factors from 5×10^9 (10 GeV) to 23 (10 TeV): the quark couplings must be smaller than the relic value by the square roots of these factors. Lepton portals give a loop-suppressed scattering that LZ excludes below about 1.3 TeV and allows above: a leptophilic χ of a TeV or more can be the thermal relic. The Higgs-portal coupling of FE-47, $|\mathfrak{g}_{h\chi}| = 2q_{\rm b}^2/(\omega_{\rm b}v)$, must stay below 3.2×10^{-3} (100 GeV) to 3.2×10^{-2} (10 TeV), so the binding must be weak: $q_{\rm b} \lesssim 0.06\,\omega_{\rm b}$ to $0.02\,\omega_{\rm b}$. Together with S-13 ($q_{\rm b} > H$) this requires a bulk scale H well below the mass of χ. Status: ESTIMATE (T3, T4).

8.3 The right-handed neutrinos

$$\frac{\mathrm{d}\rho_{\nu_{\rm R}}}{\mathrm{d}x_4} + 4H_4\,\rho_{\nu_{\rm R}} = \mathcal{C}_{\nu_{\rm R}}, \qquad \Delta N_{\rm eff} = 3\Big(\frac{10.75}{g_{\rm s}(\mathsf{T}_{\rm dec})}\Big)^{4/3} \quad \text{(three right-handed neutrinos decoupling at } \mathsf{T}_{\rm dec}\text{)} \quad \text{(FE-62)}$$

Symbols in (FE-62).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $\rho_{\nu_{\rm R}}$: The energy density of the three right-handed neutrinos. Unit: GeV^4. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- $\mathcal{C}_{\nu_{\rm R}}$: The collision term of the right-handed neutrinos: the rate at which collisions produce (or remove) their energy density. Unit: GeV^5. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows ($\Delta\rho$ a perturbation, Δm^2 a difference of squared masses, $\Delta N_{\rm eff}$ an addition to $N_{\rm eff}$). Status: STANDARD.
- $N_{\rm eff}$: The effective number of neutrino species: the energy density of the relativistic particles other than photons in units of that of one neutrino species whose temperature is $(4/11)^{1/3}$ times that of the photons. Unit: dimensionless. Value: 3.044 (Standard Model); measured 2.99 ± 0.17 (Planck 2018). Status: STANDARD.
- $g_{\rm s}$: The effective number of relativistic degrees of freedom of the entropy density. Unit: dimensionless. Value: 3.909 today. Status: STANDARD.
- $\mathsf{T}_{\rm dec}$: The temperature at which the right-handed neutrinos decouple from the plasma. Unit: GeV. Status: STANDARD.

$$\frac{n_{\chi,\rm eq}^2\,\sigma_{\rm pair}\mathsf{v}}{16\,n_{\nu_{\rm R},\rm eq}}\Big|_{\mathsf{T}=\mathsf{T}_{\rm dec}} = H_4\big(\mathsf{T}_{\rm dec}\big), \qquad n_{\nu_{\rm R},\rm eq} = \frac{9\,\zeta(3)}{2\pi^2}\,\mathsf{T}^3 \tag{S-20}$$

Symbols in (S-20).

- $n_{\chi,\rm eq}$: The number density of χ in thermal equilibrium with the plasma (non-relativistic). Unit: GeV^3. Status: STANDARD.
- $\sigma_{\rm pair}$: The cross-section of the annihilation of one particle χ_a with one antiparticle $\bar\chi_a$ of the same species, summed over the final states; $\sigma_{\rm pair}\mathsf{v}$ its product with the relative velocity (a constant for s-wave annihilation). Unit: GeV^{-2}. Status: DERIVED.
- v: The relative velocity of two annihilating particles; $\langle\sigma_{\rm ann}\mathsf{v}\rangle$ is the thermal average of the product. Unit: dimensionless. Status: STANDARD.

- $n_{\nu_R,\mathrm{eq}}$: The number density of the three right-handed neutrinos and their antiparticles in thermal equilibrium at the temperature T. Unit: GeV^3. Status: STANDARD.
- T: The temperature of the Standard-Model plasma (of the photons after the annihilation of electrons and positrons). Unit: GeV. Status: STANDARD.
- $\mathsf{T}_{\mathrm{dec}}$: The temperature at which the right-handed neutrinos decouple from the plasma. Unit: GeV. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- ζ: The Riemann zeta function; $\zeta(3) = 1.2020569$. Unit: dimensionless. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

The right-handed neutrinos. Their production by the neutrino Yukawa couplings is negligible ($\Gamma/H_4 \sim y_\nu^2 M_{\mathrm{Pl}}/\mathsf{T} = 5\times10^{-9}$ at 100 GeV for $y_\nu = 0.05$ eV$/v$). If the lepton portal includes them ($c_X^\nu \neq 0$), they stay in equilibrium with χ as long as the rate per right-handed neutrino of $\nu_R\bar{\nu}_R \to \chi\bar{\chi}$ (the left-hand side of S-20, by detailed balance: the three right-handed neutrinos take a quarter of $\sigma_{\mathrm{pair}}\mathsf{v}$, and $n_{\nu_R,\mathrm{eq}}$ counts them with their antiparticles) exceeds the expansion rate, and decouple at $\mathsf{T}_{\mathrm{dec}}$. Status: STANDARD, ESTIMATE (equal couplings of the twelve chiral lepton states).

ω_b (GeV)	10	30	100	300	1000	3000	10000
$\omega_b/\mathsf{T}_{\mathrm{dec}}$	13.4	14.0	14.6	15.2	15.8	16.4	17.0
ΔN_{eff} (FE-62)	0.236	0.199	0.188	0.182	0.155	0.143	0.141
N_{eff}	3.28	3.24	3.23	3.23	3.20	3.19	3.18
standard deviations above ACT DR6 (2.89 ± 0.11)	3.5	3.2	3.1	3.1	2.8	2.7	2.7

Thermalised right-handed neutrinos would add $\Delta N_{\mathrm{eff}} = 0.14$ to 0.24, which is 1.2 to 1.7 standard deviations above Planck 2018 ($N_{\mathrm{eff}} = 2.99\pm0.17$) and 2.7 to 3.5 above the combination of ACT DR6 with Planck, lensing, DESI BAO and nucleosynthesis (2.89 ± 0.11; without nucleosynthesis 2.86 ± 0.13). The CMB alone (the refit below) gives $\Delta\chi^2 = +1.6$ for $\Delta N_{\mathrm{eff}} = 0.141$. UM-3 therefore requires that the right-handed neutrinos never thermalise: their portal couplings must be much smaller than those of the lepton doublets, and then $N_{\mathrm{eff}} = 3.044$ as in the Standard Model. Status: DERIVED (T5), ESTIMATE (the rates).

9 The CMB, the CNB and the BAO

The CMB, the CNB and the BAO are computed with the metric of the fold as the power series FE-35: the expanding background (order 0), the cosmological perturbations (order $\epsilon_c \sim 10^{-5}$) and the response of the bulk (order $\epsilon_r \lesssim 10^{-15}$, the radion mode of S-7 to S-9); the terms of order $\epsilon_c\epsilon_r$ and ϵ_c^2 are dropped. To order ϵ_r the fold obeys general relativity with Newton's constant $8\pi G_N = \kappa_4/(1+5\epsilon_r/6)$ (FE-59, above). The content of the fold is that of UM-3: the bound states χ of dirac16complex as cold dark matter (pressureless in the dilute limit, as the Kohn-Sham equation of state shows; their drag on the baryons is negligible for the couplings that direct detection allows), the vacuum energy ρ_Λ of the fold as dark energy ($w = -1$ exactly), the baryons and photons of the Standard Model, and three Dirac neutrinos of the normal ordering with the lightest mass zero ($m_1 = 0$, $m_2 = \sqrt{\Delta m_{21}^2} = 8.61$ meV, $m_3 = \sqrt{\Delta m_{31}^2} = 50.1$ meV, $\sum_i m_i = 0.0587$ eV; ASSUMED: the lightest mass is not fixed by any measurement) whose right-handed states are not populated ($N_{\mathrm{eff}} = 3.044$). The primordial spectrum, the optical depth and the densities are the Planck 2018 values (TT,TE,EE+lowE+lensing: $\Omega_b h_{100}^2 = 0.02237$, $\Omega_\chi h_{100}^2 = 0.1200$, $H_4^{(0)} = 67.36$ km/s/Mpc, $\tau_{\mathrm{reion}} = 0.0544$, $\ln(10^{10}A_s) = 3.044$, $n_S = 0.9649$; ASSUMED: UM-3 does not compute its primordial era to this precision). The Boltzmann hierarchy and the line-of-sight integrals are solved by the code CAMB (Lewis, Challinor and Lasenby 2000); the background and the BAO ruler are computed a second time here, independently.

$$\Big(\partial_{x_4} - a_4'\,|\vec{p}|\,\frac{\partial}{\partial|\vec{p}|}\Big)\,\varrho_{\vec{p}} = -\mathrm{i}\big[\mathsf{H}_{\vec{p}},\ \varrho_{\vec{p}}\big] + \mathcal{C}_{\vec{p}},$$
$$N_{\mathrm{eff}} = \frac{8}{7}\Big(\frac{11}{4}\Big)^{4/3}\frac{\rho_\nu}{\rho_\gamma}, \qquad \mathsf{T}_\nu^{(0)} = \Big(\frac{4}{11}\Big)^{1/3}\mathsf{T}_\gamma^{(0)}, \qquad n_\nu^{(0)} = \frac{3\zeta(3)}{2\pi^2}\big(\mathsf{T}_\nu^{(0)}\big)^3 \tag{FE-53}$$

Symbols in (FE-53).

- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\vec{p}$: The momentum of a neutrino (three components along ordinary space, in the rest frame of the medium); $|\vec{p}| = E$ for a neutrino of negligible mass. Unit: GeV. Status: STANDARD.
- $\varrho_{\vec{p}}$, $\bar{\varrho}_{\vec{p}}$: The 3×3 flavour density matrices of the neutrinos and of the antineutrinos of momentum $\vec{p}$ in the basis e, μ, τ: the diagonal entries are the occupation numbers of the three flavours, the off-diagonal entries the coherence between the flavours; $\bar{\varrho}$ is defined with the flavour indices in the reversed order, so that it obeys an equation of the same form. Unit: dimensionless. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\mathsf{H}_{\vec{p}}$, $\bar{\mathsf{H}}_{\vec{p}}$: The flavour Hamiltonians (3×3 Hermitian matrices) of the neutrinos and of the antineutrinos of momentum $\vec{p}$. Unit: GeV. Status: STANDARD.
- $\mathcal{C}_{\vec{p}}$, $\bar{\mathcal{C}}_{\vec{p}}$: The collision terms ($3 \times 3$ matrices) of the neutrinos and the antineutrinos of momentum $\vec{p}$: the rates of change of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$ by emission, absorption, scattering and pair processes with the matter. Unit: GeV. Status: STANDARD.
- N_{eff}: The effective number of neutrino species: the energy density of the relativistic particles other than photons in units of that of one neutrino species whose temperature is $(4/11)^{1/3}$ times that of the photons. Unit: dimensionless. Value: 3.044 (Standard Model); measured 2.99 ± 0.17 (Planck 2018). Status: STANDARD.
- ρ_ν: The energy density of the neutrinos and antineutrinos of all flavours. Unit: GeV^4. Status: STANDARD.
- ρ_γ: The energy density of the photons. Unit: GeV^4. Status: STANDARD.
- T_ν: The temperature of the cosmic neutrino background; $\mathsf{T}_\nu^{(0)}$ its value today. Unit: GeV (or K). Value: 1.945 K today. Status: STANDARD.
- T_γ: The temperature of the photons (of the CMB); $\mathsf{T}_\gamma^{(0)}$ its value today. Unit: GeV (or K). Value: 2.7255 K today. Status: STANDARD.
- n_ν: The number density of the cosmic neutrino background per flavour (neutrinos plus antineutrinos, one helicity each); $n_\nu^{(0)}$ its value today. Unit: GeV^3. Value: 112 cm^{-3} today. Status: STANDARD.
- ζ: The Riemann zeta function; $\zeta(3) = 1.2020569$. Unit: dimensionless. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

$$\hat{h}_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu = a^2\Big[-(1+2\psi_{\mathrm{B}})\,\mathrm{d}\tau_{\mathrm{c}}^2 + (1-2\phi_{\mathrm{B}})\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big)\Big],$$
$$a = \mathrm{e}^{a_4}, \qquad \mathrm{d}\tau_{\mathrm{c}} = \frac{\mathrm{d}x_4}{a}, \qquad \mathcal{H} = \frac{\dot{a}}{a},$$
$$k^2\phi_{\mathrm{B}} + 3\mathcal{H}\big(\dot{\phi}_{\mathrm{B}} + \mathcal{H}\psi_{\mathrm{B}}\big) = -4\pi G_{\mathrm{N}} a^2\,\Delta\rho, \qquad k^2\big(\dot{\phi}_{\mathrm{B}} + \mathcal{H}\psi_{\mathrm{B}}\big) = 4\pi G_{\mathrm{N}} a^2(\rho + \mathcal{P})\,\vartheta, \tag{FE-54}$$
$$\ddot{\phi}_{\mathrm{B}} + \mathcal{H}\big(\dot{\psi}_{\mathrm{B}} + 2\dot{\phi}_{\mathrm{B}}\big) + \Big(2\frac{\ddot{a}}{a} - \mathcal{H}^2\Big)\psi_{\mathrm{B}} + \frac{k^2}{3}\big(\phi_{\mathrm{B}} - \psi_{\mathrm{B}}\big) = 4\pi G_{\mathrm{N}} a^2\,\Delta\mathcal{P},$$
$$k^2\big(\phi_{\mathrm{B}} - \psi_{\mathrm{B}}\big) = 12\pi G_{\mathrm{N}} a^2(\rho + \mathcal{P})\,\Sigma_{\mathrm{s}}$$

Symbols in (FE-54).

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}_{\mu\nu}^{(1)}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.

- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- x: The coordinates of the fold: x_i $(i = 1, 2, 3)$ of ordinary space, x_4 the time, x_r $(r = 5, 6, 7)$ the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- a, $\dot{a}$, $\ddot{a}$: The scale factor of the fold, $a = \mathrm{e}^{a_4}$ (normalised to 1 today by the choice of the comoving coordinates x_1, x_2, x_3), and its first and second derivatives with respect to the conformal time τ_c. Unit: dimensionless. Status: STANDARD.
- ψ_B, $\dot{\psi}_\mathrm{B}$: The Bardeen potential of the time-time part of the perturbed metric of the fold (the Newtonian potential) in the conformal Newtonian gauge, a function of τ_c and of the wave number k, and its derivative with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- τ_c: The conformal time of the fold, $\mathrm{d}\tau_\mathrm{c} = \mathrm{d}x_4/a$; a dot on a quantity of the cosmological perturbations is $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$. Unit: length (= time). Status: STANDARD.
- ϕ_B, $\dot{\phi}_\mathrm{B}$, $\ddot{\phi}_\mathrm{B}$: The Bardeen potential of the space-space part of the perturbed metric of the fold (the curvature potential) in the conformal Newtonian gauge, and its first and second derivatives with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\mathcal{H}$: The conformal expansion rate, $\mathcal{H} = \dot{a}/a = a\,a_4'$. Unit: 1/length. Status: STANDARD.
- k: The comoving wave number of a Fourier mode of the perturbations, $k = |\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_N: Newton's constant of four-dimensional gravity on the fold, $G_\mathrm{N} = 1/M_\mathrm{Pl}^2$. Unit: GeV^{-2}. Value: $6.708{\times}10^{-39}$ GeV^{-2}. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows ($\Delta\rho$ a perturbation, Δm^2 a difference of squared masses, ΔN_eff an addition to N_eff). Status: STANDARD.
- ρ: The total energy density of the matter of the fold (all species); $\Delta\rho$ its perturbation. Unit: GeV^4. Status: STANDARD.
- $\mathcal{P}$: The total pressure of the matter of the fold; $\Delta\mathcal{P}$ its perturbation. Unit: GeV^4. Status: STANDARD.
- ϑ: The divergence of the velocity of the total matter: $(\rho + \mathcal{P})\vartheta$ is the sum over the species of their energy density plus pressure times the divergence of their velocity. Unit: 1/length. Status: STANDARD.
- Σ_s: The anisotropic stress (shear) of the total matter: $(\rho + \mathcal{P})\Sigma_\mathrm{s}$ is the sum over the species (photons and neutrinos) of their energy density plus pressure times their shear. Unit: dimensionless. Status: STANDARD.

$$
\begin{aligned}
&\dot\delta_\chi = -\vartheta_\chi + 3\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\chi = -\mathcal{H}\vartheta_\chi + k^2\psi_\mathrm{B} + \Gamma_{\chi\mathrm{b}}\big(\vartheta_\mathrm{b} - \vartheta_\chi\big),\\
&\dot\delta_\mathrm{b} = -\vartheta_\mathrm{b} + 3\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\mathrm{b} = -\mathcal{H}\vartheta_\mathrm{b} + c_\mathrm{s}^2k^2\delta_\mathrm{b} + k^2\psi_\mathrm{B} + \frac{a\,x_\mathrm{e}n_e\sigma_\mathrm{T}}{\mathsf{R}}\big(\vartheta_\gamma - \vartheta_\mathrm{b}\big) + \frac{\rho_\chi}{\rho_\mathrm{b}}\,\Gamma_{\chi\mathrm{b}}\big(\vartheta_\chi - \vartheta_\mathrm{b}\big), \qquad \mathsf{R} = \frac{3\rho_\mathrm{b}}{4\rho_\gamma},\\
&\dot\delta_\gamma = -\frac{4}{3}\vartheta_\gamma + 4\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\gamma = k^2\Big(\frac{\delta_\gamma}{4} - \Sigma_\gamma\Big) + k^2\psi_\mathrm{B} + a\,x_\mathrm{e}n_e\sigma_\mathrm{T}\big(\vartheta_\mathrm{b} - \vartheta_\gamma\big),\\
&\dot{\mathsf{F}}_l = \frac{k}{2l+1}\big(l\,\mathsf{F}_{l-1} - (l+1)\,\mathsf{F}_{l+1}\big) - a\,x_\mathrm{e}n_e\sigma_\mathrm{T}\Big(\mathsf{F}_l - \frac{\mathsf{G}_0 + \mathsf{F}_2 + \mathsf{G}_2}{10}\,\delta_{l2}\Big) \quad (l \geq 2),\\
&\dot{\mathsf{G}}_l = \frac{k}{2l+1}\big(l\,\mathsf{G}_{l-1} - (l+1)\,\mathsf{G}_{l+1}\big) - a\,x_\mathrm{e}n_e\sigma_\mathrm{T}\Big(\mathsf{G}_l - \frac{\mathsf{G}_0 + \mathsf{F}_2 + \mathsf{G}_2}{2}\Big(\delta_{l0} + \frac{\delta_{l2}}{5}\Big)\Big) \quad (l \geq 0),\\
&\mathsf{F}_0 = \delta_\gamma, \qquad \mathsf{F}_1 = \frac{4\vartheta_\gamma}{3k}, \qquad \mathsf{F}_2 = 2\Sigma_\gamma,\\
&\dot\delta_\nu = -\frac{4}{3}\vartheta_\nu + 4\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\nu = k^2\Big(\frac{\delta_\nu}{4} - \Sigma_\nu\Big) + k^2\psi_\mathrm{B}, \qquad \dot{\mathsf{N}}_l = \frac{k}{2l+1}\big(l\,\mathsf{N}_{l-1} - (l+1)\,\mathsf{N}_{l+1}\big) \quad (l \geq 2),\\
&\mathsf{N}_1 = \frac{4\vartheta_\nu}{3k}, \qquad \mathsf{N}_2 = 2\Sigma_\nu,\\
&\dot\Upsilon_0 = -\frac{\mathsf{q}k}{\mathsf{E}}\Upsilon_1 - \dot\phi_\mathrm{B}\frac{\mathrm{d}\ln\mathcal{F}_0}{\mathrm{d}\ln\mathsf{q}}, \qquad \dot\Upsilon_1 = \frac{\mathsf{q}k}{3\mathsf{E}}\big(\Upsilon_0 - 2\Upsilon_2\big) - \frac{\mathsf{E}k}{3\mathsf{q}}\,\psi_\mathrm{B}\frac{\mathrm{d}\ln\mathcal{F}_0}{\mathrm{d}\ln\mathsf{q}},\\
&\dot\Upsilon_l = \frac{\mathsf{q}k}{(2l+1)\mathsf{E}}\big(l\,\Upsilon_{l-1} - (l+1)\,\Upsilon_{l+1}\big) \quad (l \geq 2), \qquad \mathsf{E} = \sqrt{\mathsf{q}^2 + a^2m_i^2}, \qquad \mathcal{F}_0 = \frac{1}{\mathrm{e}^{\mathsf{q}/\mathsf{T}_\nu^{(0)}} + 1}
\end{aligned}
\tag{FE-55}
$$

Symbols in (FE-55).

- δ_χ, $\dot\delta_\chi$: The density contrast of the dark matter (the bound states χ of dirac16complex), the perturbation of its energy density divided by its mean, and its derivative with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- ϑ_χ, $\dot\vartheta_\chi$: The divergence of the velocity of the dark matter (for a Fourier mode, ik times its velocity along the wave vector), and its derivative. Unit: 1/length. Status: STANDARD.
- ϕ_B, $\dot\phi_\mathrm{B}$, $\ddot\phi_\mathrm{B}$: The Bardeen potential of the space-space part of the perturbed metric of the fold (the curvature potential) in the conformal Newtonian gauge, and its first and second derivatives with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- $\mathcal{H}$: The conformal expansion rate, $\mathcal{H} = \dot a/a = a\,a_4'$. Unit: 1/length. Status: STANDARD.
- k: The comoving wave number of a Fourier mode of the perturbations, $k = |\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.
- ψ_B, $\dot\psi_\mathrm{B}$: The Bardeen potential of the time-time part of the perturbed metric of the fold (the Newtonian potential) in the conformal Newtonian gauge, a function of τ_c and of the wave number k, and its derivative with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- $\Gamma_{\chi\mathrm{b}}$: The rate (per conformal time) at which the dark matter exchanges momentum with the baryons through the portal couplings of FE-47 (elastic scattering of χ on nuclei and electrons). Unit: 1/length. Value: negligible for the couplings that direct detection allows. Status: DERIVED.
- ϑ_b, $\dot\vartheta_\mathrm{b}$: The divergence of the velocity of the baryons and its derivative. Unit: 1/length. Status: STANDARD.
- δ_b, $\dot\delta_\mathrm{b}$: The density contrast of the baryons and its derivative. Unit: dimensionless. Status: STANDARD.
- c_s: The sound speed of the baryon gas: its square is the temperature of the gas divided by the mean mass per particle, times one minus a third of the logarithmic derivative of that temperature with respect to a. Unit: dimensionless. Status: STANDARD.
- a, $\dot a$, $\ddot a$: The scale factor of the fold, $a = \mathrm{e}^{a_4}$ (normalised to 1 today by the choice of the comoving coordinates x_1, x_2, x_3), and its first and second derivatives with respect to the conformal time τ_c. Unit: dimensionless. Status: STANDARD.
- x_e: The ionisation fraction: the fraction of the electrons that are free. Unit: dimensionless. Status: STANDARD.
- n_e: The number density of electrons (bound and free; in a supernova all are free, in the recombining universe $x_\mathrm{e}n_e$ is the density of the free electrons). Unit: GeV^3 (cm^{-3}). Status: STANDARD.
- σ_T: The Thomson cross-section of the scattering of photons on free electrons. Unit: GeV^{-2}. Value: $6.6524587 \times 10^{-25}$ cm^2. Status: STANDARD.
- R: The ratio of the inertia of the baryons to that of the photons, $\mathsf{R} = 3\rho_\mathrm{b}/(4\rho_\gamma)$. Unit: dimensionless. Status:

STANDARD.

- ϑ_γ, $\dot{\vartheta}_\gamma$: The divergence of the velocity of the photons and its derivative. Unit: 1/length. Status: STANDARD.
- ρ_χ: The energy density of the dark matter, the bound states χ of dirac16complex; $\rho_\chi^{(0)}$ its value today. Unit: GeV^4. Status: DERIVED.
- ρ_b: The (rest-mass) energy density of the baryons: in a supernova the local density of the matter (quoted in $\mathrm{g/cm}^3$), in cosmology the mean density of the baryons ($\rho_\mathrm{b}^{(0)}$ today). Unit: GeV^4. Status: STANDARD.
- ρ_γ: The energy density of the photons. Unit: GeV^4. Status: STANDARD.
- δ_γ, $\dot{\delta}_\gamma$: The density contrast of the photons and its derivative. Unit: dimensionless. Status: STANDARD.
- Σ_γ: The anisotropic stress (shear) of the photons. Unit: dimensionless. Status: STANDARD.
- F_l, $\dot{\mathsf{F}}_l$: The multipoles of the temperature perturbation of the photons ($\mathsf{F}_0 = \delta_\gamma$, $\mathsf{F}_1 = 4\vartheta_\gamma/(3k)$, $\mathsf{F}_2 = 2\Sigma_\gamma$; the $F_{\gamma l}$ of Ma and Bertschinger) and their derivatives with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- l: The multipole index and number ($l = 0, 1, 2, \ldots$) of an expansion in Legendre polynomials of the direction of a photon or a neutrino relative to the wave vector, or of the CMB sky in spherical harmonics. Status: STANDARD.
- G_l, $\dot{\mathsf{G}}_l$: The multipoles of the polarisation perturbation of the photons (the $G_{\gamma l}$ of Ma and Bertschinger) and their derivatives. Unit: dimensionless. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta_N^M = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}} = 1$ for $X = \mathrm{L}$); with upper and lower index strings the generalised Kronecker delta $\delta_{N_1 \cdots N_n}^{M_1 \cdots M_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta_{N_b}^{M_a}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- δ_ν, $\dot{\delta}_\nu$: The density contrast of the relativistic neutrinos and its derivative. Unit: dimensionless. Status: STANDARD.
- ϑ_ν, $\dot{\vartheta}_\nu$: The divergence of the velocity of the relativistic neutrinos and its derivative. Unit: 1/length. Status: STANDARD.
- Σ_ν: The anisotropic stress (shear) of the relativistic neutrinos. Unit: dimensionless. Status: STANDARD.
- N_l, $\dot{\mathsf{N}}_l$: The multipoles of the perturbation of the relativistic neutrinos ($\mathsf{N}_0 = \delta_\nu$, $\mathsf{N}_1 = 4\vartheta_\nu/(3k)$, $\mathsf{N}_2 = 2\Sigma_\nu$) and their derivatives. Unit: dimensionless. Status: STANDARD.
- $\Upsilon_l(\mathsf{q})$, $\dot{\Upsilon}_l$: The multipoles of the relative perturbation of the distribution of a massive neutrino state at the comoving momentum q (the Ψ_l of Ma and Bertschinger) and their derivatives. Unit: dimensionless. Status: STANDARD.
- q: The comoving momentum of a neutrino, $a|\vec{p}|$. Unit: GeV. Status: STANDARD.
- E: The comoving energy of a massive neutrino (a times its energy). Unit: GeV. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $\mathcal{F}_0(\mathsf{q})$: The unperturbed (Fermi-Dirac) occupation number of the cosmic neutrino background at the comoving momentum q. Unit: dimensionless. Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; in Δm^2 the squared-mass difference of two neutrino mass states ($m_2^2 - m_1^2$ for the two states of a resonance, multiplied by v_loc^2/v^2 where this differs from 1). Unit: GeV. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- T_ν: The temperature of the cosmic neutrino background; $\mathsf{T}_\nu^{(0)}$ its value today. Unit: GeV (or K). Value: 1.945 K today. Status: STANDARD.

$$
\begin{aligned}
r_\mathrm{s} &= \int_0^{\tau_\mathrm{d}} c_{\gamma\mathrm{b}}\, \mathrm{d}\tau_\mathrm{c}, \qquad c_{\gamma\mathrm{b}} = \frac{1}{\sqrt{3(1+\mathsf{R})}}, \\
D_\mathrm{M}(\mathsf{z}) &= \int_0^{\mathsf{z}} \frac{\mathrm{d}\mathsf{z}'}{H_4(\mathsf{z}')}, \qquad D_\mathrm{H}(\mathsf{z}) = \frac{1}{H_4(\mathsf{z})}, \qquad D_\mathrm{V}(\mathsf{z}) = \left(\mathsf{z}\, D_\mathrm{M}^2(\mathsf{z})\, D_\mathrm{H}(\mathsf{z})\right)^{1/3}
\end{aligned}
\tag{FE-56}
$$

Symbols in (FE-56).

- r_s: The comoving sound horizon at the drag epoch: the comoving distance that a sound wave of the photon-baryon fluid travels until the baryons are released from the photons. Unit: length. Value: about 147 Mpc

(Planck 2018). Status: STANDARD.

- τ_{d}: The conformal time of the drag epoch, when the baryons are released from the photons (redshift about 1060). Unit: length (= time). Status: STANDARD.
- $c_{\gamma\mathrm{b}}$: The sound speed of the tightly coupled photon-baryon fluid. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- τ_{c}: The conformal time of the fold, $\mathrm{d}\tau_{\mathrm{c}} = \mathrm{d}x_4/a$; a dot on a quantity of the cosmological perturbations is $\mathrm{d}/\mathrm{d}\tau_{\mathrm{c}}$. Unit: length (= time). Status: STANDARD.
- R: The ratio of the inertia of the baryons to that of the photons, $\mathsf{R} = 3\rho_{\mathrm{b}}/(4\rho_\gamma)$. Unit: dimensionless. Status: STANDARD.
- $D_{\mathrm{M}}(\mathsf{z})$: The transverse comoving distance (comoving angular-diameter distance) to the redshift z, for a spatially flat fold. Unit: length. Status: STANDARD.
- z: The redshift, $1 + \mathsf{z} = 1/a$; z' an integration variable. Unit: dimensionless. Status: STANDARD.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- $D_{\mathrm{H}}(\mathsf{z})$: The Hubble distance at the redshift z. Unit: length. Status: STANDARD.
- $D_{\mathrm{V}}(\mathsf{z})$: The volume-averaged distance to the redshift z. Unit: length. Status: STANDARD.

$$
\begin{aligned}
C_l^{\mathrm{TT}} &= 4\pi \int \frac{\mathrm{d}k}{k}\, \mathcal{P}_{\mathcal{R}}(k)\, \big(\mathcal{I}_l^{\mathrm{T}}(k)\big)^2, \qquad & C_l^{\mathrm{TE}} &= 4\pi \int \frac{\mathrm{d}k}{k}\, \mathcal{P}_{\mathcal{R}}(k)\, \mathcal{I}_l^{\mathrm{T}}(k)\, \mathcal{I}_l^{\mathrm{E}}(k), \\
C_l^{\mathrm{EE}} &= 4\pi \int \frac{\mathrm{d}k}{k}\, \mathcal{P}_{\mathcal{R}}(k)\, \big(\mathcal{I}_l^{\mathrm{E}}(k)\big)^2, \qquad & \mathcal{P}_{\mathcal{R}}(k) &= A_{\mathrm{s}} \Big(\frac{k}{k_*}\Big)^{n_{\mathrm{S}}-1}
\end{aligned}
\tag{FE-57}
$$

Symbols in (FE-57).

- C_l^{TT}, C_l^{TE}, C_l^{EE}: The angular power spectra of the CMB at the multipole l: temperature with temperature, temperature with E-mode polarisation, E-mode polarisation with itself. Unit: dimensionless. Status: STANDARD.
- l: The multipole index and number ($l = 0, 1, 2, \ldots$) of an expansion in Legendre polynomials of the direction of a photon or a neutrino relative to the wave vector, or of the CMB sky in spherical harmonics. Status: STANDARD.
- T, E: The labels of the CMB observables: T the temperature anisotropy, E the E-mode (gradient) polarisation; TT, TE, EE label the spectra of their products. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- k: The comoving wave number of a Fourier mode of the perturbations, $k = |\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.
- $\mathcal{P}_{\mathcal{R}}(k)$: The primordial power spectrum of the comoving curvature perturbation, per logarithmic interval of k. Unit: dimensionless. Status: ASSUMED.
- $\mathcal{I}_l^{\mathrm{T}}(k)$, $\mathcal{I}_l^{\mathrm{E}}(k)$: The transfer functions of the CMB: the multipole l of the temperature or E-mode anisotropy today produced by a Fourier mode k of unit primordial curvature perturbation, computed from FE-54 and FE-55 (the line-of-sight integral). Unit: dimensionless. Status: STANDARD.
- A_{s}: The amplitude of the primordial curvature spectrum at the pivot k_*. Unit: dimensionless. Value: 2.10×10^{-9} (Planck 2018). Status: ASSUMED.
- k_*: The pivot wave number of the primordial spectrum. Unit: GeV (1/length). Value: 0.05 Mpc^{-1}. Status: STANDARD.
- n_{S}: The spectral index of the primordial curvature spectrum. Unit: dimensionless. Value: 0.965 (Planck 2018). Status: ASSUMED.

$$\Omega_\chi = \frac{\rho_\chi^{(0)}}{\rho_\mathrm{c}^{(0)}}, \qquad \Omega_\mathrm{b} = \frac{\rho_\mathrm{b}^{(0)}}{\rho_\mathrm{c}^{(0)}}, \qquad \Omega_\Lambda = \frac{\rho_\Lambda}{\rho_\mathrm{c}^{(0)}}, \qquad \Omega_\nu h_{100}^2 = \frac{\sum_i m_i}{93.14\ \mathrm{eV}},$$
$$\rho_\mathrm{c}^{(0)} = \frac{3\big(H_4^{(0)}\big)^2}{8\pi G_\mathrm{N}}, \qquad H_4^{(0)} = 100\,h_{100}\ \mathrm{km/s/Mpc} \tag{FE-58}$$

Symbols in (FE-58).

- Ω_χ: The density parameter of the dark matter today. Unit: dimensionless. Value: $\Omega_\chi h_{100}^2 = 0.120$ (Planck 2018). Status: DERIVED.
- ρ_χ: The energy density of the dark matter, the bound states χ of dirac16complex; $\rho_\chi^{(0)}$ its value today. Unit: GeV^4. Status: DERIVED.
- ρ_c: The critical density, the energy density of a spatially flat fold at the expansion rate H_4; $\rho_\mathrm{c}^{(0)}$ its value today. Unit: GeV^4. Value: $\rho_\mathrm{c}^{(0)} = 1.0537\times10^{-5}\,h_{100}^2$ GeV/cm^3. Status: STANDARD.
- Ω_b: The density parameter of the baryons today. Unit: dimensionless. Value: $\Omega_\mathrm{b} h_{100}^2 = 0.0224$ (Planck 2018). Status: STANDARD.
- ρ_b: The (rest-mass) energy density of the baryons: in a supernova the local density of the matter (quoted in g/cm^3), in cosmology the mean density of the baryons ($\rho_\mathrm{b}^{(0)}$ today). Unit: GeV^4. Status: STANDARD.
- Ω_Λ: The density parameter of the dark energy (the vacuum energy of the fold) today. Unit: dimensionless. Value: 0.685 (Planck 2018). Status: DERIVED.
- ρ_Λ: The vacuum energy density of the fold, the dark energy of UM-3: $\rho_\Lambda = \rho_\mathrm{ff} - 31HV_*/(4\kappa)$ (FE-59). Unit: GeV^4. Status: DERIVED.
- Ω_ν, $\Omega_\nu^{(i)}$: The density parameter of the neutrinos of the CNB: Ω_ν that of the massive states today; $\Omega_\nu^{(i)}(\mathsf{z})$ the energy density of the mass state i at the redshift z in units of the critical density today. Unit: dimensionless. Status: STANDARD.
- h_{100}: The present expansion rate of the fold in units of 100 km/s/Mpc. Unit: dimensionless. Value: 0.674 (Planck 2018). Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; in Δm^2 the squared-mass difference of two neutrino mass states ($m_2^2 - m_1^2$ for the two states of a resonance, multiplied by v_loc^2/v^2 where this differs from 1). Unit: GeV. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_N: Newton's constant of four-dimensional gravity on the fold, $G_\mathrm{N} = 1/M_\mathrm{Pl}^2$. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.

The neutrinos in the expanding universe (FE-53), the scalar perturbations of the metric of the fold in the conformal Newtonian gauge and the Einstein equations of the fold to first order (FE-54), the Boltzmann equations of the species (FE-55), the BAO ruler and distances (FE-56), the angular power spectra of the CMB (FE-57) and the density parameters (FE-58). Status: STANDARD (UM-3: the dark matter is χ, the dark energy is the vacuum energy of the fold).

9.1 The background and the BAO ruler

$$\Big(\frac{H_4(\mathsf{z})}{H_4^{(0)}}\Big)^2 = \Omega_\gamma(1+\mathsf{z})^4 + \sum_{i=1}^{3}\Omega_\nu^{(i)}(\mathsf{z}) + \big(\Omega_\mathrm{b}+\Omega_\chi\big)(1+\mathsf{z})^3 + \Omega_\Lambda,$$
$$\Omega_\nu^{(i)}(\mathsf{z}) = \frac{7}{8}\Big(\frac{\mathsf{T}_{\nu,\mathrm{eff}}}{\mathsf{T}_\gamma}\Big)^4\Omega_\gamma\,(1+\mathsf{z})^4\,\frac{\mathsf{I}\big(m_i/\mathsf{T}_{\nu,\mathrm{eff}}\big)}{\mathsf{I}(0)}, \qquad \mathsf{I}(y) = \int_0^\infty \mathrm{d}\mathsf{q}\,\frac{\mathsf{q}^2\sqrt{\mathsf{q}^2+y^2}}{\mathrm{e}^{\mathsf{q}}+1}, \qquad \mathsf{T}_{\nu,\mathrm{eff}} = \Big(\frac{N_\mathrm{eff}}{3}\Big)^{1/4}\Big(\frac{4}{11}\Big)^{1/3}\mathsf{T}_\gamma \tag{S-21}$$

Symbols in (S-21).

- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift;

$H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.

- z: The redshift, $1 + \mathsf{z} = 1/a$; z' an integration variable. Unit: dimensionless. Status: STANDARD.
- Ω_γ: The density parameter of the photons today. Unit: dimensionless. Value: 5.450×10^{-5} (for $h_{100} = 0.6736$). Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- Ω_ν, $\Omega_\nu^{(i)}$: The density parameter of the neutrinos of the CNB: Ω_ν that of the massive states today; $\Omega_\nu^{(i)}(\mathsf{z})$ the energy density of the mass state i at the redshift z in units of the critical density today. Unit: dimensionless. Status: STANDARD.
- Ω_b: The density parameter of the baryons today. Unit: dimensionless. Value: $\Omega_\mathrm{b} h_{100}^2 = 0.0224$ (Planck 2018). Status: STANDARD.
- Ω_χ: The density parameter of the dark matter today. Unit: dimensionless. Value: $\Omega_\chi h_{100}^2 = 0.120$ (Planck 2018). Status: DERIVED.
- Ω_Λ: The density parameter of the dark energy (the vacuum energy of the fold) today. Unit: dimensionless. Value: 0.685 (Planck 2018). Status: DERIVED.
- $\mathsf{T}_{\nu,\mathrm{eff}}$: The effective temperature of the neutrinos that carries N_eff (the convention of CAMB). Unit: GeV (or K). Status: STANDARD.
- T_γ: The temperature of the photons (of the CMB); $\mathsf{T}_\gamma^{(0)}$ its value today. Unit: GeV (or K). Value: 2.7255 K today. Status: STANDARD.
- $\mathsf{I}(y)$: The Fermi-Dirac energy integral of a neutrino state with the mass-to-temperature ratio y; $\mathsf{I}(0) = 7\pi^4/120$. Unit: dimensionless. Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; in Δm^2 the squared-mass difference of two neutrino mass states ($m_2^2 - m_1^2$ for the two states of a resonance, multiplied by v_loc^2/v^2 where this differs from 1). Unit: GeV. Status: STANDARD.
- y: The ratio of the mass of a neutrino state to its temperature, the argument of I. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- q: The comoving momentum of a neutrino, $a|\vec{p}|$. Unit: GeV. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.
- N_eff: The effective number of neutrino species: the energy density of the relativistic particles other than photons in units of that of one neutrino species whose temperature is $(4/11)^{1/3}$ times that of the photons. Unit: dimensionless. Value: 3.044 (Standard Model); measured 2.99 ± 0.17 (Planck 2018). Status: STANDARD.

The expansion rate of the fold as a function of the redshift z (FE-59 with the content of UM-3): photons, the three neutrino states with their exact Fermi-Dirac energy integrals (each at the temperature $\mathsf{T}_{\nu,\mathrm{eff}}$ that carries $N_\mathrm{eff} = 3.044$, the convention of CAMB), baryons, χ, and the vacuum energy Ω_Λ that makes the fold spatially flat. Computed here it agrees with CAMB to 8.7×10^{-7} for $0 \le \mathsf{z} \le 10^4$, with $\Omega_\gamma = 5.450 \times 10^{-5}$, $\Omega_\nu = 1.404 \times 10^{-3}$ today (the massless state included), $\Omega_\mathrm{b} = 0.0493$, $\Omega_\chi = 0.2645$, $\Omega_\Lambda = 0.6848$; the age of the fold universe is 13.798 Gyr. The terms of order ϵ_r change H_4, the distances and the ruler by $5\epsilon_\mathrm{r}/12 \lesssim 4 \times 10^{-16}$ at fixed κ_4, and not at all at fixed G_N. Status: DERIVED (`um3_cosmology.py`, C1).

The ruler FE-56, integrated here with the same background up to the drag epoch $\mathsf{z}_\mathrm{d} = 1059.93$ of CAMB, is $r_\mathrm{s} = 147.1047$ Mpc (CAMB: 147.1047 Mpc), and the distances D_M and D_H at the seven redshifts of DESI agree with CAMB to 8.4×10^{-7} (for example $D_\mathrm{M} = 1985.92$ Mpc and $D_\mathrm{H} = 3345.07$ Mpc at $\mathsf{z} = 0.51$). Status: DERIVED (C2).

9.2 The CMB

quantity	UM-3	Planck 2018	difference in standard deviations
redshift of last scattering z_*	1089.905	1089.92 ± 0.25	-0.06

quantity	UM-3	Planck 2018	difference in standard deviations
sound horizon r_* (Mpc)	144.445	144.43 ± 0.26	+0.06
acoustic angle $100\,\theta_*$	1.04108	1.04110 ± 0.00031	−0.06
drag redshift z_d	1059.931	1059.94 ± 0.30	−0.03
ruler $r_\mathrm{s}(\mathsf{z}_\mathrm{d})$ (Mpc)	147.105	147.09 ± 0.26	+0.06
damping wave number k_D (Mpc^{-1})	0.140865	0.14087 ± 0.00030	−0.02
matter-radiation equality z_eq	3402.9	3402 ± 26	+0.03
σ_8	0.8120	0.8111 ± 0.0060	+0.15
age (Gyr)	13.7977	13.797 ± 0.023	+0.03

The derived quantities of the CMB of UM-3 with the Planck 2018 parameters (the column TT,TE,EE+lowE+lensing of Table 2 of Planck 2018 VI) lie within 0.2 standard deviations of the Planck values; the small differences come from the three neutrino states of UM-3 in place of Planck's single state of 0.06 eV. Status: DERIVED (C3).

$$\chi^2_\mathrm{fit} = \left(\mathsf{X}_\mathrm{model} - \mathsf{X}_\mathrm{data}\right)^T \mathsf{C}^{-1} \left(\mathsf{X}_\mathrm{model} - \mathsf{X}_\mathrm{data}\right), \qquad \mathsf{X}_b = \sum_{l \text{ in the bin } b} \mathsf{w}_l\, C_l \tag{S-22}$$

Symbols in (S-22).

- χ^2_fit: The chi-square of a comparison of a model with data (written χ^2 in the text): the squared deviations weighted with the inverse covariance. Unit: dimensionless. Status: STANDARD.
- $\mathsf{X}_\mathrm{model}$, X_data: The vectors of the model values and of the measured values of a data set (binned CMB spectra, BAO distances). Unit: as the data. Status: STANDARD.
- ^{T}: The transpose of a matrix. Status: STANDARD.
- C: The covariance matrix of the measured values; C^{-1} its inverse. Unit: as the data squared. Status: STANDARD.
- X_b: A model value of the binned CMB spectrum, for the bin b of the plik-lite binning. Unit: $\mu\mathrm{K}^2$. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- l: The multipole index and number ($l = 0, 1, 2, \ldots$) of an expansion in Legendre polynomials of the direction of a photon or a neutrino relative to the wave vector, or of the CMB sky in spherical harmonics. Status: STANDARD.
- w_l: The Planck weight of the multipole l in its bin. Unit: dimensionless. Status: STANDARD.
- C_l^TT, C_l^TE, C_l^EE: The angular power spectra of the CMB at the multipole l: temperature with temperature, temperature with E-mode polarisation, E-mode polarisation with itself. Unit: dimensionless. Status: STANDARD.

The comparison with measurements, χ^2_fit (the usual χ^2): the measured values X_data (binned spectra or BAO distances), their covariance matrix C, and the model values $\mathsf{X}_\mathrm{model}$; for the Planck 2018 high-l likelihood (plik-lite: foreground-marginalised spectra in 215 bins of TT for $30 \le l \le 2508$ and 199 bins each of TE and EE for $30 \le l \le 1996$, with the Planck calibration fixed at 1) a model bin is the weighted sum X_b of the C_l of the multipoles of the bin with the Planck weights w_l; two Gaussian bins of the low-l temperature spectrum can be added (Prince and Dunkley 2019). The implementation reproduces the reference log-likelihoods of the plik-lite code on its test spectrum to 10^{-14} (-101.5812307 for TT, -293.9558650 for TT, TE, EE with the low-l bins). Status: STANDARD (the likelihood), DERIVED (its implementation, C4).

model	N_eff	χ^2 high-l TT, TE, EE (613 bins)	χ^2 TT (215 bins)	refitted χ^2 (615 bins)	$H_4^{(0)}$ after the refit
UM-3	3.044	583.8	206.2	587.9	67.31

model	N_{eff}	χ^2 high-l TT, TE, EE (613 bins)	χ^2 TT (215 bins)	refitted χ^2 (615 bins)	$H_4^{(0)}$ after the refit
UM-3 with thermal ν_{R}, same θ_*	3.185	604.3	215.1	589.5	68.23
Planck's convention (one state of 0.06 eV)	3.044	583.9	206.1		

The CMB spectra of UM-3 against the Planck 2018 data: $\chi^2 = 583.8$ for 613 bins (expected 613 ± 35). Three thermal right-handed neutrinos at the same acoustic angle and the same other parameters raise χ^2 by 20.5 (the damping tail); when $\Omega_{\mathrm{b}}h^2_{100}$, $\Omega_\chi h^2_{100}$, θ_*, A_{s} and n_{S} are refitted (Gauss-Newton with the Jacobian of CAMB, τ_{reion} fixed) to the 613 high-l bins and the two low-l bins, the difference shrinks to $\Delta\chi^2 = +1.6$, in line with the Planck bound $N_{\mathrm{eff}} = 2.99 \pm 0.17$ (the refit shifts n_{S} from 0.9650 to 0.9702 and $H_4^{(0)}$ from 67.31 to 68.23 km/s/Mpc; the errors from the Fisher matrix, $\sigma(\Omega_{\mathrm{b}}h^2_{100}) = 0.00015$, $\sigma(\Omega_\chi h^2_{100}) = 0.0013$, $\sigma(100\theta_*) = 0.0003$, $\sigma(n_{\mathrm{S}}) = 0.0042$, match Planck's). The spectra are shown in the figure. Status: DERIVED (C4).

9.3 The BAO

tracer	z	measured (DESI DR2)	UM-3	χ^2
BGS	0.295	$D_{\mathrm{V}}/r_{\mathrm{s}} = 7.942 \pm 0.075$	8.057	2.34
LRG1	0.510	$D_{\mathrm{M}}/r_{\mathrm{s}} = 13.588 \pm 0.167$, $D_{\mathrm{H}}/r_{\mathrm{s}} = 21.863 \pm 0.425$	13.500, 22.739	4.48
LRG2	0.706	17.351 ± 0.177, 19.455 ± 0.330	17.700, 20.169	14.36
LRG3+ELG1	0.934	21.576 ± 0.152, 17.641 ± 0.193	21.994, 17.569	8.29
ELG2	1.321	27.601 ± 0.318, 14.176 ± 0.221	28.081, 14.065	2.31
QSO	1.484	30.512 ± 0.760, 12.817 ± 0.516	30.275, 12.880	0.10
Lyα	2.330	38.988 ± 0.531, 8.632 ± 0.101	39.182, 8.616	0.13

The BAO distances of UM-3 with the Planck parameters ($\Omega_{\mathrm{m}} = 0.3152$, $h_{100}r_{\mathrm{s}} = 99.09$ Mpc) against DESI DR2 (DESI Collaboration 2025, Table IV; χ^2 with the correlation of D_{M} and D_{H} of each tracer): $\chi^2 = 32.0$ for 13 measurements. Fitted to DESI alone, $\Omega_{\mathrm{m}} = 0.2971$ and $h_{100}r_{\mathrm{s}} = 101.56$ Mpc with $\chi^2 = 10.5$ for 11 degrees of freedom ($p = 0.48$), as DESI finds for the cosmological-constant model; the Planck point lies $\Delta\chi^2 = 21.5$ above this optimum when its own errors are neglected (with the CMB errors DESI quotes a 2.3 standard-deviation tension between BAO and CMB). The dark energy of UM-3 is a vacuum energy with $w = -1$ to order ϵ_{r}: UM-3 inherits this tension and cannot produce the evolving dark energy ($w_0 > -1$, $w_a < 0$) that DESI DR2 with the CMB prefers at 3.1 standard deviations. Status: DERIVED (C5).

9.4 The CNB

$$\Gamma_{\mathrm{C}\nu\mathrm{B}} = N_{\mathrm{T}}\,\bar{\sigma}_{\mathrm{T}}\,\frac{n_\nu^{(0)}}{2}\sum_{i=1}^{3}|U_{\mathrm{e}i}|^2\big(1+\mathsf{s}_i\big) \qquad \text{(Dirac neutrinos)} \tag{S-23}$$

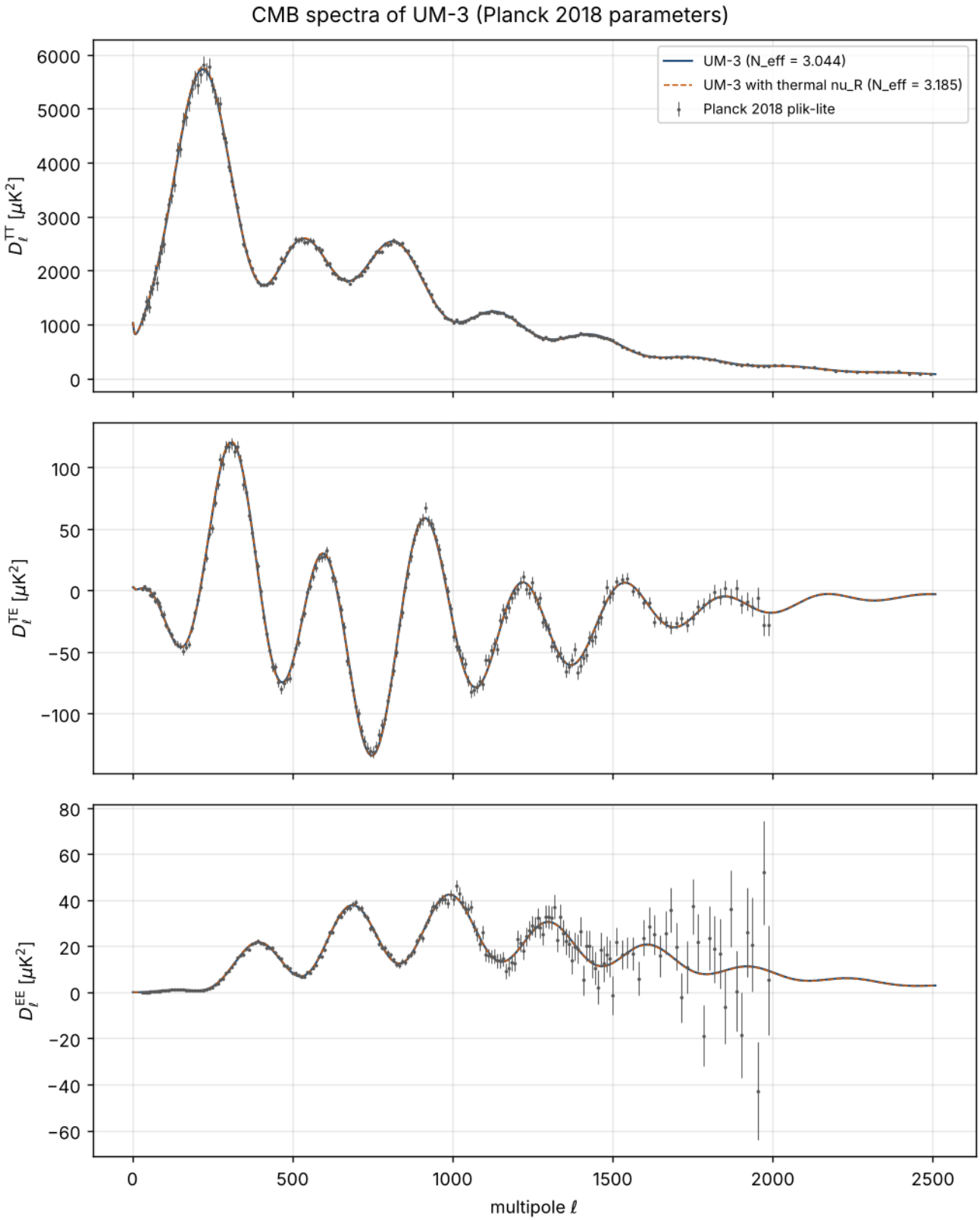


Figure 1: The CMB spectra of UM-3 with the Planck 2018 parameters ($D_l = l(l+1)C_l/(2\pi)$ of the temperature, the cross-correlation and the E-mode polarisation, lensed), with three thermal right-handed neutrinos (dashed), and the Planck 2018 plik-lite data.

Symbols in (S-23).

- $\Gamma_{\mathrm{C}\nu\mathrm{B}}$: The rate at which a tritium target captures neutrinos of the cosmic neutrino background. Unit: 1/time. Status: STANDARD.
- N_{T}: The number of tritium nuclei of the target. Unit: dimensionless. Value: 2.0×10^{25} for 100 g. Status: STANDARD.
- $\bar{\sigma}_{\mathrm{T}}$: The averaged cross-section of the capture of a non-relativistic electron neutrino on tritium, times the speed of light. Unit: cm^2. Value: 3.834×10^{-45} cm^2. Status: STANDARD.
- n_ν: The number density of the cosmic neutrino background per flavour (neutrinos plus antineutrinos, one helicity each); $n_\nu^{(0)}$ its value today. Unit: GeV^3. Value: 112 cm^{-3} today. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- U_{ei}: The electron row of the mixing matrix, $U_{\mathrm{e}i} = (U_{\mathrm{PMNS}})_{ei}$: $|U_{\mathrm{e}i}|^2$ is the electron-flavour fraction of the mass state i. Unit: dimensionless. Value: $|U_{\mathrm{e}i}|^2 = 0.681, 0.296, 0.022$. Status: STANDARD.
- s_i: The speed of the neutrino mass state i today (0 for a non-relativistic state, 1 for a massless one). Unit: dimensionless. Status: STANDARD.

The capture rate of the cosmic neutrinos on a target of N_{T} tritium nuclei ($\nu_e + {}^3\mathrm{H} \to {}^3\mathrm{He} + e^-$, the PTOLEMY idea): each mass state i contributes with its electron fraction $|U_{\mathrm{e}i}|^2$; for Dirac neutrinos only the left-handed helicity of each state is populated ($n_\nu^{(0)}/2 = 56$ cm^{-3} per state), and it is captured with the averaged cross-section $\bar{\sigma}_{\mathrm{T}} = 3.834 \times 10^{-45}$ cm^2 times $(1 + \mathsf{s}_i)$, s_i the speed of the state (Long, Lunardini and Sabancilar 2014). Status: STANDARD.

The CNB of UM-3: the temperature $\mathsf{T}_\nu^{(0)} = (4/11)^{1/3}\mathsf{T}_\gamma^{(0)} = 1.9454$ K (1.676×10^{-4} eV); 112.0 neutrinos and antineutrinos per cm^3 per flavour, 336 in all (FE-53); the states ν_2 and ν_3 became non-relativistic at $\mathsf{z} = 15.3$ and 94 (where their mass equals the mean momentum $3.15\mathsf{T}_\nu$), ν_1 is massless; $\Omega_\nu h^2_{100} = 6.31 \times 10^{-4}$ for the two massive states (FE-58: $\sum m_i/93.14$ eV $= 6.30 \times 10^{-4}$; CAMB: 6.31×10^{-4}). Because the right-handed states are not populated, the CNB of UM-3 has one helicity per state: a 100 g tritium target captures 4.1 cosmic neutrinos per year if all states are non-relativistic (Majorana neutrinos: 8.1, twice as many; Long, Lunardini and Sabancilar: about 4 and 8), and 6.8 per year with the massless ν_1 of UM-3, whose left-handed helicity is captured twice as often (these captures lie at the end point of the beta spectrum, where only a mass of ν_1 would separate them). The sum of the masses, 0.0587 eV, lies below the bounds of DESI DR2 with the CMB (0.064 eV) and of ACT DR6 (0.082 eV, 95%). Status: DERIVED (C6).

10 What the solution does not settle

- **The media of the bulk** (ASSUMED). The late universe needs a medium in the bulk that keeps the bulk static (the stabiliser) and, with frozen extra times, a medium of the frozen branch; both violate the null energy condition (S-3), so no ordinary field provides them, and UM-3 gives no field theory for them. The power series additionally needs the medium to respond to the matter of the fold with the radion mode (S-8, S-9); a rigid medium would change the expansion law in a way that observation excludes.
- **The primordial era** (OPEN). The author's vacuum is degenerate (S-1: the linear response of the bulk equations vanishes, and the transverse traceless graviton has no time-kinetic term there); a varying b_4 needs bulk sources of second order in its rate (S-2); the passage from the author's vacuum to the static block with frozen extra times is not computed, and the primordial spectrum, the optical depth and the densities are taken from Planck 2018.
- **The Krein norm** (OPEN). The charge of dirac16complex is indefinite (FE-43): of the eight bound states at $+\omega_{\mathrm{b}}$ four carry positive and four negative energy. The Kohn-Sham states occupy the four of positive energy, without the negative-energy sea; whether the quantised theory with both kinds of states is stable is not settled.
- **The extra times** (ASSUMED, OPEN). Every field that depends on the extra times is tachyonic (FE-38); UM-3 keeps only fields that do not depend on them, and the stability of the theory without this restriction is open.
- **Lorentz violation in the bulk** (DERIVED). Zero modes of bulk fields and the bound particles χ move faster than light on the fold (S-12, S-13). For the Standard Model this fixes its place on the fold; for χ it requires $q_{\mathrm{b}} > H$ and, with the direct-detection bound $q_{\mathrm{b}} \lesssim 0.03\,\omega_{\mathrm{b}}$, a bulk scale H well below the mass of χ.
- **The dark energy** (ASSUMED). The vacuum energy ρ_Λ of the fold is a free parameter: UM-3 accommodates its value but does not explain it, and it has $w = -1$, so the evolving dark energy that DESI DR2 with the CMB prefers is not available.

- **The free parameters**. The bulk mass m_Ψ, the scale H, the portal couplings, the self-coupling λ_Ψ, the amplitude $v_{\mathcal{A}}$ and rate a_{40} of the ad hoc vacuum, the lightest neutrino mass (taken as zero) are not fixed by UM-3; the relic density fixes one combination of the mass of χ and its portals.
- **Neutrinos in dense matter** (STANDARD, ESTIMATE). The non-linear evolution of the fast instability and the collective conversion deep in a supernova are not solved; their outcome is modelled by the equipartition FE-52, whose effect on the explosion is taken from the transport simulations of Gogilashvili and Tamborra.

11 What was verified

Every result labelled DERIVED rests on a computation in the folder `universe_model_3/code/` of the repository, which checks itself and writes a report to `universe_model_3/reports/` (all equations and results are written out above). Each script is run with `python universe_model_3/code/NAME.py` (Python 3 with numpy, scipy, sympy, matplotlib and CAMB 2.0.5; the cosmology script downloads the Planck 2018 plik-lite files from github.com/heatherprince/planck-lite-py into `universe_model_3/external/` once).

- `um3_dirac_matrices.py` (55 checks): the author's matrices, their reality, the Clifford relations, the chiral projectors, Pin(4,4).
- `um3_geometry.py` (26 checks): the exact curvature of the author's metric, the Lovelock tensors by two routes, the trace and Bianchi identities, the background equations FE-29, the author's vacuum and its degeneracy, the spin connection and the vielbein postulate.
- `um3_fold.py` (13 checks): the proper chart, the general warped metric FE-34, the junction potential, the fold sources, the conservation law at the fold.
- `um3_bulk.py` (10 checks): the zero modes and the speeds of bulk scalars, photons, fermions and gravitons (S-11, S-12).
- `um3_fields.py` (16 checks): the bilinears of dirac16complex, the Krein operator, the electroweak masses of the ad hoc vacuum, the anomalies with three right-handed neutrinos.
- `um3_spectra.py` (5 checks): the bound states FE-40 with the exact 16×16 matrices, their Krein signature, their speed (S-13).
- `um3_background.py` (6 checks): S-1 to S-4, the ad hoc vacuum as a gauge rotation and its residual (S-10).
- `um3_series.py` (6 checks): S-5 to S-9.
- `um3_kohnsham.py` (KS_NCHECKS checks): the Kohn-Sham states (S-24 to S-28) in three regimes, lifted to sixteen components.
- `um3_neutrino.py` (5 checks): the flavour blindness of V_Ψ, the resonance formulas, the crossing criterion, the equipartition, the CNB numbers.
- `um3_flavour.py` (4 checks): S-14 to S-17.
- `um3_thermal.py` (6 checks): the degrees of freedom, S-18 to S-20, the ad hoc vacuum before nucleosynthesis.
- `um3_cosmology.py` (6 checks): S-21 to S-23 and the comparisons with Planck 2018 and DESI DR2.